

restart

▼ Solution with two media

$$eq1 := \frac{(\epsilon_{eff} - 1)}{\epsilon_{eff} + 2} = \frac{(\epsilon_1 - 1)}{\epsilon_1 + 2} \cdot \eta_1 + \frac{(\epsilon_2 - 1)}{\epsilon_2 + 2} \cdot \eta_2$$
$$\frac{\epsilon_{eff} - 1}{\epsilon_{eff} + 2} = \frac{(\epsilon_1 - 1) \eta_1}{\epsilon_1 + 2} + \frac{(\epsilon_2 - 1) \eta_2}{\epsilon_2 + 2} \quad (1.1)$$

$$\eta_1 := (1 - \text{LiquidFraction}); \eta_2 := \text{LiquidFraction};$$
$$1 - \text{LiquidFraction}$$
$$\text{LiquidFraction} \quad (1.2)$$

Analytical solution

$$\epsilon_{eff} := \epsilon_{effR} + I \epsilon_{effC}; \epsilon_1 := \epsilon_{1R} + I \epsilon_{1C}; \epsilon_2$$
$$:= \epsilon_{2R} + I \epsilon_{2C};$$
$$\frac{\epsilon_{effR} + I \epsilon_{effC}}{\epsilon_{1R} + I \epsilon_{1C}}$$
$$\frac{\epsilon_{2R} + I \epsilon_{2C}}{\epsilon_{2R} + I \epsilon_{2C}} \quad (1.3)$$

$$eq2 := \text{evalc}(eq1)$$
$$\frac{(\epsilon_{effR} - 1)(\epsilon_{effR} + 2)}{(\epsilon_{effR} + 2)^2 + \epsilon_{effC}^2} + \frac{\epsilon_{effC}^2}{(\epsilon_{effR} + 2)^2 + \epsilon_{effC}^2} \quad (1.4)$$
$$+ I \left(\frac{\epsilon_{effC}(\epsilon_{effR} + 2)}{(\epsilon_{effR} + 2)^2 + \epsilon_{effC}^2} \right.$$
$$\left. - \frac{(\epsilon_{effR} - 1)\epsilon_{effC}}{(\epsilon_{effR} + 2)^2 + \epsilon_{effC}^2} \right) = \left(\frac{(\epsilon_{1R} - 1)(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right.$$
$$\left. + \frac{\epsilon_{1C}^2}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1 - \text{LiquidFraction})$$
$$+ \left(\frac{(\epsilon_{2R} - 1)(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right.$$
$$\left. + \frac{\epsilon_{2C}^2}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction}$$
$$+ I \left(\left(\frac{\epsilon_{1C}(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} - \frac{(\epsilon_{1R} - 1)\epsilon_{1C}}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1 - \right.$$
$$\left. - \text{LiquidFraction}) + \left(\frac{\epsilon_{2C}(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right. \right.$$
$$\left. - \frac{(\epsilon_{2R} - 1)\epsilon_{2C}}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction} \Big)$$

Two equations can be built from this

$$\begin{aligned}
eq2r &:= \frac{(\epsilon_{\text{EffR}} - 1)(\epsilon_{\text{EffR}} + 2)}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} + \frac{\epsilon_{\text{EffC}}^2}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} \\
&= \left(\frac{(\epsilon_{1R} - 1)(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} + \frac{\epsilon_{1C}^2}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1. \\
&\quad - \text{LiquidFraction}) + \left(\frac{(\epsilon_{2R} - 1)(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right. \\
&\quad \left. + \frac{\epsilon_{2C}^2}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction} \\
&\quad \frac{(\epsilon_{\text{EffR}} - 1)(\epsilon_{\text{EffR}} + 2)}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} + \frac{\epsilon_{\text{EffC}}^2}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} \tag{1.5} \\
&= \left(\frac{(\epsilon_{1R} - 1)(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} + \frac{\epsilon_{1C}^2}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1. \\
&\quad - \text{LiquidFraction}) + \left(\frac{(\epsilon_{2R} - 1)(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right. \\
&\quad \left. + \frac{\epsilon_{2C}^2}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction}
\end{aligned}$$

$$\begin{aligned}
eq2c &:= \frac{\epsilon_{\text{EffC}}(\epsilon_{\text{EffR}} + 2)}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} - \frac{(\epsilon_{\text{EffR}} - 1)\epsilon_{\text{EffC}}}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} \\
&= \left(\frac{\epsilon_{1C}(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} - \frac{(\epsilon_{1R} - 1)\epsilon_{1C}}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1. \\
&\quad - \text{LiquidFraction}) + \left(\frac{\epsilon_{2C}(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right. \\
&\quad \left. - \frac{(\epsilon_{2R} - 1)\epsilon_{2C}}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction} \\
&\quad \frac{\epsilon_{\text{EffC}}(\epsilon_{\text{EffR}} + 2)}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} - \frac{(\epsilon_{\text{EffR}} - 1)\epsilon_{\text{EffC}}}{(\epsilon_{\text{EffR}} + 2)^2 + \epsilon_{\text{EffC}}^2} \tag{1.6} \\
&= \left(\frac{\epsilon_{1C}(\epsilon_{1R} + 2)}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} - \frac{(\epsilon_{1R} - 1)\epsilon_{1C}}{(\epsilon_{1R} + 2)^2 + \epsilon_{1C}^2} \right) (1. \\
&\quad - \text{LiquidFraction}) + \left(\frac{\epsilon_{2C}(\epsilon_{2R} + 2)}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right. \\
&\quad \left. - \frac{(\epsilon_{2R} - 1)\epsilon_{2C}}{(\epsilon_{2R} + 2)^2 + \epsilon_{2C}^2} \right) \text{LiquidFraction}
\end{aligned}$$

So we can find analytical solutions for both

$sol := \text{solve}(\{eq2r, eq2c\}, \{\epsilon_{\text{EffR}}, \epsilon_{\text{EffC}}\})$

$$\left\{ \epsilon_{\text{EffC}} = (-4. \epsilon_{1C} \text{LiquidFraction} \epsilon_{2R} \right. \tag{1.7}$$

$$\left. + 4. \epsilon_{1C} \epsilon_{2R} + \epsilon_{1C} \epsilon_{2C}^2 + \epsilon_{1C} \epsilon_{2R}^2 \right\}$$

$$\begin{aligned}
& + 4. \epsilon_{2C} \text{LiquidFraction} \epsilon_{1R} \\
& + \epsilon_{2C} \text{LiquidFraction} \epsilon_{1C}^2 - 4. \epsilon_{1C} \text{LiquidFraction} \\
& + \epsilon_{2C} \text{LiquidFraction} \epsilon_{1R}^2 + 4. \epsilon_{1C} \\
& - 1. \epsilon_{1C} \text{LiquidFraction} \epsilon_{2C}^2 + 4. \epsilon_{2C} \text{LiquidFraction} \\
& - 1. \epsilon_{1C} \text{LiquidFraction} \epsilon_{2R}^2) / \\
& (2. \epsilon_{1R} \text{LiquidFraction} \epsilon_{2R} + 4. - 2. \text{LiquidFraction} \epsilon_{2R}^2 \\
& - 4. \text{LiquidFraction} \epsilon_{2R} - 2. \text{LiquidFraction} \epsilon_{2C}^2 \\
& + 4. \epsilon_{1R} \text{LiquidFraction} + \epsilon_{2R}^2 + 4. \epsilon_{2R} + \epsilon_{2C}^2 \\
& + 2. \epsilon_{1C} \epsilon_{2C} \text{LiquidFraction} + \text{LiquidFraction}^2 \epsilon_{2R}^2 \\
& + \epsilon_{1R}^2 \text{LiquidFraction}^2 - 2. \text{LiquidFraction}^2 \epsilon_{2R} \epsilon_{1R} \\
& - 2. \epsilon_{2C} \text{LiquidFraction}^2 \epsilon_{1C} + \epsilon_{1C}^2 \text{LiquidFraction}^2 \\
& + \epsilon_{2C}^2 \text{LiquidFraction}^2), \epsilon_{\text{EffR}} = \\
& - (1. (2. \epsilon_{1C}^2 \text{LiquidFraction}^2 - 2. \text{LiquidFraction} \epsilon_{2R}^2 \\
& - 4. \epsilon_{1R} - 4. \epsilon_{1R} \epsilon_{2R} - 1. \epsilon_{1R} \epsilon_{2C}^2 \\
& + 4. \epsilon_{1R} \text{LiquidFraction} \epsilon_{2R} - 4. \text{LiquidFraction} \epsilon_{2R} \\
& - 1. \epsilon_{1R}^2 \text{LiquidFraction} \epsilon_{2R} \\
& + \epsilon_{1R} \text{LiquidFraction} \epsilon_{2R}^2 - 2. \epsilon_{1R}^2 \text{LiquidFraction} \\
& + \epsilon_{1R} \text{LiquidFraction} \epsilon_{2C}^2 \\
& - 1. \epsilon_{1C}^2 \text{LiquidFraction} \epsilon_{2R} + 2. \epsilon_{2C}^2 \text{LiquidFraction}^2 \\
& - 4. \epsilon_{2C} \text{LiquidFraction}^2 \epsilon_{1C} + 4. \epsilon_{1R} \text{LiquidFraction} \\
& - 1. \epsilon_{1R} \epsilon_{2R}^2 + 2. \epsilon_{1R}^2 \text{LiquidFraction}^2 \\
& - 4. \text{LiquidFraction}^2 \epsilon_{2R} \epsilon_{1R} - 2. \text{LiquidFraction} \epsilon_{2C}^2 \\
& - 2. \epsilon_{1C}^2 \text{LiquidFraction} + 4. \epsilon_{1C} \epsilon_{2C} \text{LiquidFraction}
\end{aligned}$$

$$\begin{aligned}
& + 2. \text{LiquidFraction}^2 \text{epsilon}2R^2)) / (2. \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2R \\
& + 4. - 2. \text{LiquidFraction} \text{epsilon}2R^2 - 4. \text{LiquidFraction} \text{epsilon}2R \\
& - 2. \text{LiquidFraction} \text{epsilon}2C^2 + 4. \text{epsilon}1R \text{LiquidFraction} + \text{epsilon}2R^2 \\
& + 4. \text{epsilon}2R + \text{epsilon}2C^2 + 2. \text{epsilon}1C \text{epsilon}2C \text{LiquidFraction} \\
& + \text{LiquidFraction}^2 \text{epsilon}2R^2 + \text{epsilon}1R^2 \text{LiquidFraction}^2 \\
& - 2. \text{LiquidFraction}^2 \text{epsilon}2R \text{epsilon}1R \\
& - 2. \text{epsilon}2C \text{LiquidFraction}^2 \text{epsilon}1C + \text{epsilon}1C^2 \text{LiquidFraction}^2 \\
& + \text{epsilon}2C^2 \text{LiquidFraction}^2) \}
\end{aligned}$$

#with(codegen)

#epsilonEffR:=sol[2];#cost(epsilonEffR);

Define a function for optimizing the beast.

#epsilonEffRe:=(epsilon1R, epsilon1C, epsilon2R, epsilon2C, LiquidFraction) →

$$\begin{aligned}
& - (1. (2. \text{epsilon}1C^2 \text{LiquidFraction}^2 - 2. \text{LiquidFraction} \text{epsilon}2R^2 - 4. \text{epsilon}1R \\
& + \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2C^2 - 1. \text{epsilon}1C^2 \text{LiquidFraction} \text{epsilon}2R \\
& + 4. \text{epsilon}1R \text{LiquidFraction} - 1. \text{epsilon}1R^2 \text{LiquidFraction} \text{epsilon}2R \\
& + \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2R^2 + 4. \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2R \\
& - 1. \text{epsilon}1R \text{epsilon}2C^2 - 2. \text{epsilon}1C^2 \text{LiquidFraction} \\
& + 4. \text{epsilon}1C \text{epsilon}2C \text{LiquidFraction} + 2. \text{epsilon}2C^2 \text{LiquidFraction}^2 \\
& - 2. \text{epsilon}1R^2 \text{LiquidFraction} - 1. \text{epsilon}1R \text{epsilon}2R^2 - 4. \text{epsilon}1R \text{epsilon}2R \\
& + 2. \text{epsilon}1R^2 \text{LiquidFraction}^2 - 4. \text{LiquidFraction} \text{epsilon}2R \\
& - 2. \text{LiquidFraction} \text{epsilon}2C^2 - 4. \text{epsilon}2C \text{LiquidFraction}^2 \text{epsilon}1C \\
& + 2. \text{LiquidFraction}^2 \text{epsilon}2R^2 - 4. \text{LiquidFraction}^2 \text{epsilon}2R \text{epsilon}1R)) / \\
& (2. \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2R + 4. + 4. \text{epsilon}1R \text{LiquidFraction} \\
& - 2. \text{LiquidFraction} \text{epsilon}2R^2 - 4. \text{LiquidFraction} \text{epsilon}2R \\
& - 2. \text{LiquidFraction} \text{epsilon}2C^2 + \text{epsilon}2R^2 + 4. \text{epsilon}2R + \text{epsilon}2C^2 \\
& + 2. \text{epsilon}1C \text{epsilon}2C \text{LiquidFraction} + \text{epsilon}1R^2 \text{LiquidFraction}^2 \\
& + \text{LiquidFraction}^2 \text{epsilon}2R^2 - 2. \text{LiquidFraction}^2 \text{epsilon}2R \text{epsilon}1R \\
& - 2. \text{epsilon}2C \text{LiquidFraction}^2 \text{epsilon}1C + \text{epsilon}2C^2 \text{LiquidFraction}^2 \\
& + \text{epsilon}1C^2 \text{LiquidFraction}^2)
\end{aligned}$$

#optimize(epsilonEffRe); cost(optimize(epsilonEffRe))

#epsilonEffImag:=sol[1];#cost(epsilonEffImag);

#epsilonEffImag:=(epsilon1R, epsilon1C, epsilon2R, epsilon2C, LiquidFraction)

$$\begin{aligned}
& \rightarrow (4. \text{epsilon}2C \text{LiquidFraction} - 1. \text{epsilon}1C \text{LiquidFraction} \text{epsilon}2C^2 + 4. \text{epsilon}1C \text{epsilon}2R \\
& + 4. \text{epsilon}2C \text{LiquidFraction} \text{epsilon}1R + \text{epsilon}2C \text{LiquidFraction} \text{epsilon}1C^2 \\
& - 4. \text{epsilon}1C \text{LiquidFraction} + \text{epsilon}2C \text{LiquidFraction} \text{epsilon}1R^2 \\
& + 4. \text{epsilon}1C + \text{epsilon}1C \text{epsilon}2C^2 + \text{epsilon}1C \text{epsilon}2R^2 \\
& - 1. \text{epsilon}1C \text{LiquidFraction} \text{epsilon}2R^2) / (2. \text{epsilon}1R \text{LiquidFraction} \text{epsilon}2R \\
& + 4. + 4. \text{epsilon}1R \text{LiquidFraction} - 2. \text{LiquidFraction} \text{epsilon}2R^2 \\
& - 4. \text{LiquidFraction} \text{epsilon}2R - 2. \text{LiquidFraction} \text{epsilon}2C^2 + \text{epsilon}2R^2 \\
& + 4. \text{epsilon}2R + \text{epsilon}2C^2 + 2. \text{epsilon}1C \text{epsilon}2C \text{LiquidFraction} \\
& + \text{epsilon}1R^2 \text{LiquidFraction}^2 + \text{LiquidFraction}^2 \text{epsilon}2R^2
\end{aligned}$$

– 2. $LiquidFraction^2 \epsilon_{2R} \epsilon_{1R}$ – 2. $\epsilon_{2C} LiquidFraction^2 \epsilon_{1C}$
+ $\epsilon_{2C}^2 LiquidFraction^2 + \epsilon_{1C}^2 LiquidFraction^2$)
 $\#cost(\epsilon_{effImag}); cost(optimize(\epsilon_{effImag}))$ **Optimization would not be huge.**

VERIFICATIONS

$$\begin{aligned} \epsilon_{effRe} &:= sol[2]; \epsilon_{effImag} := sol[1]; \\ eval(\epsilon_{effRe}, LiquidFraction = 0); eval(\epsilon_{effImag}, LiquidFraction = 0) \\ \epsilon_{effR} &= - \left(1. \left(-4. \epsilon_{1R} - 4. \epsilon_{1R} \epsilon_{2R} \right. \right. \\ &\quad \left. \left. - 1. \epsilon_{1R} \epsilon_{2C}^2 - 1. \epsilon_{1R} \epsilon_{2R}^2 \right) \right) / \left(4. + \epsilon_{2R}^2 \right. \\ &\quad \left. + 4. \epsilon_{2R} + \epsilon_{2C}^2 \right) \\ \epsilon_{effC} &= \left(4. \epsilon_{1C} \epsilon_{2R} + \epsilon_{1C} \epsilon_{2C}^2 \right. \\ &\quad \left. + \epsilon_{1C} \epsilon_{2R}^2 + 4. \epsilon_{1C} \right) / \left(4. + \epsilon_{2R}^2 + 4. \epsilon_{2R} \right. \\ &\quad \left. + \epsilon_{2C}^2 \right) \end{aligned} \quad (1.8)$$

Numerical application using the found analytical formula

$$\begin{aligned} \epsilon_{Si} &:= 13.64 + I \cdot 48 \\ &13.64 + 48. I \end{aligned} \quad (1.9)$$

$$\begin{aligned} \epsilon_{LSi} &:= -17 + I \cdot 49 \\ &-17 + 49 I \end{aligned} \quad (1.10)$$

$$\begin{aligned} \epsilon_{1R} &:= \text{Re}(\epsilon_{Si}); \epsilon_{1C} := \text{Im}(\epsilon_{Si}); \epsilon_{2R} := \text{Re}(\epsilon_{LSi}); \\ \epsilon_{2C} &:= \text{Im}(\epsilon_{LSi}) \\ &13.64 \\ &48. \\ &-17 \\ &49 \end{aligned} \quad (1.11)$$

$$\begin{aligned} eval(\epsilon_{effRe}, LiquidFraction = 0.5); eval(\epsilon_{effImag}, LiquidFraction = 0.5) \\ \epsilon_{effR} &= -1.396030969 \\ \epsilon_{effC} &= 53.33594371 \end{aligned} \quad (1.12)$$

Numerical application using Maple solver, without splitting Re and Im.

$$\begin{aligned} \epsilon_1 &:= \epsilon_{Si}; \epsilon_2 := \epsilon_{LSi} \\ &13.64 + 48. I \\ &-17 + 49 I \end{aligned} \quad (1.13)$$

$$eval(solve(eq1, \epsilon_{eff}), LiquidFraction = 0.5)$$

$$-1.396030950 + 53.33594372 \text{ I} \quad (1.14)$$

OPTIMIZE the writing of analytical formula

with(*codegen*, *optimize*, *makeproc*, *cost*, *prep2trans*)

$$[\textit{optimize}, \textit{makeproc}, \textit{cost}, \textit{prep2trans}] \quad (1.15)$$

$$\begin{aligned} \textit{epsilonEffR} := & \left(4. \textit{LiquidFraction}^2 \epsilon_{r2} \epsilon_{r1} - 2. \epsilon_{r1}^2 \textit{LiquidFraction}^2 \right. \\ & - 4. \epsilon_{c2} \textit{LiquidFraction} \epsilon_{c1} + \epsilon_{r1} \epsilon_{c2}^2 - 4. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2} \\ & + 4. \epsilon_{c2} \textit{LiquidFraction}^2 \epsilon_{c1} - 2. \epsilon_{c1}^2 \textit{LiquidFraction}^2 + 2. \epsilon_{r1}^2 \textit{LiquidFraction} \\ & + 2. \textit{LiquidFraction} \epsilon_{r2}^2 - 2. \epsilon_{c2}^2 \textit{LiquidFraction}^2 + 4. \epsilon_{r1} + 4. \epsilon_{r1} \epsilon_{r2} \\ & - 4. \epsilon_{r1} \textit{LiquidFraction} - 2. \textit{LiquidFraction}^2 \epsilon_{r2}^2 + \epsilon_{r1} \epsilon_{r2}^2 + 2. \textit{LiquidFraction} \epsilon_{c2}^2 + \\ & \epsilon_{r1}^2 \textit{LiquidFraction} \epsilon_{r2} - 1. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2}^2 + 4. \textit{LiquidFraction} \epsilon_{r2} \\ & \left. - 1. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{c2}^2 + \epsilon_{c1}^2 \textit{LiquidFraction} \epsilon_{r2} + 2. \epsilon_{c1}^2 \textit{LiquidFraction} \right) / \\ & \left(\textit{LiquidFraction}^2 \epsilon_{r2}^2 + \epsilon_{r1}^2 \textit{LiquidFraction}^2 - 2. \textit{LiquidFraction}^2 \epsilon_{r2} \epsilon_{r1} + \right. \\ & \epsilon_{c2}^2 \textit{LiquidFraction}^2 + \epsilon_{c1}^2 \textit{LiquidFraction}^2 + 2. \epsilon_{c2} \textit{LiquidFraction} \epsilon_{c1} \\ & - 2. \epsilon_{c2} \textit{LiquidFraction}^2 \epsilon_{c1} + \epsilon_{r2}^2 + 4. \epsilon_{r2} + \epsilon_{c2}^2 + 4. - 2. \textit{LiquidFraction} \epsilon_{r2}^2 \\ & - 4. \textit{LiquidFraction} \epsilon_{r2} - 2. \textit{LiquidFraction} \epsilon_{c2}^2 + 4. \epsilon_{r1} \textit{LiquidFraction} \\ & \left. + 2. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2} \right) \end{aligned}$$

$$\begin{aligned} & \left(4. \textit{LiquidFraction}^2 \epsilon_{r2} \epsilon_{r1} - 2. \epsilon_{r1}^2 \textit{LiquidFraction}^2 - 4. \epsilon_{c2} \textit{LiquidFraction} \epsilon_{c1} \right. \\ & + \epsilon_{r1} \epsilon_{c2}^2 - 4. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2} + 4. \epsilon_{c2} \textit{LiquidFraction}^2 \epsilon_{c1} - 2. \\ & \epsilon_{c1}^2 \textit{LiquidFraction}^2 + 2. \epsilon_{r1}^2 \textit{LiquidFraction} + 2. \textit{LiquidFraction} \epsilon_{r2}^2 - 2. \\ & \epsilon_{c2}^2 \textit{LiquidFraction}^2 + 4. \epsilon_{r1} + 4. \epsilon_{r1} \epsilon_{r2} - 4. \epsilon_{r1} \textit{LiquidFraction} \\ & - 2. \textit{LiquidFraction}^2 \epsilon_{r2}^2 + \epsilon_{r1} \epsilon_{r2}^2 + 2. \textit{LiquidFraction} \epsilon_{c2}^2 + \\ & \epsilon_{r1}^2 \textit{LiquidFraction} \epsilon_{r2} - 1. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2}^2 + 4. \textit{LiquidFraction} \epsilon_{r2} \\ & \left. - 1. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{c2}^2 + \epsilon_{c1}^2 \textit{LiquidFraction} \epsilon_{r2} + 2. \epsilon_{c1}^2 \textit{LiquidFraction} \right) / \\ & \left(\textit{LiquidFraction}^2 \epsilon_{r2}^2 + \epsilon_{r1}^2 \textit{LiquidFraction}^2 - 2. \textit{LiquidFraction}^2 \epsilon_{r2} \epsilon_{r1} + \right. \\ & \epsilon_{c2}^2 \textit{LiquidFraction}^2 + \epsilon_{c1}^2 \textit{LiquidFraction}^2 + 2. \epsilon_{c2} \textit{LiquidFraction} \epsilon_{c1} \\ & - 2. \epsilon_{c2} \textit{LiquidFraction}^2 \epsilon_{c1} + \epsilon_{r2}^2 + 4. \epsilon_{r2} + \epsilon_{c2}^2 + 4. - 2. \textit{LiquidFraction} \epsilon_{r2}^2 \\ & - 4. \textit{LiquidFraction} \epsilon_{r2} - 2. \textit{LiquidFraction} \epsilon_{c2}^2 + 4. \epsilon_{r1} \textit{LiquidFraction} \\ & \left. + 2. \epsilon_{r1} \textit{LiquidFraction} \epsilon_{r2} \right) \end{aligned} \quad (1.16)$$

to string

$$\begin{aligned} & "(4.*\textit{LiquidFraction}^2*\textit{epsilon}[r2]*\textit{epsilon}[r1]-2.*\textit{epsilon}[r1]^2* \\ & \textit{LiquidFraction}^2-4.*\textit{epsilon}[c2]*\textit{LiquidFraction}*\textit{epsilon}[c1]*\textit{epsilon}[r1]* \end{aligned} \quad (1.17)$$

$$\begin{aligned} \epsilon_{c1}^2 \epsilon_{r2}^2 - 4. \epsilon_{c1} \text{LiquidFraction} + 4. \epsilon_{c2} \text{LiquidFraction} + \epsilon_{c1} \epsilon_{c2}^2 + 4. \epsilon_{c1} \epsilon_{r2} \\ - 1. \epsilon_{c1} \text{LiquidFraction} \epsilon_{r2}^2 + \epsilon_{c2} \text{LiquidFraction} \epsilon_{c1}^2 + 4. \epsilon_{c2} \text{LiquidFraction} \epsilon_{r1} \\ + \epsilon_{c2} \text{LiquidFraction} \epsilon_{r1}^2 - 1. \epsilon_{c1} \text{LiquidFraction} \epsilon_{c2}^2 - 4. \epsilon_{c1} \text{LiquidFraction} \epsilon_{r2} \\ + 4. \epsilon_{c1}) / (\text{LiquidFraction}^2 \epsilon_{r2}^2 + \epsilon_{r1}^2 \text{LiquidFraction}^2 - 2. \text{LiquidFraction}^2 \epsilon_{r2} \epsilon_{r1} + \\ \epsilon_{c2}^2 \text{LiquidFraction}^2 + \epsilon_{c1}^2 \text{LiquidFraction}^2 + 2. \epsilon_{c2} \text{LiquidFraction} \epsilon_{c1} \\ - 2. \epsilon_{c2} \text{LiquidFraction}^2 \epsilon_{c1} + \epsilon_{r2}^2 + 4. \epsilon_{r2} + \epsilon_{c2}^2 + 4. - 2. \text{LiquidFraction} \epsilon_{r2}^2 \\ - 4. \text{LiquidFraction} \epsilon_{r2} - 2. \text{LiquidFraction} \epsilon_{c2}^2 + 4. \epsilon_{r1} \text{LiquidFraction} \\ + 2. \epsilon_{r1} \text{LiquidFraction} \epsilon_{r2}) \end{aligned} \quad (1.18)$$

$$\begin{aligned}
& "(\epsilon[c1]\epsilon[r2]^2-4.\epsilon[c1]\text{LiquidFractionC}4.\epsilon[c2]* \\
& \text{LiquidFractionC}\epsilon[c1]\epsilon[c2]^2C4.\epsilon[c1]\epsilon[r2]-1.* \\
& \epsilon[c1]\text{LiquidFraction}*\epsilon[r2]^2C\epsilon[c2]\text{LiquidFraction}* \\
& \epsilon[c1]^2C4.\epsilon[c2]\text{LiquidFraction}*\epsilon[r1]C\epsilon[c2]* \\
& \text{LiquidFraction}*\epsilon[r1]^2-1.*\epsilon[c1]\text{LiquidFraction}*\epsilon[c2]^2 \\
& -4.\epsilon[c1]\text{LiquidFraction}*\epsilon[r2]C4.\epsilon[c1])/ \\
& (\text{LiquidFraction}^2*\epsilon[r2]^2C\epsilon[r1]^2*\text{LiquidFraction}^2-2.* \\
& \text{LiquidFraction}^2*\epsilon[r2]*\epsilon[r1]C\epsilon[c2]^2* \\
& \text{LiquidFraction}^2C\epsilon[c1]^2*\text{LiquidFraction}^2C2.*\epsilon[c2]* \\
& \text{LiquidFraction}*\epsilon[c1]-2.*\epsilon[c2]\text{LiquidFraction}^2*\epsilon[c1]C \\
& \epsilon[r2]^2C4.\epsilon[r2]C\epsilon[c2]^2C4.-2.*\text{LiquidFraction}*\epsilon[r2]^2-4.* \\
& \text{LiquidFraction}*\epsilon[r2]-2.*\text{LiquidFraction}*\epsilon[c2]^2C4.* \\
& \epsilon[r1]\text{LiquidFractionC}2.*\epsilon[r1]\text{LiquidFraction}*\epsilon[r2])"
\end{aligned}
\tag{1.19}$$

optimize(epsilonEffR)

$$\begin{aligned}
t1 &= \text{LiquidFraction}^2, t3 = t1 \epsilon_{r2} \epsilon_{r1}, t5 = \epsilon_{r1}^2, t6 = t5 t1, t9 \\
&= \epsilon_{c2} \text{LiquidFraction} \epsilon_{c1}, t11 = \epsilon_{c2}^2, t13 = \epsilon_{r1} \text{LiquidFraction}, t14 = t13 \epsilon_{r2}, t17 \\
&= \epsilon_{c2} t1 \epsilon_{c1}, t19 = \epsilon_{c1}^2, t20 = t19 t1, t22 = t5 \text{LiquidFraction}, t24 = \epsilon_{r2}^2, t26 \\
&= 2. \text{LiquidFraction} t24, t27 = t11 t1, t30 = 4. t3 - 2. t6 - 4. t9 + \epsilon_{r1} t11 \\
&- 4. t14 + 4. t17 - 2. t20 + 2. t22 + t26 - 2. t27 + 4. \epsilon_{r1}, t33 = 4. t13, t34 \\
&= t1 t24, t38 = 2. \text{LiquidFraction} t11, t43 = 4. \text{LiquidFraction} \epsilon_{r2}, t46 \\
&= t19 \text{LiquidFraction}, t49 = 4. \epsilon_{r1} \epsilon_{r2} - t33 - 2. t34 + \epsilon_{r1} t24 + t38 + t22 \epsilon_{r2} \\
&- 1. t13 t24 + t43 - 1. t13 t11 + t46 \epsilon_{r2} + 2. t46, t56 = t34 + t6 - 2. t3 + t27 \\
&+ t20 + 2. t9 - 2. t17 + t24 + 4. \epsilon_{r2} + t11 + 4. - t26 - t43 - t38 + t33 \\
&+ 2. t14, t58 = \frac{t30 + t49}{t56}
\end{aligned}
\tag{1.20}$$

optimize(epsilonEffC)

$$\begin{aligned}
t1 &= \epsilon_{r2}^2, t3 = \epsilon_{c1} \text{LiquidFraction}, t5 = \epsilon_{c2} \text{LiquidFraction}, t7 = \epsilon_{c2}^2, t13 = \epsilon_{c1}^2, t17 = \\
&\epsilon_{r1}^2, t24 = \epsilon_{c1} t1 - 4. t3 + 4. t5 + \epsilon_{c1} t7 + 4. \epsilon_{c1} \epsilon_{r2} - 1. t3 t1 + t5 t13 + 4. t5 \epsilon_{r1} \\
&+ t5 t17 - 1. t3 t7 - 4. t3 \epsilon_{r2} + 4. \epsilon_{c1}, t25 = \text{LiquidFraction}^2, t45 \\
&= \epsilon_{r1} \text{LiquidFraction}, t49 = t25 t1 + t17 t25 - 2. t25 \epsilon_{r2} \epsilon_{r1} + t7 t25 + t13 t25 \\
&+ 2. t5 \epsilon_{c1} - 2. \epsilon_{c2} t25 \epsilon_{c1} + t1 + 4. \epsilon_{r2} + t7 + 4. - 2. \text{LiquidFraction} t1 \\
&- 4. \text{LiquidFraction} \epsilon_{r2} - 2. \text{LiquidFraction} t7 + 4. t45 + 2. t45 \epsilon_{r2}, t51 = \frac{t24}{t49}
\end{aligned}
\tag{1.21}$$

Solution with 3 media

restart

$$eq1 := \frac{(\epsilon_{eff}-1)}{\epsilon_{eff}+2} = \frac{(\epsilon_1-1)}{\epsilon_1+2} \cdot \eta_1 + \frac{(\epsilon_2-1)}{\epsilon_2+2} \cdot \eta_2 + \frac{(\epsilon_3-1)}{\epsilon_3+2} \cdot \eta_3$$

$$\frac{\epsilon_{eff}-1}{\epsilon_{eff}+2} = \frac{(\epsilon_1-1) \eta_1}{\epsilon_1+2} + \frac{(\epsilon_2-1) \eta_2}{\epsilon_2+2} + \frac{(\epsilon_3-1) \eta_3}{\epsilon_3+2} \quad (2.1)$$

$$\epsilon_{eff} := \epsilon_{reff} + I \cdot \epsilon_{ceff}; \epsilon_1 := \epsilon_{r1} + I \cdot \epsilon_{c1}; \epsilon_2 := \epsilon_{r2} + I \cdot \epsilon_{c2}; \epsilon_3 := \epsilon_{r3} + I \cdot \epsilon_{c3};$$

$$\epsilon_{reff} + I \epsilon_{ceff}$$

$$\epsilon_{r1} + I \epsilon_{c1}$$

$$\epsilon_{r2} + I \epsilon_{c2}$$

$$\epsilon_{r3} + I \epsilon_{c3} \quad (2.2)$$

eq2 := evalc(eq1)

$$\frac{(\epsilon_{reff}-1) (\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + I \left(\frac{\epsilon_{ceff} (\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} - \frac{(\epsilon_{reff}-1) \epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} \right) = \left(\frac{(\epsilon_{r1}-1) (\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} + \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1$$

$$+ \left(\frac{(\epsilon_{r2}-1) (\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 + \left(\frac{(\epsilon_{r3}-1) (\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 + I \left(\left(\frac{\epsilon_{c1} (\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} - \frac{(\epsilon_{r1}-1) \epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 \right.$$

$$+ \left(\frac{\epsilon_{c2} (\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1) \epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 + \left(\frac{\epsilon_{c3} (\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} - \frac{(\epsilon_{r3}-1) \epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \Bigg)$$

$$eq2r := \frac{(\epsilon_{reff}-1) (\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{(\epsilon_{r1}-1) (\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} + \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{(\epsilon_{r2}-1) (\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2$$

$$\begin{aligned}
& + \left(\frac{(\epsilon_{r3}-1)(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \\
& \frac{(\epsilon_{reff}-1)(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} + \frac{\epsilon_{ceff}^2}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{(\epsilon_{r1}-1)(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \\
& \left. + \frac{\epsilon_{c1}^2}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{(\epsilon_{r2}-1)(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} + \frac{\epsilon_{c2}^2}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{(\epsilon_{r3}-1)(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} + \frac{\epsilon_{c3}^2}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3
\end{aligned} \tag{2.4}$$

$$\begin{aligned}
eq2c := & \left(\frac{\epsilon_{ceff}(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} - \frac{(\epsilon_{reff}-1)\epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} \right) = \left(\left(\frac{\epsilon_{c1}(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \right. \\
& \left. \left. - \frac{(\epsilon_{r1}-1)\epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{\epsilon_{c2}(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1)\epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \right. \\
& \left. + \left(\frac{\epsilon_{c3}(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} - \frac{(\epsilon_{r3}-1)\epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3 \right) \\
& \frac{\epsilon_{ceff}(\epsilon_{reff}+2)}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} - \frac{(\epsilon_{reff}-1)\epsilon_{ceff}}{(\epsilon_{reff}+2)^2 + \epsilon_{ceff}^2} = \left(\frac{\epsilon_{c1}(\epsilon_{r1}+2)}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right. \\
& \left. - \frac{(\epsilon_{r1}-1)\epsilon_{c1}}{(\epsilon_{r1}+2)^2 + \epsilon_{c1}^2} \right) \eta_1 + \left(\frac{\epsilon_{c2}(\epsilon_{r2}+2)}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} - \frac{(\epsilon_{r2}-1)\epsilon_{c2}}{(\epsilon_{r2}+2)^2 + \epsilon_{c2}^2} \right) \eta_2 \\
& + \left(\frac{\epsilon_{c3}(\epsilon_{r3}+2)}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} - \frac{(\epsilon_{r3}-1)\epsilon_{c3}}{(\epsilon_{r3}+2)^2 + \epsilon_{c3}^2} \right) \eta_3
\end{aligned} \tag{2.5}$$

$$\begin{aligned}
sol := & solve(\{eq2r, eq2c\}, \{\epsilon_{reff}, \epsilon_{ceff}\}) \\
& \left\{ \epsilon_{ceff} = (9(16\epsilon_{c1}\eta_1 + 16\epsilon_{c2}\eta_2 + 16\epsilon_{c3}\eta_3 + 16\epsilon_{c1}\eta_1\epsilon_{r2} + 16\epsilon_{c1}\eta_1\epsilon_{r3} + 4\epsilon_{c1}\eta_1 \right. \\
& \left. \epsilon_{r2}^2 + 4\epsilon_{c1}\eta_1\epsilon_{r3}^2 + 4\epsilon_{c1}\eta_1\epsilon_{c2}^2 + 4\epsilon_{c1}\eta_1\epsilon_{c3}^2 + 16\epsilon_{c2}\eta_2\epsilon_{r1} + 16\epsilon_{c2}\eta_2\epsilon_{r3} \right. \\
& \left. + 4\epsilon_{c2}\eta_2\epsilon_{r1}^2 + 4\epsilon_{c2}\eta_2\epsilon_{r3}^2 + 4\epsilon_{c2}\eta_2\epsilon_{c1}^2 + 4\epsilon_{c2}\eta_2\epsilon_{c3}^2 + 16\epsilon_{c3}\eta_3\epsilon_{r1} \right.
\end{aligned} \tag{2.6}$$

$$\begin{aligned}
& + 16 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1} \eta_1 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1} \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1} \eta_1 \\
& \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1} \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 4 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2) \Big/ \left(64 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 \right. \\
& + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{c1}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 \\
& + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\
& \left. - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \right)
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 \\
& - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \\
& \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \\
& + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \\
& + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \\
& + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1} \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 + 16 \eta_3^2 - 2 \eta_2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1^2 \epsilon_{r1}^2 + 64 \eta_1^2 \epsilon_{r2}^2 \\
& + 64 \eta_1^2 \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_1^2 \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{c1}^2 + 16 \eta_1^2 \epsilon_{c2}^2 + 16 \eta_1^2 \epsilon_{c3}^2 \\
& + 64 \eta_2^2 \epsilon_{r1}^2 - 32 \eta_2^2 \epsilon_{r2}^2 + 64 \eta_2^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1}^2 - 32 \eta_2^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_2^2 \epsilon_{c2}^2 + 16 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_3^2 \epsilon_{r1}^2 + 64 \eta_3^2 \epsilon_{r2}^2 - 32 \eta_3^2 \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1}^2 + 16 \eta_3^2 \epsilon_{r2}^2 \\
& - 32 \eta_3^2 \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{c1}^2 + 16 \eta_3^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2^2 \eta_1^2 + 32 \eta_3^2 \eta_1^2 + 32 \eta_2^2 \eta_3^2 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2^2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2
\end{aligned}$$

$$\epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2$$

$$\epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}$$

$$\epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{r1}^2$$

$$\epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1}$$

$$\epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2$$

$$\epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1}$$

$$\epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3$$

$$\epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4$$

$$\epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 +$$

$$\epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}$$

$$+ 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r1}$$

$$\epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c1}^2$$

$$\begin{aligned}
& \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{r3}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& + 2 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 2 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 \\
& + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2
\end{aligned}$$

$$\begin{aligned}
& -8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{rl} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{rl} \epsilon_{r3} + 16 \eta_2 \epsilon_{rl} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{rl}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{rl}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{rl}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{rl}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{rl}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{rl} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{rl} \epsilon_{r2} + 16 \eta_3 \epsilon_{rl} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{cl}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{rl} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{rl} \epsilon_{r3} - 32 \eta_3 \epsilon_{rl} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{rl}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{rl}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{rl}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{rl}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{cl}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{rl} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{rl} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{rl}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{rl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{cl}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{rl} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{rl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{rl} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32
\end{aligned}$$

$$\begin{aligned}
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3} \\
& - 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}
\end{aligned}$$

$$\begin{aligned}
& + 8 \eta_2 \epsilon_{cl}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{cl}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{cl}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2), \epsilon_{reff} = - \left(-64 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32 \eta_1 + 32 \eta_2 + 32 \eta_3 - 64 \epsilon_{r1} \right. \\
& - 64 \epsilon_{r2} - 64 \epsilon_{r3} - 16 \epsilon_{cl}^2 - 16 \epsilon_{c2}^2 - 16 \epsilon_{c3}^2 - 16 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 36 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r2} \eta_1 \\
& + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 16 \eta_3^2 \\
& \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 16 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \\
& \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \\
& \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + 2 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{r3}^2 \\
& + 16 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \\
& \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 32 \epsilon_{c1}^2 \eta_1^2 + 32 \eta_3^2 \epsilon_{r1}^2 + 32 \eta_3^2 \epsilon_{r2}^2 - 64 \eta_3^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_3^2 \epsilon_{r2}^2 + 8 \\
& \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3^2 \epsilon_{c2}^2 + 32 \eta_2^2 \epsilon_{r1}^2 - 64 \eta_2^2 \epsilon_{r2}^2 + 32 \eta_2^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{r1}^2 + 8 \eta_2^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \\
& + 8 \eta_2^2 \epsilon_{c3}^2 + 32 \epsilon_{c3}^2 \eta_3^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r2}^2 - 64 \epsilon_{r1} \epsilon_{r2} \\
& - 16 \epsilon_{r1} \epsilon_{c2}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{r1}^2 - 16 \epsilon_{r2}^2 - 16 \epsilon_{r3}^2 + 32 \eta_1^2 + 32 \eta_2^2 \\
& + 32 \eta_3^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c3}^2 - 16 \epsilon_{r2} \epsilon_{r3}^2 - 64 \epsilon_{r2} \epsilon_{r3}^2 - 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1} + 32 \eta_1 \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& + 32 \eta_1 \epsilon_{r3} - 16 \eta_1 \epsilon_{r1}^2 + 8 \eta_1 \epsilon_{r2}^2 + 8 \eta_1 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 + 8 \eta_1 \epsilon_{c2}^2 + 8 \eta_1 \epsilon_{c3}^2 \\
& + 32 \eta_2 \epsilon_{r1} - 16 \eta_2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} + 8 \eta_2 \epsilon_{r1}^2 - 16 \eta_2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{c3}^2 + 32 \eta_3 \epsilon_{r1} + 32 \eta_3 \epsilon_{r2} - 16 \eta_3 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 + 8 \eta_3 \epsilon_{r2}^2 \\
& - 16 \eta_3 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 - 16 \eta_3 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 + 32 \eta_2^2 \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \\
& + 64 \eta_2 \eta_1 + 64 \eta_3 \eta_1 + 64 \eta_2 \eta_3 + 32 \eta_3^2 \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 8 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1} \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 \\
& + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 16 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2
\end{aligned}$$

$$\begin{aligned}
& +4\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2\eta_3+16\eta_2\epsilon_{r3}\eta_1\epsilon_{r1}^2\epsilon_{c2}^2+16\eta_2\epsilon_{r3}\eta_1\epsilon_{cl}^2\epsilon_{r2}^2+16\eta_2\epsilon_{r3}\eta_1\epsilon_{cl}^2\epsilon_{r3}^2 \\
& +4\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_2+4\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_3+4\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}\eta_1+16\eta_3 \\
& \epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+4\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{c3}^2\eta_1+4\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}^2\eta_1+16\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}\eta_1 \\
& +4\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2\eta_1+4\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2\eta_1+4\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_1+16\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_1 \\
& +4\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_1+16\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+16\eta_3\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2+4\eta_3\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 \\
& -8\eta_2\epsilon_{r2}\epsilon_{c3}^2\eta_1-64\eta_1^2\epsilon_{r1}+32\eta_1^2\epsilon_{r2}+32\eta_1^2\epsilon_{r3}+8\eta_1^2\epsilon_{r2}^2+8\eta_1^2\epsilon_{r3}^2+8\eta_1^2 \\
& \epsilon_{c2}^2+8\eta_1^2\epsilon_{c3}^2+8\eta_2^2\epsilon_{r3}^2\epsilon_{r2}^2-16\eta_1\epsilon_{r1}\epsilon_{r3}-4\eta_1\epsilon_{r1}\epsilon_{c3}^2+8\eta_1\epsilon_{r2}\epsilon_{r3}^2 \\
& +32\eta_1\epsilon_{r2}\epsilon_{r3}+8\eta_1\epsilon_{r2}\epsilon_{c3}^2-4\eta_1\epsilon_{r1}^2\epsilon_{r3}^2-16\eta_1\epsilon_{r1}^2\epsilon_{r3}-4\eta_1\epsilon_{r1}^2\epsilon_{c3}^2+2\eta_1\epsilon_{r2}^2 \\
& \epsilon_{r3}^2+8\eta_1\epsilon_{r2}^2\epsilon_{r3}+2\eta_1\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_1\epsilon_{cl}^2\epsilon_{r3}^2-16\eta_1\epsilon_{cl}^2\epsilon_{r3}-4\eta_1\epsilon_{cl}^2\epsilon_{c3}^2+2\eta_1 \\
& \epsilon_{c2}^2\epsilon_{r3}^2+8\eta_1\epsilon_{c2}^2\epsilon_{r3}+2\eta_1\epsilon_{c2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r1}^2\epsilon_{r2}^2-4\eta_2\epsilon_{r1}^2\epsilon_{r2}-4\eta_2\epsilon_{r1}^2\epsilon_{c2}^2 \\
& -16\eta_2\epsilon_{r1}\epsilon_{r2}^2-16\eta_2\epsilon_{r1}\epsilon_{r2}-16\eta_2\epsilon_{r1}\epsilon_{c2}^2-4\eta_2\epsilon_{cl}^2\epsilon_{r2}^2-4\eta_2\epsilon_{cl}^2\epsilon_{r2}-4\eta_2 \\
& \epsilon_{cl}^2\epsilon_{c2}^2+8\eta_2\epsilon_{r1}\epsilon_{r3}^2+32\eta_2\epsilon_{r1}\epsilon_{r3}+8\eta_2\epsilon_{r1}\epsilon_{c3}^2-4\eta_2\epsilon_{r2}\epsilon_{r3}^2-16\eta_2\epsilon_{r2}\epsilon_{r3} \\
& -4\eta_2\epsilon_{r2}\epsilon_{c3}^2+2\eta_2\epsilon_{r1}^2\epsilon_{r3}^2+8\eta_2\epsilon_{r1}^2\epsilon_{r3}+2\eta_2\epsilon_{r1}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r2}^2\epsilon_{r3}^2-16\eta_2 \\
& \epsilon_{r2}^2\epsilon_{r3}-4\eta_2\epsilon_{r2}^2\epsilon_{c3}^2+2\eta_2\epsilon_{cl}^2\epsilon_{r3}^2+8\eta_2\epsilon_{cl}^2\epsilon_{r3}+2\eta_2\epsilon_{cl}^2\epsilon_{c3}^2-4\eta_2\epsilon_{c2}^2\epsilon_{r3}^2 \\
& -16\eta_2\epsilon_{c2}^2\epsilon_{r3}-4\eta_2\epsilon_{c2}^2\epsilon_{c3}^2+2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2+8\eta_3\epsilon_{r1}^2\epsilon_{r2}+2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2+8\eta_3\epsilon_{r1} \\
& \epsilon_{r2}^2+32\eta_3\epsilon_{r1}\epsilon_{r2}+8\eta_3\epsilon_{r1}\epsilon_{c2}^2+2\eta_3\epsilon_{cl}^2\epsilon_{r2}^2+8\eta_3\epsilon_{cl}^2\epsilon_{r2}+2\eta_3\epsilon_{cl}^2\epsilon_{c2}^2 \\
& -16\eta_3\epsilon_{r1}\epsilon_{r3}^2-16\eta_3\epsilon_{r1}\epsilon_{r3}-16\eta_3\epsilon_{r1}\epsilon_{c3}^2-16\eta_3\epsilon_{r2}\epsilon_{r3}^2-16\eta_3\epsilon_{r2}\epsilon_{r3} \\
& -16\eta_3\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r3}-4\eta_3\epsilon_{r1}^2\epsilon_{c3}^2-4\eta_3\epsilon_{r2}^2\epsilon_{r3}^2-4\eta_3 \\
& \epsilon_{r2}^2\epsilon_{r3}-4\eta_3\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_3\epsilon_{cl}^2\epsilon_{r3}^2-4\eta_3\epsilon_{cl}^2\epsilon_{r3}-4\eta_3\epsilon_{cl}^2\epsilon_{c3}^2-4\eta_3\epsilon_{c2}^2\epsilon_{r3}^2-4\eta_3 \\
& \epsilon_{c2}^2\epsilon_{r3}-4\eta_3\epsilon_{c2}^2\epsilon_{c3}^2+8\eta_3^2\epsilon_{c2}^2\epsilon_{r3}^2+8\eta_3^2\epsilon_{r1}^2\epsilon_{r3}^2+8\eta_3^2\epsilon_{r2}^2\epsilon_{r3}^2+8\eta_3^2\epsilon_{cl}^2\epsilon_{r3}^2+32 \\
& \eta_3^2\epsilon_{r1}\epsilon_{r3}^2+32\eta_3^2\epsilon_{r2}\epsilon_{r3}^2-32\eta_3\epsilon_{r3}\eta_1+16\eta_3\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+4\eta_3\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 \\
& +4\eta_3\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+4\eta_3\eta_2\epsilon_{cl}^2\epsilon_{r2}^2\epsilon_{r3}^2+16\eta_3\epsilon_{r2}\eta_1\epsilon_{cl}^2\epsilon_{c3}^2+4\eta_3\epsilon_{r2}^2\eta_1\epsilon_{cl}^2\epsilon_{c3}^2 \\
& +4\eta_3\epsilon_{cl}^2\eta_1\epsilon_{c2}^2\epsilon_{r3}^2+4\eta_3\epsilon_{cl}^2\eta_1\epsilon_{c2}^2\epsilon_{c3}^2+16\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2\eta_3+4\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}^2\eta_3 \\
& +4\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{r3}\eta_3+4\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{c3}^2\eta_3+16\eta_3\epsilon_{r1}^2\eta_1\epsilon_{r2}\epsilon_{r3}^2+16\eta_3\epsilon_{cl}^2\eta_1\epsilon_{r2}\epsilon_{r3}^2 \\
& +16\eta_3\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2+4\eta_3\eta_2\epsilon_{cl}^2\epsilon_{r2}\epsilon_{c3}^2+4\eta_3\eta_2\epsilon_{cl}^2\epsilon_{c2}^2\epsilon_{r3}^2+4\eta_3\eta_2\epsilon_{cl}^2\epsilon_{c2}^2\epsilon_{c3}^2 \\
& +16\eta_3\epsilon_{r2}^2\eta_2\epsilon_{r1}\epsilon_{c3}^2+4\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_1\epsilon_{r3}^2+4\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_1\epsilon_{c3}^2+16\eta_3\epsilon_{r1}^2\epsilon_{r2}\eta_1\epsilon_{c3}^2 \\
& +4\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\eta_1\epsilon_{r3}^2+4\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\eta_1\epsilon_{c3}^2+4\eta_3\epsilon_{cl}^2\epsilon_{r2}^2\eta_1\epsilon_{r3}^2+16\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}\eta_2
\end{aligned}$$

$$\begin{aligned}
& + 16 \eta_3 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 + 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_2 - 32 \eta_3 \varepsilon_{r1} \eta_1 - 32 \eta_2 \varepsilon_{c1}^2 \eta_1 + 16 \eta_2 \\
& \varepsilon_{c3}^2 \eta_1 - 32 \eta_3 \varepsilon_{c1}^2 \eta_1 + 16 \eta_3 \varepsilon_{c2}^2 \eta_1 - 32 \eta_2 \varepsilon_{r1} \eta_1 + 8 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1}^2 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \\
& + 32 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1} + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c3}^2 + 2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 + 8 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + 2 \eta_1^2 \varepsilon_{r2}^2 \\
& \varepsilon_{c3}^2 + 8 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3} + 2 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 + 32 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} + 8 \eta_1^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 8 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 + 32 \varepsilon_{c2}^2 \\
& \eta_2^2 \varepsilon_{r3} + 32 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r3} - 32 \eta_2 \varepsilon_{r2} \eta_3 - 32 \eta_2 \varepsilon_{r3} \eta_3 + 16 \eta_2 \varepsilon_{r1}^2 \eta_3 - 32 \eta_2 \varepsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2} + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 8 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c2}^2 + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 32 \eta_1 \varepsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \varepsilon_{r3}^2 \eta_3 - 32 \eta_1 \varepsilon_{c2}^2 \eta_2 - 32 \eta_1 \varepsilon_{c3}^2 \eta_3 + 32 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r2} + 64 \eta_2 \varepsilon_{r3} \eta_1 + 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \\
& + 8 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2} + 2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 - 8 \eta_2 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_1 - 8 \eta_2 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_1 - 64 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 16 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 16 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 8 \eta_2 \\
& \varepsilon_{c1}^2 \varepsilon_{r3} \eta_3 - 8 \eta_2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 \eta_3 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 \eta_3 - 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_3 \\
& - 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 \eta_3 - 8 \eta_2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 \eta_3 + 16 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_1 + 16 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_1 + 16 \eta_2 \varepsilon_{r2} \\
& \varepsilon_{r3}^2 \eta_3 + 16 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_3 + 16 \eta_2 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 32 \\
& \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 64 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 16 \eta_1^2 \varepsilon_{r1} \\
& \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{c3}^2 - 8 \eta_3 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_1 \\
& + 16 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_1 - 8 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_1 - 32 \eta_3 \varepsilon_{c1} \varepsilon_{r2} \eta_1 - 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \varepsilon_{r1} \varepsilon_{r3} \eta_1 - 32 \eta_3 \varepsilon_{r2} \varepsilon_{r3} \eta_1 + 16 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_1 - 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \eta_1 + 2 \varepsilon_{c2}^2 \eta_2^2 \\
& \varepsilon_{c1}^2 \varepsilon_{r3}^2 - 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - 16 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 32 \eta_1 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 + 16 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 + 16 \eta_1 \varepsilon_{r1} \varepsilon_{r3}^2 \eta_3 + 16 \eta_1 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_3 + 8 \varepsilon_{c3}^2 \eta_3^2 \\
& \varepsilon_{c1}^2 \varepsilon_{r2} + 16 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 16 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 2 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 8 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 4 \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 16 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} - 4 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + 2 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 \\
& - 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_3 - 32 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_3 - \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 16 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 32 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \\
& + 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 + 2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 8 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{c2}^2 - 64 \eta_3^2 \varepsilon_{r1} \varepsilon_{r3} - 64 \\
& \eta_3^2 \varepsilon_{r2} \varepsilon_{r3} + 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3} + 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 + 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3} + 2 \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 + 2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 16 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} - 64 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2} - 16 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 8 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3}^2 \\
& + 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3} + 8 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 16 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3} - 64 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3} - 16 \eta_2^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 32 \\
& \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} - 32 \eta_3 \varepsilon_{r2}^2 \eta_2 - 32 \eta_3 \varepsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 8 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1}^2 + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 \\
& + 32 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3} + 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2}^2 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 + 32 \eta_2^2 \varepsilon_{r3} \varepsilon_{r2}^2 + 64 \eta_2 \varepsilon_{r1} \eta_3 - 16 \eta_3^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c2}^2 \epsilon_{r3} + 16 \eta_2 \epsilon_{c1}^2 \eta_3 - 32 \eta_2 \epsilon_{c3}^2 \eta_3 + 64 \eta_3 \epsilon_{r2} \eta_1 - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2) / (64 + 4 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{c1}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \\
& \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& + 4 \eta_3^2 \epsilon_{c2}^2 + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \\
& \eta_2^2 \epsilon_{c3}^2 + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& + 16 \epsilon_{r1} \epsilon_{c2}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 \\
& + 16 \eta_3^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32
\end{aligned}$$

$$\begin{aligned}
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3} \\
& - 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3}^2) \}
\end{aligned}$$

$eps_c := rhs(sol[1])$

$$\begin{aligned}
& (9 \left(16 \varepsilon_{cl} \eta_1 + 16 \varepsilon_{c2} \eta_2 + 16 \varepsilon_{c3} \eta_3 + 16 \varepsilon_{cl} \eta_1 \varepsilon_{r2} + 16 \varepsilon_{cl} \eta_1 \varepsilon_{r3} + 4 \varepsilon_{cl} \eta_1 \varepsilon_{r2}^2 \right. \\
& \quad + 4 \varepsilon_{cl} \eta_1 \varepsilon_{r3}^2 + 4 \varepsilon_{cl} \eta_1 \varepsilon_{c2}^2 + 4 \varepsilon_{cl} \eta_1 \varepsilon_{c3}^2 + 16 \varepsilon_{c2} \eta_2 \varepsilon_{r1} + 16 \varepsilon_{c2} \eta_2 \varepsilon_{r3} + 4 \varepsilon_{c2} \eta_2 \\
& \quad \varepsilon_{r1}^2 + 4 \varepsilon_{c2} \eta_2 \varepsilon_{r3}^2 + 4 \varepsilon_{c2} \eta_2 \varepsilon_{cl}^2 + 4 \varepsilon_{c2} \eta_2 \varepsilon_{c3}^2 + 16 \varepsilon_{c3} \eta_3 \varepsilon_{r1} + 16 \varepsilon_{c3} \eta_3 \varepsilon_{r2} \\
& \quad + 4 \varepsilon_{c3} \eta_3 \varepsilon_{r1}^2 + 4 \varepsilon_{c3} \eta_3 \varepsilon_{r2}^2 + 4 \varepsilon_{c3} \eta_3 \varepsilon_{cl}^2 + 4 \varepsilon_{c3} \eta_3 \varepsilon_{c2}^2 + 4 \varepsilon_{cl} \eta_1 \varepsilon_{r2} \varepsilon_{r3}^2 \\
& \quad + 16 \varepsilon_{cl} \eta_1 \varepsilon_{r2} \varepsilon_{r3} + 4 \varepsilon_{cl} \eta_1 \varepsilon_{r2} \varepsilon_{c3}^2 + \varepsilon_{cl} \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3}^2 + 4 \varepsilon_{cl} \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3} + \varepsilon_{cl} \eta_1 \varepsilon_{r2}^2 \varepsilon_{c3}^2 \\
& \quad + \varepsilon_{cl} \eta_1 \varepsilon_{c2}^2 \varepsilon_{r3}^2 + 4 \varepsilon_{cl} \eta_1 \varepsilon_{c2}^2 \varepsilon_{r3} + \varepsilon_{cl} \eta_1 \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 4 \varepsilon_{c2} \eta_2 \varepsilon_{r1} \varepsilon_{r3}^2 + 16 \varepsilon_{c2} \eta_2 \varepsilon_{r1} \varepsilon_{r3} \\
& \quad + 4 \varepsilon_{c2} \eta_2 \varepsilon_{r1} \varepsilon_{c3}^2 + \varepsilon_{c2} \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 4 \varepsilon_{c2} \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3} + \varepsilon_{c2} \eta_2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + \varepsilon_{c2} \eta_2 \varepsilon_{cl}^2 \varepsilon_{r3}^2 \\
& \quad + 4 \varepsilon_{c2} \eta_2 \varepsilon_{cl}^2 \varepsilon_{r3} + \varepsilon_{c2} \eta_2 \varepsilon_{cl}^2 \varepsilon_{c3}^2 + \varepsilon_{c3} \eta_3 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 4 \varepsilon_{c3} \eta_3 \varepsilon_{r1}^2 \varepsilon_{r2} + \varepsilon_{c3} \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \\
& \quad + 4 \varepsilon_{c3} \eta_3 \varepsilon_{r1} \varepsilon_{r2}^2 + 16 \varepsilon_{c3} \eta_3 \varepsilon_{r1} \varepsilon_{r2} + 4 \varepsilon_{c3} \eta_3 \varepsilon_{r1} \varepsilon_{c2}^2 + \varepsilon_{c3} \eta_3 \varepsilon_{cl}^2 \varepsilon_{r2}^2 + 4 \varepsilon_{c3} \eta_3 \\
& \quad \varepsilon_{cl}^2 \varepsilon_{r2} + \varepsilon_{c3} \eta_3 \varepsilon_{cl}^2 \varepsilon_{c2}^2 \left. \right) \Big/ \left(64 + 4 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \varepsilon_{r1} \right. \\
& \quad + 64 \varepsilon_{r2} + 64 \varepsilon_{r3} + 16 \varepsilon_{cl}^2 + 16 \varepsilon_{c2}^2 + 16 \varepsilon_{c3}^2 - 8 \eta_3^2 \varepsilon_{cl}^2 \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r2}^2 \varepsilon_{r3} \\
& \quad + 18 \varepsilon_{cl} \eta_1 \varepsilon_{c3} \eta_3 \varepsilon_{r2}^2 + 4 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r3}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r2}^2 + 4 \eta_2^2 \varepsilon_{cl}^2 \varepsilon_{r2}^2 - 16 \eta_2 \varepsilon_{r2} \eta_1 \\
& \quad + 8 \eta_3 \varepsilon_{r1}^2 \eta_1 \varepsilon_{r3}^2 - 16 \eta_3 \varepsilon_{r1} \varepsilon_{r2}^2 \eta_2 - 16 \eta_3 \varepsilon_{r1} \varepsilon_{c2}^2 \eta_2 - 4 \eta_3 \varepsilon_{cl}^2 \varepsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \\
& \quad \varepsilon_{cl}^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_3^2 \varepsilon_{cl}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3}^2 \eta_2 - 4 \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_2 \\
& \quad + 8 \eta_3 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 + 16 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3} + 4 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3} + 16 \varepsilon_{c3}^2 \\
& \quad \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} + 8 \eta_2 \eta_1 \varepsilon_{cl}^2 \varepsilon_{r2}^2 + 8 \eta_2 \eta_1 \varepsilon_{cl}^2 \varepsilon_{c2}^2 - 4 \eta_2 \varepsilon_{cl}^2 \varepsilon_{r2} \eta_3 - 16 \eta_2 \varepsilon_{r1} \varepsilon_{r3}^2 \eta_3 \\
& \quad - 16 \eta_2 \varepsilon_{r1} \varepsilon_{r3} \eta_3 - 16 \eta_2 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_3 + \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + 8 \eta_2 \varepsilon_{cl}^2 \varepsilon_{r2} \eta_1 - 4 \eta_2 \varepsilon_{r1} \\
& \quad \varepsilon_{r3}^2 \eta_1 - 16 \eta_2 \varepsilon_{r1} \varepsilon_{r3} \eta_1 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{cl}^2 + 8 \eta_3 \varepsilon_{c2}^2 \eta_2 \varepsilon_{r3}^2 - 4 \eta_2 \varepsilon_{cl}^2 \varepsilon_{r3}^2 \eta_1 - 16 \eta_2 \\
& \quad \varepsilon_{cl}^2 \varepsilon_{r3} \eta_1 - 4 \eta_2 \varepsilon_{cl}^2 \varepsilon_{c3}^2 \eta_1 + 8 \eta_3 \varepsilon_{r2}^2 \eta_2 \varepsilon_{c3}^2 + 8 \eta_3 \varepsilon_{r2}^2 \eta_2 \varepsilon_{r3}^2 - 4 \eta_1 \varepsilon_{r2}^2 \varepsilon_{c3}^2 \eta_2 - 4 \eta_1 \\
& \quad \varepsilon_{r2}^2 \varepsilon_{c3}^2 \eta_3 - 4 \eta_1 \varepsilon_{c2}^2 \varepsilon_{r3}^2 \eta_2 - 4 \eta_1 \varepsilon_{c2}^2 \varepsilon_{r3}^2 \eta_3 - 16 \eta_1 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 - 4 \eta_1 \varepsilon_{c2}^2 \varepsilon_{c3}^2 \eta_2 \\
& \quad - 4 \eta_1 \varepsilon_{c2}^2 \varepsilon_{c3}^2 \eta_3 - 4 \eta_3 \varepsilon_{cl}^2 \varepsilon_{c2}^2 \eta_2 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 \eta_1 - 16 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_1 - 4 \eta_2 \varepsilon_{r1}^2 \\
& \quad \varepsilon_{c3}^2 \eta_1 + \varepsilon_{cl}^2 \eta_1^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 + \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 8 \eta_3 \eta_1 \varepsilon_{cl}^2 \varepsilon_{r3}^2 + 8 \eta_3 \eta_1 \varepsilon_{cl}^2 \\
& \quad \varepsilon_{c3}^2 + 16 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 + 4 \eta_3^2 \varepsilon_{cl}^2 \varepsilon_{r2} \varepsilon_{r3}^2 + \eta_3^2 \varepsilon_{cl}^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 4 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \eta_2 - 4 \eta_3 \varepsilon_{r1}^2 \\
& \quad \varepsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \varepsilon_{cl}^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 8 \eta_3 \varepsilon_{r1}^2 \eta_1 \varepsilon_{c3}^2 + 4 \eta_2^2 \\
& \quad \varepsilon_{c3}^2 \varepsilon_{r2}^2 + 4 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - 16 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_1 - 4 \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \eta_1 - 4 \eta_3 \varepsilon_{r1} \varepsilon_{r2}^2 \eta_1 \\
& \quad - 16 \eta_3 \varepsilon_{r1} \varepsilon_{r2} \eta_1 + 72 \varepsilon_{cl} \eta_1 \varepsilon_{c3} \eta_3 + 72 \varepsilon_{c2} \eta_2 \varepsilon_{c3} \eta_3 - 4 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \eta_1 \\
& \quad \left. + 72 \varepsilon_{cl} \eta_1 \varepsilon_{c2} \eta_2 - 2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} + 4 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 + 8 \eta_3 \varepsilon_{c2}^2 \eta_2 \varepsilon_{c3}^2 + 16 \varepsilon_{cl}^2 \eta_1^2 \right)
\end{aligned}
\tag{2.7}$$

$$\begin{aligned}
& + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \\
& + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \\
& + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1} \epsilon_{c2}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 + 16 \eta_3^2 - 2 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} \\
& + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{c1}^2 \epsilon_{r3} + 4 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c1}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2
\end{aligned}$$

[illegible]

$$\begin{aligned}
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 32 \eta_1^2 \epsilon_{r1} + 16 \eta_1^2 \epsilon_{r2} + 16 \eta_1^2 \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2}^2 \\
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_2^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4
\end{aligned}$$

$$\begin{aligned}
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3} \\
& - 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2
\end{aligned}$$

$$\begin{aligned} & \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\ & \epsilon_{r1}^2 \epsilon_{r3}^2) \end{aligned}$$

$eps_r := rhs(sol[2])$

$$\begin{aligned} & - \left(-64 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32 \eta_1 + 32 \eta_2 + 32 \eta_3 - 64 \epsilon_{r1} - 64 \epsilon_{r2} - 64 \epsilon_{r3} - 16 \epsilon_{c1}^2 \right. \\ & \quad - 16 \epsilon_{c2}^2 - 16 \epsilon_{c3}^2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 \\ & \quad + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r2} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 \\ & \quad - 32 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r2}^2 \\ & \quad \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 32 \\ & \quad \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \\ & \quad \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 2 \epsilon_{c2}^2 \\ & \quad \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \\ & \quad + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \\ & \quad \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\ & \quad - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_3 \epsilon_{c1}^2 \\ & \quad \epsilon_{c2}^2 \eta_2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c2}^2 \\ & \quad \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\ & \quad + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_3^2 \epsilon_{c1}^2 \\ & \quad \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 8 \eta_1^2 \\ & \quad \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 \\ & \quad + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 4 \eta_3^2 \\ & \quad \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 32 \epsilon_{c1}^2 \eta_1^2 + 32 \eta_3^2 \epsilon_{r1} + 32 \eta_3^2 \epsilon_{r2} \\ & \quad - 64 \eta_3^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3^2 \epsilon_{c2}^2 + 32 \eta_2^2 \epsilon_{r1} - 64 \eta_2^2 \epsilon_{r2} \\ & \quad + 32 \eta_2^2 \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 + 8 \eta_2^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 + 8 \eta_2^2 \epsilon_{c3}^2 + 32 \epsilon_{c3}^2 \eta_3^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \\ & \quad \epsilon_{r1}^2 \epsilon_{r2} - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r2}^2 - 64 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1} \epsilon_{c2}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{c1}^2 \epsilon_{r2} - 16 \\ & \quad \epsilon_{r1}^2 - 16 \epsilon_{r2}^2 - 16 \epsilon_{r3}^2 + 32 \eta_1^2 + 32 \eta_2^2 + 32 \eta_3^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\ & \quad - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\ & \quad - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \end{aligned} \quad (2.8)$$

$$\begin{aligned}
& \epsilon_{c3}^2 - 4\eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4\eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4\eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4\eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 4\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 36\epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 144\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 36\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 36\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4\eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 144\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 36\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 144\epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& - 4\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 4\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8\eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32\eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 8\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2\epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 8\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 16\eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4\epsilon_{c1}^2 \epsilon_{c2}^2 - 16\epsilon_{r1} \epsilon_{r3}^2 - 64\epsilon_{r1} \epsilon_{r3} - 16\epsilon_{r1} \epsilon_{c3}^2 - 16\epsilon_{r2} \epsilon_{r3}^2 - 64\epsilon_{r2} \epsilon_{r3} - 16\epsilon_{r2} \\
& \epsilon_{c3}^2 - 4\epsilon_{r1}^2 \epsilon_{r3}^2 - 16\epsilon_{r1}^2 \epsilon_{r3} - 4\epsilon_{r1}^2 \epsilon_{c3}^2 - 4\epsilon_{r2}^2 \epsilon_{r3}^2 - 16\epsilon_{r2}^2 \epsilon_{r3} - 4\epsilon_{r2}^2 \epsilon_{c3}^2 - 4\epsilon_{c1}^2 \epsilon_{r3}^2 \\
& - 16\epsilon_{c1}^2 \epsilon_{r3} - 4\epsilon_{c1}^2 \epsilon_{c3}^2 - 4\epsilon_{c2}^2 \epsilon_{r3}^2 - 16\epsilon_{c2}^2 \epsilon_{r3} - 4\epsilon_{c2}^2 \epsilon_{c3}^2 - 16\eta_1 \epsilon_{r1} + 32\eta_1 \epsilon_{r2} \\
& + 32\eta_1 \epsilon_{r3} - 16\eta_1 \epsilon_{r1}^2 + 8\eta_1 \epsilon_{r2}^2 + 8\eta_1 \epsilon_{r3}^2 - 16\eta_1 \epsilon_{c1}^2 + 8\eta_1 \epsilon_{c2}^2 + 8\eta_1 \epsilon_{c3}^2 \\
& + 32\eta_2 \epsilon_{r1} - 16\eta_2 \epsilon_{r2} + 32\eta_2 \epsilon_{r3} + 8\eta_2 \epsilon_{r1}^2 - 16\eta_2 \epsilon_{r2}^2 + 8\eta_2 \epsilon_{r3}^2 + 8\eta_2 \epsilon_{c1}^2 \\
& - 16\eta_2 \epsilon_{c2}^2 + 8\eta_2 \epsilon_{c3}^2 + 32\eta_3 \epsilon_{r1} + 32\eta_3 \epsilon_{r2} - 16\eta_3 \epsilon_{r3} + 8\eta_3 \epsilon_{r1}^2 + 8\eta_3 \epsilon_{r2}^2 \\
& - 16\eta_3 \epsilon_{r3}^2 + 8\eta_3 \epsilon_{c1}^2 + 8\eta_3 \epsilon_{c2}^2 - 16\eta_3 \epsilon_{c3}^2 + 32\eta_1^2 \epsilon_{r1}^2 + 32\eta_2^2 \epsilon_{r2}^2 + 32\epsilon_{c2}^2 \eta_2^2 \\
& + 64\eta_2 \eta_1 + 64\eta_3 \eta_1 + 64\eta_2 \eta_3 + 32\eta_3^2 \epsilon_{r3}^2 - 64\eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 8\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 8\eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + 2\epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2\epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8\epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 8\epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 2\epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8\eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 2\eta_2^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8\eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 8\epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 16\eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 16\eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1} \epsilon_{r2} + 2\eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 8\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 2\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8\eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2\eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8\eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2\eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 4\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1
\end{aligned}$$

$$\begin{aligned}
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 \\
& + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 16 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 \\
& + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 64 \eta_1^2 \epsilon_{r1} + 32 \eta_1^2 \epsilon_{r2} + 32 \eta_1^2 \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r3}^2 + 8 \eta_1^2 \\
& \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 32 \eta_1 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 4 \eta_2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{cl}^2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 + 32 \eta_3 \epsilon_{r1} \epsilon_{r2} + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \\
& - 16 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 32 \eta_3 \epsilon_{r1} \eta_1 - 32 \eta_2 \epsilon_{cl}^2 \eta_1 + 16 \eta_2 \\
& \epsilon_{c3}^2 \eta_1 - 32 \eta_3 \epsilon_{cl}^2 \eta_1 + 16 \eta_3 \epsilon_{c2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \\
& + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 32 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r3} + 32 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r3} - 32 \eta_2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \eta_3 - 32 \eta_2 \epsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \eta_2 - 32 \eta_1 \epsilon_{c3}^2 \eta_3 + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} \eta_1 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3} \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& \varepsilon_{r3}^2 \eta_3 + 16 \eta_2 \varepsilon_{r2} \varepsilon_{r3} \eta_3 + 16 \eta_2 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 + 32 \\
& \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3} - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 64 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - 16 \eta_1^2 \varepsilon_{r1} \\
& \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 4 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 16 \eta_1^2 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{c3}^2 - 8 \eta_3 \varepsilon_{r2}^2 \varepsilon_{r3} \eta_1 \\
& + 16 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r3} \eta_1 - 8 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_1 - 32 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2} \eta_1 - 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \varepsilon_{r1} \varepsilon_{r3} \eta_1 - 32 \eta_3 \varepsilon_{r2} \varepsilon_{r3} \eta_1 + 16 \eta_3 \varepsilon_{r1}^2 \varepsilon_{r3} \eta_1 - 8 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \eta_1 + 2 \varepsilon_{c2}^2 \eta_2^2 \\
& \varepsilon_{c1}^2 \varepsilon_{r3}^2 - 4 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - 16 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 32 \eta_1 \varepsilon_{r2} \varepsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \varepsilon_{r2} \varepsilon_{c3}^2 \eta_3 + 16 \eta_3 \varepsilon_{c2}^2 \varepsilon_{r3} \eta_2 + 16 \eta_1 \varepsilon_{r1} \varepsilon_{r3}^2 \eta_3 + 16 \eta_1 \varepsilon_{r1} \varepsilon_{c3}^2 \eta_3 + 8 \varepsilon_{c3}^2 \eta_3^2 \\
& \varepsilon_{c1}^2 \varepsilon_{r2} + 16 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 + 16 \eta_2 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 + 2 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 8 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 4 \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 16 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} - 4 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{r3} + 2 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 \\
& - 8 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \eta_3 - 32 \eta_2 \varepsilon_{r1} \varepsilon_{r2} \eta_3 - \eta_3 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 16 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} + 32 \eta_3^2 \varepsilon_{r1} \varepsilon_{r2} \\
& + 8 \eta_3^2 \varepsilon_{r1} \varepsilon_{c2}^2 + 2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 + 8 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{c2}^2 - 64 \eta_3^2 \varepsilon_{r1} \varepsilon_{r3} - 64 \\
& \eta_3^2 \varepsilon_{r2} \varepsilon_{r3} + 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 8 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r3} + 2 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 + 2 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3}^2 + 8 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r3} + 2 \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{c3}^2 + 2 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r2}^2 - 16 \eta_2^2 \varepsilon_{r1}^2 \varepsilon_{r2} - 64 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2} - 16 \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2} + 8 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3}^2 \\
& + 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r3} + 8 \eta_2^2 \varepsilon_{r1} \varepsilon_{c3}^2 - 16 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 64 \eta_2^2 \varepsilon_{r2} \varepsilon_{r3} - 16 \eta_2^2 \varepsilon_{r2} \varepsilon_{c3}^2 + 32 \\
& \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2} - 32 \eta_3 \varepsilon_{r2}^2 \eta_2 - 32 \eta_3 \varepsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3}^2 + 8 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r1}^2 + 8 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{c3}^2 \\
& + 32 \eta_1^2 \varepsilon_{r1}^2 \varepsilon_{r3} + 32 \eta_2^2 \varepsilon_{r1} \varepsilon_{r2}^2 + 8 \varepsilon_{c1}^2 \eta_1^2 \varepsilon_{r2}^2 + 32 \eta_2^2 \varepsilon_{r3} \varepsilon_{r2}^2 + 64 \eta_2 \varepsilon_{r1} \eta_3 - 16 \eta_3^2 \\
& \varepsilon_{c2}^2 \varepsilon_{r3} + 16 \eta_2 \varepsilon_{c1}^2 \eta_3 - 32 \eta_2 \varepsilon_{c3}^2 \eta_3 + 64 \eta_3 \varepsilon_{r2} \eta_1 - \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 4 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 \\
& - 4 \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3}^2 - \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 4 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{c3}^2 \\
& - \eta_1 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 16 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3} - \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 - 4 \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{r3} - \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{r3}^2 \\
& - 4 \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - \eta_1 \varepsilon_{r1} \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 4 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{r3} - 4 \eta_1 \varepsilon_{r1} \varepsilon_{r2} \varepsilon_{r3}^2 \\
& - 4 \eta_1 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} - \eta_1 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{c3}^2 - \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \varepsilon_{c3}^2 - 4 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3}^2 - 16 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} \\
& - 4 \eta_1 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - \eta_1 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \varepsilon_{r3}^2 - 4 \eta_1 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - \eta_1 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \varepsilon_{c3}^2 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{r3} \\
& - 4 \eta_2 \varepsilon_{r1} \varepsilon_{c2}^2 \varepsilon_{r3}^2 - \eta_2 \varepsilon_{r1}^2 \varepsilon_{r2} \varepsilon_{c3}^2 - 4 \eta_2 \varepsilon_{r1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - 16 \eta_3^2 \varepsilon_{r1}^2 \varepsilon_{r3}) / (64 + 4 \varepsilon_{c2}^2 \eta_2^2 \\
& \varepsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \varepsilon_{r1} + 64 \varepsilon_{r2} + 64 \varepsilon_{r3} + 16 \varepsilon_{c1}^2 + 16 \varepsilon_{c2}^2 + 16 \varepsilon_{c3}^2 \\
& - 8 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r3} - 8 \eta_3^2 \varepsilon_{r2}^2 \varepsilon_{r3} + 18 \varepsilon_{c1} \eta_1 \varepsilon_{c3} \eta_3 \varepsilon_{r2}^2 + 4 \varepsilon_{c2}^2 \eta_2^2 \varepsilon_{r3}^2 + 4 \varepsilon_{c3}^2 \eta_3^2 \varepsilon_{r2}^2 + 4 \\
& \eta_2^2 \varepsilon_{c1}^2 \varepsilon_{r2}^2 - 16 \eta_2 \varepsilon_{r2} \eta_1 + 8 \eta_3 \varepsilon_{r1}^2 \eta_1 \varepsilon_{r3}^2 - 16 \eta_3 \varepsilon_{r1} \varepsilon_{r2}^2 \eta_2 - 16 \eta_3 \varepsilon_{r1} \varepsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \varepsilon_{c1}^2 \varepsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{r2} \varepsilon_{r3} - 2 \eta_3^2 \varepsilon_{c1}^2 \varepsilon_{c2}^2 \varepsilon_{r3} - 4 \eta_1 \varepsilon_{r2}^2 \varepsilon_{r3}^2 \eta_2 - 4 \eta_1 \varepsilon_{r2}^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \\
& \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& + 4 \eta_3^2 \epsilon_{c2}^2 + 16 \eta_2^2 \epsilon_{r1} - 32 \eta_2^2 \epsilon_{r2} + 16 \eta_2^2 \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 + 4 \eta_2^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 + 4 \\
& \eta_2^2 \epsilon_{c3}^2 + 16 \epsilon_{c3}^2 \eta_3^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \epsilon_{r1}^2 \epsilon_{r2} + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 + 64 \epsilon_{r1} \epsilon_{r2} \\
& + 16 \epsilon_{r1} \epsilon_{c2}^2 + 4 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \epsilon_{c1}^2 \epsilon_{r2} + 16 \epsilon_{r1}^2 + 16 \epsilon_{r2}^2 + 16 \epsilon_{r3}^2 + 16 \eta_1^2 + 16 \eta_2^2 \\
& + 16 \eta_3^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 72 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2
\end{aligned}$$

[illegible]

[illegible]

$$\begin{aligned}
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 16 \epsilon_{c1}^2 \eta_2^2 \epsilon_{r3}^2 \\
& \eta_1^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 - 16 \eta_1 \epsilon_{c3}^2 \eta_3 \\
& + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_3 \\
& + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 \\
& - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2
\end{aligned}$$

$$\begin{aligned}
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3} \\
& - 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3})
\end{aligned}$$

eps_r

$$\begin{aligned}
& - \left(-64 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 32 \eta_1 + 32 \eta_2 + 32 \eta_3 - 64 \epsilon_{r1} - 64 \epsilon_{r2} - 64 \epsilon_{r3} - 16 \epsilon_{c1}^2 \right. \\
& - 16 \epsilon_{c2}^2 - 16 \epsilon_{c3}^2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 \\
& + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r2} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 \\
& - 32 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_2 - 16 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 32 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 2 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \\
& + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_1 + 16 \eta_3 \\
& \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\
& \left. - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 8 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_3 \epsilon_{c1}^2 \right)
\end{aligned} \tag{2.9}$$

$$\begin{aligned}
& \epsilon_{c2}^2 \eta_2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_3 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 8 \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 32 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 - 4 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{c3}^2 + 32 \epsilon_{c1}^2 \eta_1^2 + 32 \eta_3^2 \epsilon_{r1} + 32 \eta_3^2 \epsilon_{r2} \\
& - 64 \eta_3^2 \epsilon_{r3} + 8 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_3^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 + 8 \eta_3^2 \epsilon_{c2}^2 + 32 \eta_2^2 \epsilon_{r1} - 64 \eta_2^2 \epsilon_{r2} \\
& + 32 \eta_2^2 \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 + 8 \eta_2^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 + 8 \eta_2^2 \epsilon_{c3}^2 + 32 \epsilon_{c3}^2 \eta_3^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r2}^2 - 64 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1} \epsilon_{c2}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2}^2 - 16 \epsilon_{c1}^2 \epsilon_{r2} - 16 \\
& \epsilon_{r1}^2 - 16 \epsilon_{r2}^2 - 16 \epsilon_{r3}^2 + 32 \eta_1^2 + 32 \eta_2^2 + 32 \eta_3^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 144 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 36 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 144 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 36 \epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 144 \epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 8 \epsilon_{c3}^2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 - 16 \epsilon_{r1} \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{c3}^2 - 16 \epsilon_{r2} \epsilon_{r3}^2 - 64 \epsilon_{r2} \epsilon_{r3} - 16 \epsilon_{r2}
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{c1}^2 \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{c3}^2 - 4 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1} + 32 \eta_1 \epsilon_{r2} \\
& + 32 \eta_1 \epsilon_{r3} - 16 \eta_1 \epsilon_{r1}^2 + 8 \eta_1 \epsilon_{r2}^2 + 8 \eta_1 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{c1}^2 + 8 \eta_1 \epsilon_{c2}^2 + 8 \eta_1 \epsilon_{c3}^2 \\
& + 32 \eta_2 \epsilon_{r1} - 16 \eta_2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} + 8 \eta_2 \epsilon_{r1}^2 - 16 \eta_2 \epsilon_{r2}^2 + 8 \eta_2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{c1}^2 \\
& - 16 \eta_2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{c3}^2 + 32 \eta_3 \epsilon_{r1} + 32 \eta_3 \epsilon_{r2} - 16 \eta_3 \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 + 8 \eta_3 \epsilon_{r2}^2 \\
& - 16 \eta_3 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 + 8 \eta_3 \epsilon_{c2}^2 - 16 \eta_3 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 + 32 \eta_2^2 \epsilon_{r2}^2 + 32 \epsilon_{c2}^2 \eta_2^2 \\
& + 64 \eta_2 \eta_1 + 64 \eta_3 \eta_1 + 64 \eta_2 \eta_3 + 32 \eta_3^2 \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 8 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 8 \\
& \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 \\
& + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 \\
& + 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 \\
& + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \\
& - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} - 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 16 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r3}^2 \eta_1 + 32 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 16 \eta_3 \epsilon_{r2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \\
& - 32 \eta_3 \epsilon_{r1}^2 \eta_1 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3} - 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \\
& \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{c1}^2
\end{aligned}$$

$$\begin{aligned} & \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{cl}^2 \\ & \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4 \eta_2 \epsilon_{c3}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 \\ & + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3} \eta_1 \epsilon_{cl}^2 \\ & \epsilon_{c2}^2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 + 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 16 \eta_3 \\ & \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 \\ & + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 + 4 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 \\ & + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\ & - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_1 - 64 \eta_1^2 \epsilon_{r1} + 32 \eta_1^2 \epsilon_{r2} + 32 \eta_1^2 \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2}^2 + 8 \eta_1^2 \epsilon_{r3}^2 + 8 \eta_1^2 \\ & \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{c3}^2 + 8 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\ & + 32 \eta_1 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_1 \epsilon_{r2}^2 \\ & \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{cl}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 2 \eta_1 \\ & \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \\ & - 16 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} - 4 \eta_2 \\ & \epsilon_{cl}^2 \epsilon_{c2}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \\ & - 4 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_2 \\ & \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + 2 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\ & - 16 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3 \epsilon_{r1} \\ & \epsilon_{r2}^2 + 32 \eta_3 \epsilon_{r1} \epsilon_{r2} + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \\ & - 16 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r3} - 16 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \\ & - 16 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\ & \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_3 \\ & \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3}^2 + 32 \\ & \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r3} \eta_1 + 16 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\ & + 4 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 \\ & + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\ & + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{cl}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \\ & + 16 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3 \eta_2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \end{aligned}$$

$$\begin{aligned}
& + 16 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 32 \eta_3 \epsilon_{r1} \eta_1 - 32 \eta_2 \epsilon_{c1}^2 \eta_1 + 16 \eta_2 \\
& \epsilon_{c3}^2 \eta_1 - 32 \eta_3 \epsilon_{c1}^2 \eta_1 + 16 \eta_3 \epsilon_{c2}^2 \eta_1 - 32 \eta_2 \epsilon_{r1} \eta_1 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \\
& + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \\
& \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} + 2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 32 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 32 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r3} + 32 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r3} - 32 \eta_2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \eta_3 - 32 \eta_2 \epsilon_{r3}^2 \eta_3 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r2}^2 \eta_2 - 32 \eta_1 \\
& \epsilon_{r3}^2 \eta_3 - 32 \eta_1 \epsilon_{c2}^2 \eta_2 - 32 \eta_1 \epsilon_{c3}^2 \eta_3 + 32 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} \eta_1 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 - 8 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 \\
& - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 16 \eta_2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 16 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 32 \\
& \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 64 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 16 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 \\
& + 16 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 \\
& + 16 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + 2 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c1}^2 \epsilon_{r3}^2 - 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 \\
& - 32 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 16 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 16 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 16 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + 8 \epsilon_{c3}^2 \eta_3^2 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + 2 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 4 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 8 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + 2 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 16 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 32 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \\
& + 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 8 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 64 \eta_3^2 \epsilon_{r1} \epsilon_{r3} - 64 \\
& \eta_3^2 \epsilon_{r2} \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + 2 \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 64 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 16 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 \\
& + 32 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 8 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 16 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 64 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 16 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 32
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} - 32 \eta_3 \epsilon_{r2}^2 \eta_2 - 32 \eta_3 \epsilon_{c2}^2 \eta_2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 8 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 8 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 \\
& + 32 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3} + 32 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 8 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 + 32 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 + 64 \eta_2 \epsilon_{r1} \eta_3 - 16 \eta_3^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + 16 \eta_2 \epsilon_{cl}^2 \eta_3 - 32 \eta_2 \epsilon_{c3}^2 \eta_3 + 64 \eta_3 \epsilon_{r2} \eta_1 - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 16 \eta_1 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 4 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 16 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2) / (64 + 4 \epsilon_{c2}^2 \eta_2^2 \\
& \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{cl}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} + 18 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \\
& \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 \\
& - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 \\
& - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \\
& \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 \\
& - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \eta_2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_1 \\
& - 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_1 + \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \\
& + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{r3}^2 + 8 \eta_3 \eta_1 \epsilon_{cl}^2 \epsilon_{c3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_2 - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_2 - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{c3}^2 \epsilon_{r2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 - 4 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_1 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2} \eta_1 + 72 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 + 72 \epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \\
& - 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 + 72 \epsilon_{cl} \eta_1 \epsilon_{c2} \eta_2 - 2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \\
& \epsilon_{c3}^2 + 16 \epsilon_{cl}^2 \eta_1^2 + 16 \eta_3^2 \epsilon_{r1} + 16 \eta_3^2 \epsilon_{r2} - 32 \eta_3^2 \epsilon_{r3} + 4 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{cl}^2
\end{aligned}$$

$$\begin{aligned}
& + 4\eta_3^2 \epsilon_{c2}^2 + 16\eta_2^2 \epsilon_{r1}^2 - 32\eta_2^2 \epsilon_{r2}^2 + 16\eta_2^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{r1}^2 + 4\eta_2^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{c1}^2 + 4 \\
& \eta_2^2 \epsilon_{c3}^2 + 16\epsilon_{c3}^2 \eta_3^2 + 4\epsilon_{r1}^2 \epsilon_{r2}^2 + 16\epsilon_{r1}^2 \epsilon_{r2}^2 + 4\epsilon_{r1}^2 \epsilon_{c2}^2 + 16\epsilon_{r1}^2 \epsilon_{r2}^2 + 64\epsilon_{r1} \epsilon_{r2} \\
& + 16\epsilon_{r1} \epsilon_{c2}^2 + 4\epsilon_{c1}^2 \epsilon_{r2}^2 + 16\epsilon_{c1}^2 \epsilon_{r2}^2 + 16\epsilon_{r1}^2 + 16\epsilon_{r2}^2 + 16\epsilon_{r3}^2 + 16\eta_1^2 + 16\eta_2^2 \\
& + 16\eta_3^2 - 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 2\eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& - 2\eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32\eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2\eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 32\eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8\eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 8\eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2\eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2\eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2\eta_2 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2\eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 - 32\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + 18\epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \\
& + 72\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1} + 18\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 18\epsilon_{c2} \eta_2 \epsilon_{c3} \eta_3 \epsilon_{c1}^2 + 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 \eta_3 + 72\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3} + 18\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 18\epsilon_{c1} \eta_1 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 \\
& + 72\epsilon_{c1} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2} + 2\eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_1 + 8\eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 - 2\eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 \\
& - 32\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2\eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8\eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 32\eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2\eta_3 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 8\eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2\eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \\
& - 8\eta_3 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2\eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 2\eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2\eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \\
& + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4\eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16\eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 8\eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4\epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 - 8\eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \\
& + 4\epsilon_{c1}^2 \epsilon_{c2}^2 + 16\epsilon_{r1} \epsilon_{r3}^2 + 64\epsilon_{r1} \epsilon_{r3} + 16\epsilon_{r1} \epsilon_{c3}^2 + 16\epsilon_{r2} \epsilon_{r3}^2 + 64\epsilon_{r2} \epsilon_{r3} + 16\epsilon_{r2} \\
& \epsilon_{c3}^2 + 4\epsilon_{r1}^2 \epsilon_{r3}^2 + 16\epsilon_{r1}^2 \epsilon_{r3} + 4\epsilon_{r1}^2 \epsilon_{c3}^2 + 4\epsilon_{r2}^2 \epsilon_{r3}^2 + 16\epsilon_{r2}^2 \epsilon_{r3} + 4\epsilon_{r2}^2 \epsilon_{c3}^2 + 4\epsilon_{c1}^2 \epsilon_{r3}^2 \\
& + 16\epsilon_{c1}^2 \epsilon_{r3} + 4\epsilon_{c1}^2 \epsilon_{c3}^2 + 4\epsilon_{c2}^2 \epsilon_{r3}^2 + 16\epsilon_{c2}^2 \epsilon_{r3} + 4\epsilon_{c2}^2 \epsilon_{c3}^2 - 32\eta_1 \epsilon_{r1} + 64\eta_1 \epsilon_{r2} \\
& + 64\eta_1 \epsilon_{r3} - 32\eta_1 \epsilon_{r1}^2 + 16\eta_1 \epsilon_{r2}^2 + 16\eta_1 \epsilon_{r3}^2 - 32\eta_1 \epsilon_{c1}^2 + 16\eta_1 \epsilon_{c2}^2 + 16\eta_1 \epsilon_{c3}^2 \\
& + 64\eta_2 \epsilon_{r1} - 32\eta_2 \epsilon_{r2} + 64\eta_2 \epsilon_{r3} + 16\eta_2 \epsilon_{r1}^2 - 32\eta_2 \epsilon_{r2}^2 + 16\eta_2 \epsilon_{r3}^2 + 16\eta_2 \epsilon_{c1}^2 \\
& - 32\eta_2 \epsilon_{c2}^2 + 16\eta_2 \epsilon_{c3}^2 + 64\eta_3 \epsilon_{r1} + 64\eta_3 \epsilon_{r2} - 32\eta_3 \epsilon_{r3} + 16\eta_3 \epsilon_{r1}^2 + 16\eta_3 \epsilon_{r2}^2 \\
& - 32\eta_3 \epsilon_{r3}^2 + 16\eta_3 \epsilon_{c1}^2 + 16\eta_3 \epsilon_{c2}^2 - 32\eta_3 \epsilon_{c3}^2 + 16\eta_1^2 \epsilon_{r1}^2 + 16\eta_2^2 \epsilon_{r2}^2 + 16\epsilon_{c2}^2 \eta_2^2 \\
& + 32\eta_2 \eta_1 + 32\eta_3 \eta_1 + 32\eta_2 \eta_3 + 16\eta_3^2 \epsilon_{r3}^2 - 32\eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4\eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}
\end{aligned}$$

$$\begin{aligned}
& +4\eta_2^2\epsilon_{r2}^2\epsilon_{r1}\epsilon_{c3}^2+\epsilon_{c3}^2\eta_3^2\epsilon_{c1}^2\epsilon_{c2}^2+\epsilon_{c1}^2\eta_1^2\epsilon_{c2}^2\epsilon_{r3}^2+\eta_3^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2+\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2+4 \\
& \epsilon_{c2}^2\eta_2^2\epsilon_{r1}\epsilon_{r3}^2+4\epsilon_{c1}^2\eta_1^2\epsilon_{r2}\epsilon_{c3}^2+\epsilon_{c1}^2\eta_1^2\epsilon_{r2}^2\epsilon_{c3}^2+4\eta_2^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2+\eta_2^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2+4 \\
& \eta_3^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2+4\epsilon_{c2}^2\eta_2^2\epsilon_{c1}^2\epsilon_{r3}^2+8\eta_1\epsilon_{r1}\epsilon_{r2}^2\eta_2+8\eta_1\epsilon_{r1}\epsilon_{c2}^2\eta_2+4\eta_3^2\epsilon_{r3}^2\epsilon_{r1}^2\epsilon_{r2}^2+ \\
& \eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2+\eta_3^2\epsilon_{r2}^2\epsilon_{r3}^2\epsilon_{r1}^2-4\eta_3\epsilon_{r1}\epsilon_{c2}^2\eta_1+4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+\eta_1^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+\eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2+16\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2+\eta_1^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2+\eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2+8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_3+2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2\eta_1+8\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1+2\eta_2\eta_3\epsilon_{r1}^2 \\
& \epsilon_{c2}^2\epsilon_{c3}^2+2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_3+2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1+8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+2\eta_3\epsilon_{c1}^2 \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_1+2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_1+2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_3+2\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}\eta_1+2\eta_3\epsilon_{r1} \\
& \epsilon_{r2}^2\epsilon_{r3}\eta_1+8\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_1+2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}\eta_2+2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}\eta_2+2\eta_3\epsilon_{r1}^2 \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_2+8\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2\eta_3+2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2\eta_2+2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2\eta_3+8\eta_1\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{r3}\eta_2+8\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}\eta_2+8\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_3+2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2\eta_2+2\eta_1\epsilon_{r1}\epsilon_{r2}^2 \\
& \epsilon_{r3}^2\eta_3+2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2-8\eta_1^2\epsilon_{r1}\epsilon_{r2}^2-32\eta_1^2\epsilon_{r1}\epsilon_{r2}+16\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}-8\eta_1^2\epsilon_{r1} \\
& \epsilon_{c2}^2-8\eta_1^2\epsilon_{r1}\epsilon_{r3}^2-32\eta_1^2\epsilon_{r1}\epsilon_{r3}-8\eta_1^2\epsilon_{r1}\epsilon_{c3}^2+8\eta_2\epsilon_{r3}^2\eta_1+16\epsilon_{c2}^2\eta_2^2\epsilon_{r1}+8\eta_3 \\
& \epsilon_{r2}^2\eta_1-16\eta_2\epsilon_{r1}^2\eta_1+4\epsilon_{c2}^2\eta_2^2\epsilon_{c1}^2-16\eta_3\epsilon_{r1}^2\eta_1+4\epsilon_{c1}^2\eta_1^2\epsilon_{r3}^2+4\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2+4 \\
& \epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2+4\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2+\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2+4\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2+16\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2+ \\
& \epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2+64\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2+\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+16\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2+16\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2 \\
& +4\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2+4\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2+4\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2+16\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2+4\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+4\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{c3}^2+\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2+4\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2+16\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2+4\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2+\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2+4\epsilon_{c1}^2 \\
& \epsilon_{c2}^2\epsilon_{r3}^2+\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{c3}^2-8\eta_1\epsilon_{r1}^2\epsilon_{r2}^2-32\eta_1\epsilon_{r1}^2\epsilon_{r2}-8\eta_1\epsilon_{r1}^2\epsilon_{c2}^2-8\eta_1\epsilon_{r1}\epsilon_{r2}^2 \\
& -32\eta_1\epsilon_{r1}\epsilon_{r2}-8\eta_1\epsilon_{r1}\epsilon_{c2}^2-8\eta_1\epsilon_{c1}^2\epsilon_{r2}^2-32\eta_1\epsilon_{c1}^2\epsilon_{r2}-8\eta_1\epsilon_{c1}^2\epsilon_{c2}^2-8\eta_1\epsilon_{r1} \\
& \epsilon_{r3}^2+2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{c1}^2\epsilon_{r2}^2+2\eta_2\epsilon_{r3}^2\eta_1\epsilon_{c1}^2\epsilon_{c2}^2+2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2+2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{c1}^2\epsilon_{r2}^2 \\
& +2\eta_2\epsilon_{c3}^2\eta_1\epsilon_{c1}^2\epsilon_{c2}^2+2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2\eta_2+2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2\eta_3+8\eta_2\epsilon_{r3}\eta_1\epsilon_{r1}^2\epsilon_{c2}^2 \\
& +8\eta_2\epsilon_{r3}\eta_1\epsilon_{c1}^2\epsilon_{r2}^2+8\eta_2\epsilon_{r3}\eta_1\epsilon_{c1}^2\epsilon_{c2}^2+2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_2+2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2\eta_3 \\
& +2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}\eta_1+8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+2\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_1+2\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2\eta_1 \\
& +8\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+2\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2\eta_1+2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2\eta_1+2\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_1 \\
& +8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}\eta_1+2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2\eta_1+8\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}\eta_1+8\eta_3\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2 \\
& +2\eta_3\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r2}^2\epsilon_{c3}^2\eta_1-32\eta_1^2\epsilon_{r1}+16\eta_1^2\epsilon_{r2}+16\eta_1^2\epsilon_{r3}+4\eta_1^2\epsilon_{r2}^2
\end{aligned}$$

$$\begin{aligned}
& + 4 \eta_1^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{r3}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1 \epsilon_{r1} \epsilon_{c3}^2 \\
& + 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 + 64 \eta_1 \epsilon_{r2} \epsilon_{r3} + 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_1 \\
& \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2 \\
& \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_2 \epsilon_{r1} \epsilon_{r2} - 32 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \\
& - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} - 8 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 64 \eta_2 \epsilon_{r1} \epsilon_{r3} + 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \\
& - 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + 4 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} + 4 \eta_2 \\
& \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 16 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \\
& + 4 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 + 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 64 \eta_3 \epsilon_{r1} \epsilon_{r2} + 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 + 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_3 \epsilon_{r1} \epsilon_{r3} - 32 \eta_3 \epsilon_{r1} \epsilon_{c3}^2 - 32 \eta_3 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 32 \eta_3 \epsilon_{r2} \epsilon_{r3} - 32 \eta_3 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c1}^2 \epsilon_{c3}^2 - 8 \eta_3 \epsilon_{c2}^2 \\
& \epsilon_{r3}^2 - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \eta_3^2 \epsilon_{r2} \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r3} \eta_1 + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r2} \eta_1 \epsilon_{c1}^2 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r2}^2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 \\
& + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{c1}^2 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& + 8 \eta_3 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 2 \eta_3 \eta_2 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 \\
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_2^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2
\end{aligned}$$

$$\begin{aligned}
& -16\eta_1\epsilon_{c3}^2\eta_3+16\epsilon_{c3}^2\eta_3^2\epsilon_{r2}+32\eta_2\epsilon_{r3}\eta_1+4\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2+4\eta_3^2\epsilon_{r1}^2\epsilon_{r2}+ \eta_3^2\epsilon_{r1}^2 \\
& \epsilon_{c2}^2+4\eta_3^2\epsilon_{r1}\epsilon_{r2}^2-4\eta_2\epsilon_{r1}\epsilon_{c3}^2\eta_1-4\eta_2\epsilon_{r2}\epsilon_{r3}^2\eta_1-16\eta_2\epsilon_{r2}\epsilon_{r3}\eta_1-32 \\
& \eta_3^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-8\eta_3^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-8\eta_3^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-4\eta_2\epsilon_{c1}^2\epsilon_{r3}\eta_3-4\eta_2\epsilon_{c1}^2\epsilon_{c3}^2\eta_3 \\
& +4\epsilon_{c1}^2\eta_1^2\epsilon_{r2}^2\epsilon_{r3}-4\eta_2\epsilon_{r1}^2\epsilon_{r3}^2\eta_3-4\eta_2\epsilon_{r1}^2\epsilon_{r3}\eta_3-4\eta_2\epsilon_{r1}^2\epsilon_{c3}^2\eta_3-4\eta_2\epsilon_{c1}^2 \\
& \epsilon_{r3}^2\eta_3+8\eta_2\epsilon_{r1}^2\epsilon_{r2}\eta_1+8\eta_2\epsilon_{r1}\epsilon_{r2}\eta_1+8\eta_2\epsilon_{r2}\epsilon_{r3}^2\eta_3+8\eta_2\epsilon_{r2}\epsilon_{r3}\eta_3+8\eta_2\epsilon_{r2} \\
& \epsilon_{c3}^2\eta_3-8\eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-2\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2+16\epsilon_{c1}^2\eta_1^2\epsilon_{r2}\epsilon_{r3}-2\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-32 \\
& \eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-8\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-8\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-2\eta_1^2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-2\eta_1^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2 \\
& -8\eta_1^2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2-4\eta_3\epsilon_{r2}^2\epsilon_{r3}\eta_1+8\eta_3\epsilon_{c1}^2\epsilon_{r3}\eta_1-4\eta_3\epsilon_{c2}^2\epsilon_{r3}\eta_1-16\eta_3 \\
& \epsilon_{c1}^2\epsilon_{r2}\eta_1-4\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\eta_1+8\eta_3\epsilon_{r1}\epsilon_{r3}\eta_1-16\eta_3\epsilon_{r2}\epsilon_{r3}\eta_1+8\eta_3\epsilon_{r1}^2\epsilon_{r3}\eta_1 \\
& -4\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\eta_1+\epsilon_{c2}^2\eta_2^2\epsilon_{c1}^2\epsilon_{r3}^2-2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-8\eta_2^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}+4\epsilon_{c1}^2\eta_1^2\epsilon_{r2}\epsilon_{r3}^2 \\
& -16\eta_1\epsilon_{r2}\epsilon_{r3}^2\eta_3-16\eta_1\epsilon_{r2}\epsilon_{c3}^2\eta_3+8\eta_3\epsilon_{c2}^2\epsilon_{r3}\eta_2+8\eta_1\epsilon_{r1}\epsilon_{r3}^2\eta_3+8\eta_1\epsilon_{r1} \\
& \epsilon_{c3}^2\eta_3+4\epsilon_{c3}^2\eta_3^2\epsilon_{c1}^2\epsilon_{r2}+8\eta_2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2+8\eta_2\eta_1\epsilon_{r1}^2\epsilon_{r2}^2+\epsilon_{c3}^2\eta_3^2\epsilon_{c1}^2\epsilon_{r2}^2+4\epsilon_{c2}^2 \\
& \eta_2^2\epsilon_{r1}\epsilon_{c3}^2-2\eta_2^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_2^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}-2\eta_2^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2+4\epsilon_{c1}^2\eta_1^2\epsilon_{c2}^2\epsilon_{r3}+\epsilon_{c1}^2 \\
& \eta_1^2\epsilon_{c2}^2\epsilon_{c3}^2-4\eta_2\epsilon_{r1}^2\epsilon_{r2}\eta_3-16\eta_2\epsilon_{r1}\epsilon_{r2}\eta_3-2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-32\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3} \\
& +16\eta_3^2\epsilon_{r1}\epsilon_{r2}+4\eta_3^2\epsilon_{r1}\epsilon_{c2}^2+\eta_3^2\epsilon_{c1}^2\epsilon_{r2}^2+4\eta_3^2\epsilon_{c1}^2\epsilon_{r2}+\eta_3^2\epsilon_{c1}^2\epsilon_{c2}^2-32\eta_3^2\epsilon_{r1}\epsilon_{r3} \\
& -32\eta_3^2\epsilon_{r2}\epsilon_{r3}+\eta_2^2\epsilon_{r1}^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{r1}^2\epsilon_{r3}+\eta_2^2\epsilon_{r1}^2\epsilon_{c3}^2+\eta_2^2\epsilon_{c1}^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{c1}^2\epsilon_{r3}+ \\
& \eta_2^2\epsilon_{c1}^2\epsilon_{c3}^2+\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2-8\eta_2^2\epsilon_{r1}^2\epsilon_{r2}-32\eta_2^2\epsilon_{r1}\epsilon_{r2}-8\eta_2^2\epsilon_{c1}^2\epsilon_{r2}+4\eta_2^2\epsilon_{r1}\epsilon_{r3}^2+16 \\
& \eta_2^2\epsilon_{r1}\epsilon_{r3}+4\eta_2^2\epsilon_{r1}\epsilon_{c3}^2-8\eta_2^2\epsilon_{r2}\epsilon_{r3}^2-32\eta_2^2\epsilon_{r2}\epsilon_{r3}-8\eta_2^2\epsilon_{r2}\epsilon_{c3}^2+16\epsilon_{c1}^2\eta_1^2\epsilon_{r2} \\
& -16\eta_3\epsilon_{r2}^2\eta_2-16\eta_3\epsilon_{c2}^2\eta_2+4\eta_1^2\epsilon_{r1}^2\epsilon_{r3}^2+4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{c3}^2+16\eta_1^2 \\
& \epsilon_{r1}^2\epsilon_{r3}+16\eta_2^2\epsilon_{r1}\epsilon_{r2}^2+4\epsilon_{c1}^2\eta_1^2\epsilon_{r2}^2+16\eta_2^2\epsilon_{r3}\epsilon_{r2}^2+32\eta_2\epsilon_{r1}\eta_3-8\eta_3^2\epsilon_{c2}^2\epsilon_{r3} \\
& +8\eta_2\epsilon_{c1}^2\eta_3-16\eta_2\epsilon_{c3}^2\eta_3+32\eta_3\epsilon_{r2}\eta_1-2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2 \\
& -8\eta_1\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_1\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-8\eta_1\epsilon_{r1}\epsilon_{r2} \\
& \epsilon_{c3}^2-2\eta_1\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2-32\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-2\eta_1\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2-8\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-2\eta_1\epsilon_{c1}^2 \\
& \epsilon_{r2}^2\epsilon_{r3}^2-8\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}-8\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2-2\eta_1\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_1\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3} \\
& -8\eta_1\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-8\eta_1\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}-2\eta_1\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2-2\eta_1\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_1\epsilon_{c1}^2\epsilon_{r2} \\
& \epsilon_{r3}^2-32\eta_1\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}-8\eta_1\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-2\eta_1\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_1\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_1\epsilon_{c1}^2 \\
& \epsilon_{c2}^2\epsilon_{c3}^2-8\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}-8\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-8\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-8\eta_2^2
\end{aligned}$$

$\epsilon_{r1}^2 \epsilon_{r3})$
 $\xrightarrow{\text{to string}}$

$$\begin{aligned} & "- (-64C 8*\epsilon[c2]^2*\eta[2]^2*\epsilon[c3]^2C 32*\eta[1]C 32*\eta[2]C 32*\eta[3] -64*\epsilon[r1] -64*\epsilon[r2] -64*\epsilon[r3] -16*\epsilon[c1]^2 -16* \\ & \epsilon[c2]^2 -16*\epsilon[c3]^2 -16*\eta[3]^2*\epsilon[c1]^2*\epsilon[r3] \\ & -16*\eta[3]^2*\epsilon[r2]^2*\epsilon[r3]C 36*\epsilon[c1]*\eta[1]*\epsilon[c3] \\ & *\eta[3]*\epsilon[r2]^2C 8*\epsilon[c2]^2*\eta[2]^2*\epsilon[r3]^2C 8*\epsilon[c3]^2*\eta[3]^2*\epsilon[r2]^2C 8*\eta[2]^2*\epsilon[c1]^2*\epsilon[r2]^2 \\ & -32*\eta[2]*\epsilon[r2]*\eta[1]C 16*\eta[3]*\epsilon[r1]^2*\eta[1]*\epsilon[r3] \\ & ^2 -32*\eta[3]*\epsilon[r1]*\epsilon[r2]^2*\eta[2] -32*\eta[3]*\epsilon[r1]* \\ & \epsilon[c2]^2*\eta[2] -8*\eta[3]*\epsilon[c1]^2*\epsilon[r2]^2*\eta[2] -16*\eta \\ & [3]^2*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3] -4*\eta[3]^2*\epsilon[c1]^2* \\ & \epsilon[c2]^2*\epsilon[r3] -8*\eta[1]*\epsilon[r2]^2*\epsilon[r3]^2*\eta[2] -8* \\ & \eta[1]*\epsilon[r2]^2*\epsilon[r3]^2*\eta[3] -32*\eta[1]*\epsilon[r2]^2*\epsilon[r3] \\ & *\eta[2]C 16*\eta[3]*\epsilon[r2]^2*\epsilon[r3]*\eta[2]C 8*\epsilon[c3]^2* \\ & \eta[3]^2*\epsilon[r1]*\epsilon[r2]^2C 32*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1] \\ & *\epsilon[r3]C 8*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]^2*\epsilon[r3]C 32* \\ & \epsilon[c3]^2*\eta[3]^2*\epsilon[r1]*\epsilon[r2]C 16*\eta[2]*\eta[1]*\epsilon[c1]^2*\epsilon[c2]^2 -8*\eta \\ & [2]*\epsilon[c1]^2*\epsilon[r2]*\eta[3] -32*\eta[2]*\epsilon[r1]*\epsilon[r3]^2* \\ & \eta[3] -32*\eta[2]*\epsilon[r1]*\epsilon[r3]*\eta[3] -32*\eta[2]*\epsilon[r1]* \\ & \epsilon[c3]^2*\eta[3]C 2*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1]^2*\epsilon[c3] \\ & ^2C 16*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\eta[1] -8*\eta[2]*\epsilon[r1]* \\ & \epsilon[r3]^2*\eta[1] -32*\eta[2]*\epsilon[r1]*\epsilon[r3]*\eta[1]C 8*\epsilon[c3] \\ & ^2*\eta[3]^2*\epsilon[c1]^2C 16*\eta[3]*\epsilon[c2]^2*\eta[2]*\epsilon[r3] \\ & ^2 -8*\eta[2]*\epsilon[c1]^2*\epsilon[r3]^2*\eta[1] -32*\eta[2]*\epsilon[c1]^2* \\ & \epsilon[r3]*\eta[1] -8*\eta[2]*\epsilon[c1]^2*\epsilon[c3]^2*\eta[1]C 16*\eta[3]* \\ & \epsilon[r2]^2*\eta[2]*\epsilon[c3]^2C 16*\eta[3]*\epsilon[r2]^2*\eta[2]*\epsilon[r3] \\ & ^2 -8*\eta[1]*\epsilon[r2]^2*\epsilon[c3]^2*\eta[2] -8*\eta[1]*\epsilon[r2] \\ & ^2*\epsilon[c3]^2*\eta[3] -8*\eta[1]*\epsilon[c2]^2*\epsilon[r3]^2*\eta[2] -8*\eta \\ & [1]*\epsilon[c2]^2*\epsilon[r3]^2*\eta[3] -32*\eta[1]*\epsilon[c2]^2*\epsilon[r3] \\ & *\eta[2] -8*\eta[1]*\epsilon[c2]^2*\epsilon[c3]^2*\eta[2] -8*\eta[1]*\epsilon[c2] \\ & ^2*\epsilon[c3]^2*\eta[3] -8*\eta[3]*\epsilon[c1]^2*\epsilon[c2]^2*\eta[2] -8* \\ & \eta[2]*\epsilon[r1]^2*\epsilon[r3]^2*\eta[1] -32*\eta[2]*\epsilon[r1]^2*\epsilon[r3] \\ & *\eta[1] -8*\eta[2]*\epsilon[r1]^2*\epsilon[c3]^2*\eta[1]C 2*\epsilon[c1]^2* \\ & \eta[1]^2*\epsilon[r2]^2*\epsilon[r3]^2C 2*\epsilon[c2]^2*\eta[2]^2*\epsilon[r1] \\ & ^2*\epsilon[r3]^2C 2*\epsilon[c3]^2*\eta[3]^2*\epsilon[r1]^2*\epsilon[r2] \end{aligned} \quad (2.10)$$

$$\begin{aligned}
& ^2C_{16}\eta[3]\eta[1]\epsilon[c1]^2\epsilon[r3]^2C_{16}\eta[3]\eta[1]\epsilon[c1]^2\epsilon[c3]^2C_{32}\eta[3]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2C_{8}\eta[3]^2\epsilon[c1]^2\epsilon[r2]\epsilon[r3]^2C_2\eta[3]^2\epsilon[c1]^2\epsilon[c2]^2\epsilon[r3]^2-8\eta[3]\epsilon[r1]^2\epsilon[r2]^2\eta[2]-4\eta[3]^2\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]-4\eta[3]^2\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]-16\eta[3]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]C_{16}\eta[3]\epsilon[r1]^2\eta[1]\epsilon[c3]^2C_{8}\eta[2]^2\epsilon[c3]^2\epsilon[r2]^2C_{8}\eta[1]^2\epsilon[r1]^2\epsilon[r2]\epsilon[c3]^2-32\eta[3]\epsilon[r1]^2\epsilon[r2]\eta[1]-8\eta[3]\epsilon[r1]^2\epsilon[c2]^2\eta[1]-8\eta[3]\epsilon[r1]\epsilon[r2]^2\eta[1]-32\eta[3]\epsilon[r1]\epsilon[r2]\eta[1]C_{144}\epsilon[c1]\eta[1]\epsilon[c3]\eta[3]C_{144}\epsilon[c2]\eta[2]\epsilon[c3]\eta[3]-8\eta[3]\epsilon[r1]^2\epsilon[r2]^2\eta[1]C_{144}\epsilon[c1]\eta[1]\epsilon[c2]\eta[2]-4\eta[3]^2\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]C_8\eta[3]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2C_{16}\eta[3]\epsilon[c2]^2\eta[2]\epsilon[c3]^2C_{32}\epsilon[c1]^2\eta[1]^2C_{32}\eta[3]^2\epsilon[r1]C_{32}\eta[3]^2\epsilon[r2]-64\eta[3]^2\epsilon[r3]C_8\eta[3]^2\epsilon[r1]^2C_8\eta[3]^2\epsilon[r2]^2C_{8}\eta[3]^2\epsilon[c1]^2C_{8}\eta[3]^2\epsilon[c2]^2C_{32}\eta[2]^2\epsilon[r1]-64\eta[2]^2\epsilon[r2]C_{32}\eta[2]^2\epsilon[r3]C_8\eta[2]^2\epsilon[r1]^2C_{8}\eta[2]^2\epsilon[r3]^2C_{8}\eta[2]^2\epsilon[c1]^2C_{8}\eta[2]^2\epsilon[c3]^2C_{32}\epsilon[c3]^2\eta[3]^2-4\epsilon[r1]^2\epsilon[r2]^2-16\epsilon[r1]^2\epsilon[r2]-4\epsilon[r1]^2\epsilon[c2]^2-16\epsilon[r1]\epsilon[r2]^2-64\epsilon[r1]\epsilon[r2]-16\epsilon[r1]\epsilon[c2]^2-4\epsilon[c1]^2\epsilon[r2]^2-16\epsilon[c1]^2\epsilon[r2]-16\epsilon[r1]^2-16\epsilon[r2]^2-16\epsilon[r3]^2C_{32}\eta[1]^2C_{32}\eta[2]^2C_{32}\eta[3]^2-\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[c3]^2-4\eta[2]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2-\eta[2]\epsilon[r1]^2\epsilon[c2]^2\epsilon[c3]^2-4\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[c3]^2-\eta[2]\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]^2-16\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[r3]-\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2-16\eta[2]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]-\eta[2]\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]^2-16\eta[2]\epsilon[r1]\epsilon[c2]^2\epsilon[r3]-\eta[2]\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2-4\eta[2]\epsilon[r1]\epsilon[r2]^2\epsilon[c3]^2-4\eta[2]\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]-4\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[r3]^2-4\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]-4\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[c2]^2\epsilon[c3]^2-\eta[2]\epsilon[c1]^2\epsilon[r2]^2\epsilon[c3]^2-\eta[2]\epsilon[c1]^2\epsilon[r2]
\end{aligned}$$

$[r2]^{\epsilon} [r3]^2 - 4 \eta^2 \epsilon [c1]^2 \epsilon [r2]^{\epsilon} [r3]^{\epsilon} - \eta^2 \epsilon [c1]^2 \epsilon [r2]^{\epsilon} \epsilon [c3]^2 - \eta^2 \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [r3]^2 - 4 \eta^2 \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - \eta^2 \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [c3]^2 - 4 \eta^3 \epsilon [r1]^2 \epsilon [r2]^{\epsilon} \epsilon [r3]^{\epsilon} - 4 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [c2]^2 \epsilon [r3]^2 - 4 \eta^3 \epsilon [r1]^2 \epsilon [r2]^{\epsilon} \epsilon [c3]^2 - \eta^3 \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - \eta^3 \epsilon [r1]^2 \epsilon [r2]^2 \epsilon [c3]^2 - 4 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [r2]^2 \epsilon [r3]^2 - 16 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [r2]^{\epsilon} \epsilon [c3]^2 C 36 \epsilon [c1]^{\epsilon} \eta [1]^{\epsilon} \epsilon [c3]^{\epsilon} \eta [3]^{\epsilon} \epsilon [c2]^{\epsilon} \eta^2 C 144 \epsilon [c2]^{\epsilon} \eta^2 \epsilon [c3]^{\epsilon} \eta [3]^{\epsilon} \epsilon [r1]^{\epsilon} C 36 \epsilon [c2]^{\epsilon} \eta^2 \epsilon [c3]^{\epsilon} \eta [3]^{\epsilon} \epsilon [r1]^2 C 36 \epsilon [c2]^{\epsilon} \eta^2 \epsilon [c3]^{\epsilon} \eta [3]^{\epsilon} \epsilon [c1]^2 C 4 \eta^2 \epsilon [r1]^2 \epsilon [r2]^{\epsilon} \epsilon [r3]^2 \eta [3]^{\epsilon} C 144 \epsilon [c1]^{\epsilon} \eta [1]^{\epsilon} \epsilon [c2]^{\epsilon} \eta^2 \epsilon [r3]^{\epsilon} C 36 \epsilon [c1]^{\epsilon} \eta [1]^{\epsilon} \epsilon [c2]^{\epsilon} \eta^2 \epsilon [r3]^2 C 36 \epsilon [c1]^{\epsilon} \eta [1]^{\epsilon} \epsilon [c2]^{\epsilon} \eta^2 \epsilon [c3]^2 C 144 \epsilon [c1]^{\epsilon} \eta [1]^{\epsilon} \epsilon [c3]^{\epsilon} \eta [3]^{\epsilon} \epsilon [r2]^{\epsilon} C 4 \eta^2 \epsilon [r1]^2 \epsilon [r2]^2 \epsilon [r3]^2 \eta [1]^{\epsilon} C 16 \eta^2 \epsilon [r1]^{\epsilon} \epsilon [r2]^{\epsilon} \epsilon [c3]^2 \eta [3]^{\epsilon} - \eta [3]^{\epsilon} \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^2 - 16 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [r2]^{\epsilon} \epsilon [r3]^{\epsilon} - \eta [3]^{\epsilon} \epsilon [r1]^2 \epsilon [r2]^2 \epsilon [r3]^2 - 4 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [r2]^2 \epsilon [r3]^{\epsilon} - \eta [3]^{\epsilon} \epsilon [c1]^2 \epsilon [r2]^2 \epsilon [r3]^2 - 4 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [c2]^2 \epsilon [r3]^2 - 4 \eta^3 \epsilon [r1]^{\epsilon} \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - 4 \eta^3 \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - 4 \eta^3 \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^2 - 4 \eta^3 \epsilon [r1]^2 \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - \eta [3]^{\epsilon} \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [r3]^{\epsilon} - \eta [3]^{\epsilon} \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [r3]^2 - \eta [3]^{\epsilon} \epsilon [c1]^2 \epsilon [c2]^2 \epsilon [c3]^2 C 2 \eta^2 \epsilon [r1]^2 \epsilon [r2]^2 \epsilon [r3]^2 C 2 \eta^2 \epsilon [c1]^2 \epsilon [r2]^2 \epsilon [r3]^2 C 32 \eta^2 \epsilon [r1]^{\epsilon} \epsilon [r2]^2 \epsilon [r3]^{\epsilon} - 4 \eta^2 \epsilon [r1]^2 \epsilon [r2]^{\epsilon} \epsilon [r3]^2 - 16 \eta^2 \epsilon [r1]^{\epsilon} \epsilon [r2]^{\epsilon} \epsilon [r3]^2 C 8 \epsilon [c3]^2 \eta [3]^2 \epsilon [r1]^2 \epsilon [r2]^{\epsilon} C 2 \epsilon [c2]^2 \eta^2 \epsilon [c1]^2 \epsilon [c3]^2 C 8 \epsilon [c3]^2 \eta [3]^2 \epsilon [r1]^{\epsilon} \epsilon [c2]^2 - 16 \eta^2 \epsilon [r1]^{\epsilon} \epsilon [r2]^{\epsilon} \epsilon [c3]^2 - 4 \epsilon [c1]^2 \epsilon [c2]^2 - 16 \epsilon [r1]^{\epsilon}$

$$\begin{aligned}
& \epsilon[r_3]^2 - 64\epsilon[r_1]\epsilon[r_3] - 16\epsilon[r_1]\epsilon[c_3]^2 \\
& - 16\epsilon[r_2]\epsilon[r_3]^2 - 64\epsilon[r_2]\epsilon[r_3] - 16\epsilon[r_2]\epsilon[c_3]^2 - 4\epsilon[r_1]^2\epsilon[r_3]^2 - 16\epsilon[r_1]^2\epsilon[r_3] \\
& - 4\epsilon[r_1]^2\epsilon[c_3]^2 - 4\epsilon[r_2]^2\epsilon[r_3]^2 - 16\epsilon[r_2]^2\epsilon[r_3] - 4\epsilon[r_2]^2\epsilon[c_3]^2 - 4\epsilon[c_1]^2\epsilon[r_3]^2 - 16\epsilon[c_1]^2\epsilon[r_3] \\
& - 4\epsilon[c_1]^2\epsilon[c_3]^2 - 4\epsilon[c_2]^2\epsilon[r_3]^2 - 16\epsilon[c_2]^2\epsilon[r_3] - 4\epsilon[c_2]^2\epsilon[c_3]^2 - 16\epsilon[\eta_1]\epsilon[r_1]C_{32}\epsilon[\eta_1]\epsilon[r_2]C_{32}\epsilon[\eta_1]^* \\
& \epsilon[r_3] - 16\epsilon[\eta_1]\epsilon[r_1]^2C_{8}\epsilon[\eta_1]\epsilon[r_2]^2C_{8}\epsilon[\eta_1]^* \\
& \epsilon[r_3]^2 - 16\epsilon[\eta_1]\epsilon[c_1]^2C_{8}\epsilon[\eta_1]\epsilon[c_2]^2C_{8}\epsilon[\eta_1]^* \\
& \epsilon[c_3]^2C_{32}\epsilon[\eta_2]\epsilon[r_1] - 16\epsilon[\eta_2]\epsilon[r_2]C_{32}\epsilon[\eta_2]^* \\
& \epsilon[r_3]C_{8}\epsilon[\eta_2]\epsilon[r_1]^2 - 16\epsilon[\eta_2]\epsilon[r_2]^2C_{8}\epsilon[\eta_2]^* \\
& \epsilon[r_3]^2C_{8}\epsilon[\eta_2]\epsilon[c_1]^2 - 16\epsilon[\eta_2]\epsilon[c_2]^2C_{8}\epsilon[\eta_2]^* \\
& \epsilon[c_3]^2C_{32}\epsilon[\eta_3]\epsilon[r_1]C_{32}\epsilon[\eta_3]\epsilon[r_2] - 16\epsilon[\eta_3]^* \\
& \epsilon[r_3]C_{8}\epsilon[\eta_3]\epsilon[r_1]^2C_{8}\epsilon[\eta_3]\epsilon[r_2]^2 - 16\epsilon[\eta_3]^* \\
& \epsilon[r_3]^2C_{8}\epsilon[\eta_3]\epsilon[c_1]^2C_{8}\epsilon[\eta_3]\epsilon[c_2]^2 - 16\epsilon[\eta_3]^* \\
& \epsilon[c_3]^2C_{32}\epsilon[\eta_1]^2\epsilon[r_1]^2C_{32}\epsilon[\eta_2]^2\epsilon[r_2]^2C_{32} \\
& \epsilon[c_2]^2\epsilon[\eta_2]^2C_{64}\epsilon[\eta_2]\epsilon[\eta_1]C_{64}\epsilon[\eta_3]\epsilon[\eta_1]C_{64}\epsilon[\eta_2]\epsilon[\eta_3]C_{32}\epsilon[\eta_3]^2\epsilon[r_3]^2 - 64\epsilon[\eta_2]^2\epsilon[r_1]\epsilon[r_2]^* \\
& \epsilon[r_3]C_{8}\epsilon[\eta_2]^2\epsilon[r_1]^2\epsilon[r_2]^2\epsilon[r_3]C_{8}\epsilon[\eta_2]^2\epsilon[r_2]^2\epsilon[r_1]\epsilon[c_3]^2C_{2}\epsilon[c_3]^2\epsilon[\eta_3]^2 \\
& \epsilon[c_1]^2\epsilon[c_2]^2C_{2}\epsilon[c_1]^2\epsilon[\eta_1]^2\epsilon[c_2]^2 \\
& \epsilon[r_3]^2C_{2}\epsilon[\eta_3]^2\epsilon[c_1]^2\epsilon[r_2]^2\epsilon[r_3]^2C_{2} \\
& \epsilon[c_3]^2\epsilon[\eta_3]^2\epsilon[r_1]^2\epsilon[c_2]^2C_{8}\epsilon[c_2]^2\epsilon[\eta_2]^2\epsilon[r_1]\epsilon[r_3]^2C_{8}\epsilon[c_1]^2\epsilon[\eta_1]^2\epsilon[r_2]^* \\
& \epsilon[c_3]^2C_{2}\epsilon[c_1]^2\epsilon[\eta_1]^2\epsilon[r_2]^2\epsilon[c_3]^2C_{8} \\
& \epsilon[\eta_2]^2\epsilon[c_1]^2\epsilon[r_2]^2\epsilon[r_3]C_{2}\epsilon[\eta_2]^2\epsilon[c_1]^2\epsilon[r_2]^2\epsilon[c_3]^2C_{8}\epsilon[\eta_3]^2\epsilon[r_1]\epsilon[c_2]^2 \\
& \epsilon[r_3]^2C_{8}\epsilon[c_2]^2\epsilon[\eta_2]^2\epsilon[c_1]^2\epsilon[r_3]C_{16} \\
& \epsilon[\eta_1]\epsilon[r_1]\epsilon[r_2]^2\epsilon[\eta_2]C_{16}\epsilon[\eta_1]\epsilon[r_1]\epsilon[c_2]^2\epsilon[\eta_2]C_{8}\epsilon[\eta_3]^2\epsilon[r_3]^2\epsilon[r_1]^2\epsilon[r_2]C_{2}\epsilon[\eta_3]^2\epsilon[r_1]^2\epsilon[c_2]^2\epsilon[r_3]^2C_{2}\epsilon[\eta_3]^2\epsilon[r_2]^2 \\
& \epsilon[r_3]^2\epsilon[r_1]^2 - 8\epsilon[\eta_3]\epsilon[r_1]\epsilon[c_2]^2\epsilon[\eta_1]C_{8} \\
& \epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[r_2]^2\epsilon[r_3]C_{2}\epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[r_2]^2\epsilon[r_3]^2C_{2}\epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[r_2]^2 \\
& \epsilon[c_3]^2C_{32}\epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[r_2]\epsilon[r_3]C_{8}\epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[r_2]\epsilon[r_3]^2C_{2}\epsilon[\eta_1]^2\epsilon[r_1]^2 \\
& \epsilon[c_2]^2\epsilon[c_3]^2C_{8}\epsilon[\eta_1]^2\epsilon[r_1]^2\epsilon[c_2]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_3] C^2 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C^{16} \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] \eta[3] C^4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[1] C^{16} \eta[2] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C^4 \eta[2] \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 C^4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \eta[3] C^4 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C^{16} \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C^4 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C^4 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C^4 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[3] C^4 \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C^4 \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C^{16} \eta[3] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] \eta[1] C^4 \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C^4 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C^4 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C^{16} \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[3] C^4 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[2] C^4 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 \eta[3] C^{16} \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C^{16} \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C^{16} \eta[1] \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[3] C^4 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 \eta[2] C^4 \eta[2] \epsilon[r_3]^2 \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 - 16 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 - 64 \eta[1]^2 \epsilon[r_1] \epsilon[r_2] - 16 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] - 16 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 - 16 \eta[1]^2 \epsilon[r_1] \epsilon[r_3]^2 - 64 \eta[1]^2 \epsilon[r_1] \epsilon[r_3] - 16 \eta[1]^2 \epsilon[r_1] \epsilon[c_3]^2 C^{16} \eta[2] \epsilon[r_3]^2 \eta[1] C^{32} \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] C^{16} \eta[3] \epsilon[r_2]^2 \eta[1] - 32 \eta[2] \epsilon[r_1]^2 \eta[1] C^8 \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_1]^2 - 32 \eta[3] \epsilon[r_1]^2 \eta[1] C^8 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_3]^2 - 4 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 4 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 4 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 4 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 16 \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 - \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 - 64 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 16 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 16 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 4 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 4 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 4 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 16 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 4 \epsilon[r_1]
\end{aligned}$$

$$\begin{aligned}
&^2*\epsilon[r2]^2*\epsilon[r3]-4*\epsilon[r1]*\epsilon[c2]^2*\epsilon[c3]^2- \\
&\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[c3]^2-4*\epsilon[c1]^2*\epsilon[r2]* \\
&\epsilon[r3]^2-16*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]-4*\epsilon[c1]^2* \\
&\epsilon[r2]*\epsilon[c3]^2-\epsilon[c1]^2*\epsilon[c2]^2*\epsilon[r3]^2-4* \\
&\epsilon[c1]^2*\epsilon[c2]^2*\epsilon[r3]-\epsilon[c1]^2*\epsilon[c2]^2* \\
&\epsilon[c3]^2-4*\eta[1]*\epsilon[r1]^2*\epsilon[r2]^2-16*\eta[1]*\epsilon[r1] \\
&^2*\epsilon[r2]-4*\eta[1]*\epsilon[r1]^2*\epsilon[c2]^2-4*\eta[1]*\epsilon[r1]* \\
&\epsilon[r2]^2-16*\eta[1]*\epsilon[r1]*\epsilon[r2]-4*\eta[1]*\epsilon[r1]* \\
&\epsilon[c2]^2-4*\eta[1]*\epsilon[c1]^2*\epsilon[r2]^2-16*\eta[1]*\epsilon[c1] \\
&^2*\epsilon[r2]-4*\eta[1]*\epsilon[c1]^2*\epsilon[c2]^2-4*\eta[1]*\epsilon[r1]* \\
&\epsilon[r3]^2C 4*\eta[2]*\epsilon[r3]^2*\eta[1]*\epsilon[c1]^2*\epsilon[r2] \\
&^2C 4*\eta[2]*\epsilon[r3]^2*\eta[1]*\epsilon[c1]^2*\epsilon[c2]^2C 4*\eta[2]* \\
&\epsilon[c3]^2*\eta[1]*\epsilon[r1]^2*\epsilon[c2]^2C 4*\eta[2]*\epsilon[c3]^2* \\
&\eta[1]*\epsilon[c1]^2*\epsilon[r2]^2C 4*\eta[2]*\epsilon[c3]^2*\eta[1]*\epsilon[c1] \\
&^2*\epsilon[c2]^2C 4*\eta[1]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3]^2* \\
&\eta[2]C 4*\eta[1]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3]^2*\eta[3]C 16*\eta[2]* \\
&\epsilon[r3]*\eta[1]*\epsilon[r1]^2*\epsilon[c2]^2C 16*\eta[2]*\epsilon[r3]*\eta[1] \\
&*\epsilon[c1]^2*\epsilon[r2]^2C 16*\eta[2]*\epsilon[r3]*\eta[1]*\epsilon[c1] \\
&^2*\epsilon[c2]^2C 4*\eta[1]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[c3]^2*\eta[2] \\
&C 4*\eta[1]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[c3]^2*\eta[3]C 4*\eta[3]* \\
&\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[r3]*\eta[1]C 16*\eta[3]*\epsilon[r1]^2* \\
&\epsilon[r2]*\epsilon[r3]*\eta[1]C 4*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3] \\
&^2*\eta[1]C 4*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]^2*\eta[1] \\
&C 16*\eta[2]*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]*\eta[1]C 4*\eta[2]*\epsilon[r1] \\
&*\epsilon[r2]*\epsilon[r3]^2*\eta[1]C 4*\eta[2]*\epsilon[r1]^2*\epsilon[r2]* \\
&\epsilon[r3]^2*\eta[1]C 4*\eta[2]*\epsilon[r1]*\epsilon[r2]*\epsilon[c3]^2*\eta[1] \\
&C 16*\eta[2]*\epsilon[r1]*\epsilon[r2]*\epsilon[r3]*\eta[1]C 4*\eta[2]*\epsilon[r1] \\
&^2*\epsilon[r2]*\epsilon[c3]^2*\eta[1]C 16*\eta[2]*\epsilon[r1]^2*\epsilon[r2]* \\
&\epsilon[r3]*\eta[1]C 16*\eta[3]*\eta[2]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3] \\
&^2C 4*\eta[3]*\eta[2]*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[c3]^2-8*\eta[2]* \\
&\epsilon[r2]*\epsilon[c3]^2*\eta[1]-64*\eta[1]^2*\epsilon[r1]C 32*\eta[1]^2* \\
&\epsilon[r2]C 32*\eta[1]^2*\epsilon[r3]C 8*\eta[1]^2*\epsilon[r2]^2C 8*\eta[1] \\
&^2*\epsilon[r3]^2C 8*\eta[1]^2*\epsilon[c2]^2C 8*\eta[1]^2*\epsilon[c3]^2C 8* \\
&\eta[2]^2*\epsilon[r3]^2*\epsilon[r2]^2-16*\eta[1]*\epsilon[r1]*\epsilon[r3]-4* \\
&\eta[1]*\epsilon[r1]*\epsilon[c3]^2C 8*\eta[1]*\epsilon[r2]*\epsilon[r3]^2C 32* \\
&\eta[1]*\epsilon[r2]*\epsilon[r3]C 8*\eta[1]*\epsilon[r2]*\epsilon[c3]^2-4*\eta[1] \\
&*\epsilon[r1]^2*\epsilon[r3]^2-16*\eta[1]*\epsilon[r1]^2*\epsilon[r3]-4*\eta[1] \\
&*\epsilon[r1]^2*\epsilon[c3]^2C 2*\eta[1]*\epsilon[r2]^2*\epsilon[r3]^2C 8*\eta[1]
\end{aligned}$$

$$\begin{aligned} & [1]^* \epsilon[r_2]^2 \epsilon[r_3] C^2 \epsilon[1]^* \epsilon[r_2]^2 \epsilon[c_3]^2 - 4^* \epsilon[1]^* \epsilon[c_1]^2 \epsilon[r_3]^2 - 16^* \epsilon[1]^* \epsilon[c_1]^2 \epsilon[r_3] - 4^* \epsilon[1]^* \epsilon[c_1]^2 \epsilon[c_3]^2 C^2 \epsilon[1]^* \epsilon[c_2]^2 \epsilon[r_3]^2 C^8 \epsilon[1]^* \epsilon[c_2]^2 \epsilon[r_3] C^2 \epsilon[1]^* \epsilon[c_2]^2 \epsilon[c_3]^2 - 4^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[r_2]^2 - 4^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[r_2] - 4^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[c_2]^2 - 16^* \epsilon[2]^* \epsilon[r_1]^* \epsilon[r_2]^2 - 16^* \epsilon[2]^* \epsilon[r_1]^* \epsilon[r_2] - 16^* \epsilon[2]^* \epsilon[r_1]^* \epsilon[c_2]^2 - 4^* \epsilon[2]^* \epsilon[c_1]^2 \epsilon[r_2]^2 - 4^* \epsilon[2]^* \epsilon[c_1]^2 \epsilon[r_2] - 4^* \epsilon[2]^* \epsilon[c_1]^2 \epsilon[c_2]^2 C^8 \epsilon[2]^* \epsilon[r_1]^* \epsilon[r_3] \wedge C^2 C^2 \epsilon[2]^* \epsilon[r_1]^* \epsilon[r_3] C^8 \epsilon[2]^* \epsilon[r_1]^* \epsilon[c_3]^2 - 4^* \epsilon[2]^* \epsilon[r_2]^* \epsilon[r_3]^2 - 16^* \epsilon[2]^* \epsilon[r_2]^* \epsilon[r_3] - 4^* \epsilon[2]^* \epsilon[r_2]^* \epsilon[c_3]^2 C^2 \epsilon[2]^* \epsilon[r_1]^2 \epsilon[r_3]^2 C^8 \epsilon[2]^* \epsilon[r_1]^2 \epsilon[r_3] C^2 \epsilon[2]^* \epsilon[r_1]^2 \epsilon[c_3]^2 - 4^* \epsilon[2]^* \epsilon[r_2]^2 \epsilon[r_3]^2 - 16^* \epsilon[2]^* \epsilon[r_2]^2 \epsilon[r_3] - 4^* \epsilon[2]^* \epsilon[r_2]^2 \epsilon[c_3]^2 C^2 \epsilon[2]^* \epsilon[c_1]^2 \epsilon[r_3] \wedge C^2 C^8 \epsilon[2]^* \epsilon[c_1]^2 \epsilon[r_3] C^2 \epsilon[2]^* \epsilon[c_1]^2 \epsilon[c_3]^2 - 4^* \epsilon[2]^* \epsilon[c_2]^2 \epsilon[r_3]^2 - 16^* \epsilon[2]^* \epsilon[c_2]^2 \epsilon[r_3] - 4^* \epsilon[2]^* \epsilon[c_2]^2 \epsilon[c_3]^2 C^2 \epsilon[3]^* \epsilon[r_1]^2 \epsilon[r_2]^2 C^8 \epsilon[3]^* \epsilon[r_1]^2 \epsilon[r_2] C^2 \epsilon[3]^* \epsilon[r_1]^2 \epsilon[c_2]^2 C^8 \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_2]^2 C^2 \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_2] C^8 \epsilon[3]^* \epsilon[r_1]^* \epsilon[c_2]^2 C^2 \epsilon[3]^* \epsilon[r_1]^* \epsilon[c_1]^2 \epsilon[r_2]^2 C^8 \epsilon[3]^* \epsilon[r_1]^* \epsilon[c_1]^2 \epsilon[r_2] C^2 \epsilon[3]^* \epsilon[r_1]^* \epsilon[c_1]^2 \epsilon[c_2]^2 - 16^* \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_3]^2 - 16^* \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_3] - 16^* \epsilon[3]^* \epsilon[r_1]^* \epsilon[c_3]^2 - 16^* \epsilon[3]^* \epsilon[r_2]^* \epsilon[r_3]^2 - 16^* \epsilon[3]^* \epsilon[r_2]^* \epsilon[r_3] - 16^* \epsilon[3]^* \epsilon[r_2]^* \epsilon[c_3]^2 - 4^* \epsilon[3]^* \epsilon[r_1]^2 \epsilon[r_3]^2 - 4^* \epsilon[3]^* \epsilon[r_1]^2 \epsilon[c_3]^2 - 4^* \epsilon[3]^* \epsilon[r_2]^2 \epsilon[r_3]^2 - 4^* \epsilon[3]^* \epsilon[r_2]^2 \epsilon[r_3] - 4^* \epsilon[3]^* \epsilon[r_2]^2 \epsilon[c_3]^2 - 4^* \epsilon[3]^* \epsilon[c_1]^2 \epsilon[r_3]^2 - 4^* \epsilon[3]^* \epsilon[c_1]^2 \epsilon[r_3] \wedge C^2 \epsilon[3]^* \epsilon[c_1]^2 \epsilon[c_3]^2 - 4^* \epsilon[3]^* \epsilon[c_2]^2 \epsilon[r_3]^2 - 4^* \epsilon[3]^* \epsilon[c_2]^2 \epsilon[r_3] \wedge C^2 C^8 \epsilon[3]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \wedge C^2 C^8 \epsilon[3]^* \epsilon[c_1]^2 \epsilon[r_3]^2 C^2 \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_3]^2 - 32^* \epsilon[3]^* \epsilon[r_1]^* \epsilon[r_3]^* \epsilon[1] C^16 \epsilon[3]^* \epsilon[2]^* \epsilon[r_1]^* \epsilon[r_2]^2 \epsilon[r_3]^2 C^4 \epsilon[3]^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C^4 \epsilon[3]^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C^4 \epsilon[3]^* \epsilon[2]^* \epsilon[r_1]^2 \epsilon[c_1]^2 \epsilon[r_3]^2 \end{aligned}$$

$$\begin{aligned}
& \epsilon[r_2]^2 \epsilon[r_3]^{2C_{16}} \eta[3] \epsilon[r_2] \eta[1] \epsilon[c_1]^2 \epsilon[c_3]^{2C_4} \eta[3] \epsilon[r_2]^2 \eta[1] \epsilon[c_1]^2 \epsilon[c_3]^{2C_4} \eta[3] \epsilon[c_1]^2 \eta[1] \epsilon[c_2]^2 \epsilon[r_3]^{2C_4} \eta[3] \epsilon[c_1]^2 \eta[1] \epsilon[c_2]^2 \epsilon[c_3]^{2C_{16}} \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[3] C_4 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[3] C_4 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \eta[3] C_{16} \eta[3] \epsilon[r_1]^2 \eta[1] \epsilon[r_2] \epsilon[r_3]^{2C_{16}} \eta[3] \epsilon[c_1]^2 \eta[1] \epsilon[r_2] \epsilon[r_3]^{2C_{16}} \eta[3] \eta[2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^{2C_4} \eta[3] \eta[2] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^{2C_4} \eta[3] \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^{2C_4} \eta[3] \eta[2] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^{2C_{16}} \eta[3] \epsilon[r_2]^2 \eta[2] \epsilon[r_1] \epsilon[c_3]^{2C_4} \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \eta[1] \epsilon[r_3]^{2C_4} \eta[3] \epsilon[r_1]^2 \epsilon[r_2]^2 \eta[1] \epsilon[c_3]^{2C_{16}} \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \eta[1] \epsilon[c_3]^{2C_4} \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \eta[1] \epsilon[c_3]^{2C_4} \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \eta[1] \epsilon[r_3]^{2C_{16}} \eta[3] \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta[2] C_{16} \eta[3] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C_4 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] \eta[2] - 32 \eta[3] \epsilon[r_1] \eta[1] - 32 \eta[2] \epsilon[c_1]^2 \eta[1] C_{16} \eta[2] \epsilon[c_3]^2 \eta[1] - 32 \eta[3] \epsilon[c_1]^2 \eta[1] C_{16} \eta[3] \epsilon[c_2]^2 \eta[1] - 32 \eta[2] \epsilon[r_1] \eta[1] C_8 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1]^{2C_8} \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 C_{32} \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_1] C_8 \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_3]^{2C_2} \eta[1]^2 \epsilon[c_2]^2 \epsilon[c_3]^{2C_2} \eta[1]^2 \epsilon[c_2]^2 \epsilon[r_3]^{2C_8} \eta[1]^2 \epsilon[c_2]^2 \epsilon[r_3] C_2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[c_3]^{2C_8} \eta[1]^2 \epsilon[r_2]^2 \epsilon[r_3] C_2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[r_3]^{2C_{32}} \eta[1]^2 \epsilon[r_2] \epsilon[r_3] C_8 \eta[1]^2 \epsilon[r_2] \epsilon[c_3]^{2C_8} \eta[1]^2 \epsilon[r_2] \epsilon[r_3]^{2C_{32}} \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_3] C_{32} \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_3] - 32 \eta[2] \epsilon[r_2] \eta[3] - 32 \eta[2] \epsilon[r_3] \eta[3] C_{16} \eta[2] \epsilon[r_1]^2 \eta[3] - 32 \eta[2] \epsilon[r_3]^2 \eta[3] C_{32} \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2] C_8 \eta[1]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 C_8 \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_2]^2 \epsilon[r_1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 - 32 \eta[1] \epsilon[r_2]^2 \eta[2] - 32 \eta[1] \epsilon[r_3]^2 \eta[3] - 32 \eta[1] \epsilon[c_2]^2 \eta[2] - 32 \eta[1] \epsilon[c_3]^2 \eta[3] C_{32} \epsilon[c_3]^2 \eta[3]^2 \epsilon[r_2] C_{64} \eta[2] \epsilon[r_3] \eta[1] C_8 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2]^{2C_8} \eta[3]^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_1]^2 \epsilon[r_2] C^2 \eta[3]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 C^8 \eta[3]^2 \epsilon[r_1] \epsilon[r_2]^2 - 8 \eta[2] \epsilon[r_1] \epsilon[c_3]^2 \eta[1] \\
& - 8 \eta[2] \epsilon[r_2] \epsilon[r_3]^2 \eta[1] - 32 \eta[2] \epsilon[r_2] \epsilon[r_3] \eta[1] - 64 \eta[3]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 16 \eta[3]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \\
& - 16 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_3] \eta[3] - 8 \eta[2] \epsilon[c_1]^2 \epsilon[c_3]^2 \eta[3] C^8 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[r_3] \\
& - 8 \eta[2] \epsilon[r_1]^2 \epsilon[r_3]^2 \eta[3] - 8 \eta[2] \epsilon[r_1]^2 \epsilon[c_3]^2 \eta[3] - 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_3]^2 \eta[3] C^{16} \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \eta[1] C^{16} \eta[2] \epsilon[r_2] \epsilon[r_3]^2 \eta[3] \\
& C^{16} \eta[2] \epsilon[r_2] \epsilon[r_3] \eta[3] C^{16} \eta[2] \epsilon[r_2] \epsilon[r_3] \eta[3] C^{16} \eta[2] \epsilon[r_2] \epsilon[c_3]^2 \eta[3] - 16 \eta[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 4 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[c_3]^2 C^{32} \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2] \epsilon[r_3] \\
& - 4 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 64 \eta[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 16 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] - 16 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] - 4 \eta[1]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 4 \eta[1]^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 16 \eta[1]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \\
& - 8 \eta[3] \epsilon[r_2]^2 \epsilon[r_3] \eta[1] C^{16} \eta[3] \epsilon[c_1]^2 \epsilon[r_3] \eta[1] - 8 \eta[3] \epsilon[c_2]^2 \epsilon[r_3] \eta[1] - 32 \eta[3] \epsilon[c_1]^2 \epsilon[r_2] \eta[1] - 8 \eta[3] \epsilon[c_1]^2 \epsilon[c_2]^2 \eta[1] C^{16} \eta[3] \epsilon[r_1] \epsilon[r_3] \eta[1] - 32 \eta[3] \epsilon[r_2] \epsilon[r_3] \eta[1] C^{16} \eta[3] \epsilon[r_1]^2 \epsilon[r_3] \eta[1] - 8 \eta[3] \epsilon[c_1]^2 \epsilon[r_2]^2 \eta[1] C^2 \epsilon[c_2]^2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3]^2 - 4 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 16 \eta[2]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] C^8 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 32 \eta[1] \epsilon[r_2] \epsilon[r_3]^2 \eta[3] - 32 \eta[1] \epsilon[r_2] \epsilon[c_3]^2 \eta[3] C^{16} \eta[3] \epsilon[c_2]^2 \epsilon[r_3] \eta[2] C^{16} \eta[1] \epsilon[r_1] \epsilon[r_3]^2 \eta[3] C^{16} \eta[1] \epsilon[r_1] \epsilon[c_3]^2 \eta[3] C^8 \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2] C^{16} \eta[2] \eta[1] \epsilon[r_1]^2 \epsilon[c_2]^2 C^{16} \eta[2] \eta[1] \epsilon[r_1]^2 \epsilon[r_2]^2 C^2 \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 C^8 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \epsilon[c_3]^2 - 4 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 16 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 4 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 C^8 \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 \epsilon[r_3] C^2 \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2
\end{aligned}$$

$$\begin{aligned}
&^2*\epsilon[c3]^2-8*\eta[2]*\epsilon[r1]^2*\epsilon[r2]*\eta[3]-32*\eta[2]* \\
&\epsilon[r1]*\epsilon[r2]*\eta[3]-\eta[3]*\epsilon[r1]^2*\epsilon[c2]^2*\epsilon \\
&[c3]^2-16*\eta[1]*\epsilon[r1]^2*\epsilon[r2]*\epsilon[r3]C\ 32*\eta[3]^2* \\
&\epsilon[r1]*\epsilon[r2]C\ 8*\eta[3]^2*\epsilon[r1]*\epsilon[c2]^2C\ 2*\eta[3]^2* \\
&\epsilon[c1]^2*\epsilon[r2]^2C\ 8*\eta[3]^2*\epsilon[c1]^2*\epsilon[r2]C\ 2*\eta \\
&[3]^2*\epsilon[c1]^2*\epsilon[c2]^2-64*\eta[3]^2*\epsilon[r1]*\epsilon[r3] \\
&-64*\eta[3]^2*\epsilon[r2]*\epsilon[r3]C\ 2*\eta[2]^2*\epsilon[r1]^2*\epsilon \\
&[r3]^2C\ 8*\eta[2]^2*\epsilon[r1]^2*\epsilon[r3]C\ 2*\eta[2]^2*\epsilon[r1]^2* \\
&\epsilon[c3]^2C\ 2*\eta[2]^2*\epsilon[c1]^2*\epsilon[r3]^2C\ 8*\eta[2]^2* \\
&\epsilon[c1]^2*\epsilon[r3]C\ 2*\eta[2]^2*\epsilon[c1]^2*\epsilon[c3]^2C\ 2*\eta \\
&[3]^2*\epsilon[r1]^2*\epsilon[r2]^2-16*\eta[2]^2*\epsilon[r1]^2*\epsilon[r2] \\
&-64*\eta[2]^2*\epsilon[r1]*\epsilon[r2]-16*\eta[2]^2*\epsilon[c1]^2*\epsilon \\
&[r2]C\ 8*\eta[2]^2*\epsilon[r1]*\epsilon[r3]^2C\ 32*\eta[2]^2*\epsilon[r1]* \\
&\epsilon[r3]C\ 8*\eta[2]^2*\epsilon[r1]*\epsilon[c3]^2-16*\eta[2]^2*\epsilon[r2] \\
&*\epsilon[r3]^2-64*\eta[2]^2*\epsilon[r2]*\epsilon[r3]-16*\eta[2]^2*\epsilon \\
&[r2]*\epsilon[c3]^2C\ 32*\epsilon[c1]^2*\eta[1]^2*\epsilon[r2]-32*\eta[3]* \\
&\epsilon[r2]^2*\eta[2]-32*\eta[3]*\epsilon[c2]^2*\eta[2]C\ 8*\eta[1]^2*\epsilon \\
&[r1]^2*\epsilon[r3]^2C\ 8*\epsilon[c3]^2*\eta[3]^2*\epsilon[r1]^2C\ 8*\eta[1]^2* \\
&\epsilon[r1]^2*\epsilon[c3]^2C\ 32*\eta[1]^2*\epsilon[r1]^2*\epsilon[r3]C\ 32* \\
&\eta[2]^2*\epsilon[r1]*\epsilon[r2]^2C\ 8*\epsilon[c1]^2*\eta[1]^2*\epsilon[r2] \\
&^2C\ 32*\eta[2]^2*\epsilon[r3]*\epsilon[r2]^2C\ 64*\eta[2]*\epsilon[r1]*\eta[3] \\
&-16*\eta[3]^2*\epsilon[c2]^2*\epsilon[r3]C\ 16*\eta[2]*\epsilon[c1]^2*\eta[3] \\
&-32*\eta[2]*\epsilon[c3]^2*\eta[3]C\ 64*\eta[3]*\epsilon[r2]*\eta[1]-\eta[1]* \\
&\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3]^2-4*\eta[1]*\epsilon[r1]^2*\epsilon \\
&[r2]*\epsilon[c3]^2-4*\eta[1]*\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[r3]-\eta[1] \\
&*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[c3]^2-\eta[1]*\epsilon[r1]*\epsilon[r2] \\
&^2*\epsilon[r3]^2-\eta[1]*\epsilon[r1]^2*\epsilon[c2]^2*\epsilon[c3]^2-4*\eta \\
&[1]*\epsilon[r1]*\epsilon[r2]*\epsilon[c3]^2-\eta[1]*\epsilon[r1]^2*\epsilon[c2] \\
&^2*\epsilon[r3]^2-16*\eta[1]*\epsilon[r1]*\epsilon[r2]*\epsilon[r3]-\eta[1]* \\
&\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[r3]^2-4*\eta[1]*\epsilon[r1]*\epsilon \\
&[r2]^2*\epsilon[r3]-\eta[1]*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]^2-4*\eta \\
&[1]*\epsilon[r1]*\epsilon[c2]^2*\epsilon[r3]-4*\eta[1]*\epsilon[r1]^2*\epsilon \\
&[r2]*\epsilon[r3]^2-\eta[1]*\epsilon[r1]*\epsilon[r2]^2*\epsilon[c3]^2-4*\eta[1] \\
&*\epsilon[c1]^2*\epsilon[r2]^2*\epsilon[r3]-4*\eta[1]*\epsilon[r1]*\epsilon[r2]* \\
&\epsilon[r3]^2-4*\eta[1]*\epsilon[r1]^2*\epsilon[r2]^2*\epsilon[r3]-\eta[1]* \\
&\epsilon[r1]*\epsilon[c2]^2*\epsilon[c3]^2-\eta[1]*\epsilon[c1]^2*\epsilon[r2] \\
&^2*\epsilon[c3]^2-4*\eta[1]*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]^2-16*\eta \\
&[1]*\epsilon[c1]^2*\epsilon[r2]*\epsilon[r3]-4*\eta[1]*\epsilon[c1]^2*\epsilon
\end{aligned}$$

$$\begin{aligned}
& [r2]^2 \epsilon[c3]^2 - \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 4 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3] - \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 4 \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] - 4 \eta[2] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3]^2 - \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[r3]^2 - 4 \eta[2] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] - 16 \eta[3]^2 \epsilon[r1]^2 \epsilon[r3]) / (64 C^4 \epsilon[c2]^2 \eta[2]^2 \epsilon[c3]^2 C^6 \eta[1] C^6 \eta[2] C^6 \eta[3] C^6 \epsilon[r1] C^6 \epsilon[r2] C^6 \epsilon[r3] C^6 \epsilon[c1]^2 C^6 \epsilon[c2]^2 C^6 \epsilon[c3]^2 \\
& - 8 \eta[3]^2 \epsilon[c1]^2 \epsilon[r3] - 8 \eta[3]^2 \epsilon[r2]^2 \epsilon[r3] C^6 \epsilon[c1] \eta[1] \epsilon[c3] \eta[3] \epsilon[r2]^2 C^4 \epsilon[c2]^2 \eta[2]^2 \epsilon[r3]^2 C^4 \epsilon[c3]^2 \eta[3]^2 \epsilon[r2]^2 C^4 \eta[2]^2 \epsilon[c1]^2 \epsilon[r2]^2 - 16 \eta[2] \epsilon[r2] \eta[1] C^8 \eta[3] \epsilon[r1]^2 \eta[1] \epsilon[r3]^2 - 16 \eta[3] \epsilon[r1] \epsilon[r2]^2 \eta[2] - 16 \eta[3] \epsilon[r1] \epsilon[c2]^2 \eta[2] - 4 \eta[3] \epsilon[c1]^2 \epsilon[r2]^2 \eta[2] - 8 \eta[3]^2 \epsilon[c1]^2 \epsilon[r2] \epsilon[r3] - 2 \eta[3]^2 \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3] - 4 \eta[1] \epsilon[r2]^2 \epsilon[r3]^2 \eta[2] - 4 \eta[1] \epsilon[r2]^2 \epsilon[r3]^2 \eta[3] - 16 \eta[1] \epsilon[r2]^2 \epsilon[r3] \eta[2] C^8 \eta[3] \epsilon[r2]^2 \epsilon[r3] \eta[2] C^4 \epsilon[c3]^2 \eta[3]^2 \epsilon[r1] \epsilon[r2]^2 C^6 \epsilon[c2]^2 \eta[2]^2 \epsilon[r1]^2 \epsilon[r3] C^4 \epsilon[c2]^2 \eta[2]^2 \epsilon[r1]^2 \epsilon[r3] C^6 \epsilon[c3]^2 \eta[3]^2 \epsilon[r1] \epsilon[r2] C^8 \eta[2] \eta[1] \epsilon[c1]^2 \epsilon[r2]^2 C^8 \eta[2] \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 - 4 \eta[2] \epsilon[c1]^2 \epsilon[r2] \eta[3] - 16 \eta[2] \epsilon[r1] \epsilon[r3]^2 \eta[3] - 16 \eta[2] \epsilon[r1] \epsilon[r3] \eta[3] - 16 \eta[2] \epsilon[r1] \epsilon[c3]^2 \eta[3] C^6 \epsilon[c2]^2 \eta[2]^2 \epsilon[r1]^2 \epsilon[c3]^2 C^8 \eta[2] \epsilon[c1]^2 \epsilon[r2] \eta[1] - 4 \eta[2] \epsilon[r1] \epsilon[r3]^2 \eta[1] - 16 \eta[2] \epsilon[r1] \epsilon[r3] \eta[1] C^4 \epsilon[c3]^2 \eta[3]^2 \epsilon[c1]^2 C^8 \eta[3] \epsilon[c2]^2 \eta[2] \epsilon[r3]^2 - 4 \eta[2] \epsilon[c1]^2 \epsilon[r3]^2 \eta[1] - 16 \eta[2] \epsilon[c1]^2 \epsilon[r3]^2 \eta[1] C^8 \eta[3] \epsilon[r2]^2 \eta[2] \epsilon[c3]^2 C^8 \eta[3] \epsilon[r2]^2 \eta[2] \epsilon[r3]^2 - 4 \eta[1] \epsilon[r2]^2 \epsilon[c3]^2 \eta[3] - 4 \eta[1] \epsilon[c2]^2 \epsilon[r3]^2 \eta[2] - 4 \eta[1] \epsilon[c2]^2 \epsilon[r3]^2 \eta[3] - 16 \eta[1] \epsilon[c2]^2 \epsilon[r3]^2 \eta[2] - 4 \eta[1] \epsilon[c2]^2 \epsilon[c3]^2 \eta[2] - 4 \eta[1] \epsilon[c2]^2 \epsilon[c3]^2 \eta[3] - 4 \eta[3] \epsilon[c1]^2 \epsilon[c2]^2 \eta[2] - 4 \eta[2] \epsilon[r1]^2 \epsilon[r3]^2 \eta[1] - 16 \eta[2] \epsilon[r1] \epsilon[r3]^2 \epsilon[c1]^2 - 4 \eta[2] \epsilon[r1]^2 \epsilon[r3]^2 \epsilon[c3]^2 \eta[1] C^6 \epsilon[c1] \epsilon[c3]
\end{aligned}$$

$$\begin{aligned}
& [c1]^2 \cdot \eta[1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3]^2 C \epsilon[c2]^2 \cdot \eta[2]^2 \cdot \\
& \epsilon[r1]^2 \cdot \epsilon[r3]^2 C \epsilon[c3]^2 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 \cdot \\
& \epsilon[r2]^2 C 8 \cdot \eta[3] \cdot \eta[1] \cdot \epsilon[c1]^2 \cdot \epsilon[r3]^2 C 8 \cdot \eta[3] \cdot \eta[1] \cdot \epsilon[c1]^2 \cdot \epsilon[c3]^2 C 16 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \epsilon[r2] \cdot \\
& \epsilon[r3]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3]^2 C \eta[3]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[c2]^2 \cdot \epsilon[r3]^2 - 4 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \\
& \epsilon[r2]^2 \cdot \eta[2] - 4 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \epsilon[c2]^2 \cdot \eta[2] - 2 \cdot \eta[3]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3] - 2 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 \cdot \\
& \epsilon[r2]^2 \cdot \epsilon[r3] - 8 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3] C 8 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \eta[1] \cdot \epsilon[c3]^2 C 4 \cdot \eta[2]^2 \cdot \epsilon[c3]^2 \cdot \\
& \epsilon[r2]^2 C 4 \cdot \eta[1]^2 \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \epsilon[c3]^2 - 16 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \eta[1] - 4 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \epsilon[c2]^2 \cdot \eta[1] - 4 \cdot \eta[3] \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \eta[1] - 16 \cdot \eta[3] \cdot \epsilon[r1] \cdot \\
& \epsilon[r2] \cdot \eta[1] C 72 \cdot \epsilon[c1] \cdot \eta[1] \cdot \epsilon[c3] \cdot \eta[3] C 72 \cdot \epsilon[c2] \cdot \eta[2] \cdot \epsilon[c3] \cdot \eta[3] - 4 \cdot \eta[3] \cdot \epsilon[r1]^2 \cdot \epsilon[r2]^2 \cdot \eta[1] C 72 \cdot \epsilon[c1] \cdot \eta[1] \cdot \epsilon[c2] \cdot \eta[2] - 2 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 \cdot \\
& \epsilon[c2]^2 \cdot \epsilon[r3] C 4 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3]^2 C 8 \cdot \eta[3] \cdot \epsilon[c2]^2 \cdot \eta[2] \cdot \epsilon[c3]^2 C 16 \cdot \epsilon[c1]^2 \cdot \eta[1]^2 C 16 \cdot \eta[3]^2 \cdot \epsilon[r1] C 16 \cdot \eta[3]^2 \cdot \epsilon[r2] - 32 \cdot \eta[3]^2 \cdot \\
& \epsilon[r3] C 4 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[r2]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[c1]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[c2]^2 C 16 \cdot \eta[2]^2 \cdot \epsilon[r1] - 32 \cdot \eta[2]^2 \cdot \epsilon[r2] C 16 \cdot \eta[2]^2 \cdot \epsilon[r3] C 4 \cdot \eta[2]^2 \cdot \epsilon[r1]^2 C 4 \cdot \eta[2]^2 \cdot \epsilon[c1]^2 C 4 \cdot \eta[2]^2 \cdot \\
& \epsilon[c3]^2 C 16 \cdot \epsilon[c3]^2 \cdot \eta[3]^2 C 4 \cdot \epsilon[r1]^2 \cdot \epsilon[r2]^2 C 16 \cdot \epsilon[r1]^2 \cdot \epsilon[r2] C 4 \cdot \epsilon[r1]^2 \cdot \epsilon[c2]^2 C 16 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 C 64 \cdot \epsilon[r1] \cdot \epsilon[r2] C 16 \cdot \epsilon[r1] \cdot \\
& \epsilon[c2]^2 C 4 \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 C 16 \cdot \epsilon[c1]^2 \cdot \epsilon[r2] C 16 \cdot \epsilon[r1]^2 C 16 \cdot \epsilon[r2]^2 C 16 \cdot \epsilon[r3]^2 C 16 \cdot \eta[1]^2 C 16 \cdot \eta[2]^2 C 16 \cdot \eta[3]^2 - 2 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r2]^2 \cdot \\
& \epsilon[c3]^2 - 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3]^2 - 2 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[c2]^2 \cdot \epsilon[c3]^2 - 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2] \cdot \epsilon[c3]^2 - 2 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[c2]^2 \cdot \epsilon[r3]^2 \\
& - 32 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2] \cdot \epsilon[r3] - 2 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3]^2 - 2 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3] - 2 \cdot \eta[2] \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3]^2 - 32 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[c2]^2 \cdot \epsilon[r3]^2 - 2 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3]^2 - 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[c3]^2 - 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3] - 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2] \cdot \epsilon[r3]
\end{aligned}$$

$$\begin{aligned}
& [r3]^2 - 8\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3] - 8\eta[2]\epsilon[r1]\epsilon[c2]^2\epsilon[c3]^2 - 2\eta[2]\epsilon[c1]^2\epsilon[r2]^2\epsilon[c3]^2 - 2\eta[2]\epsilon[c1]^2\epsilon[r2]\epsilon[r3]^2 - 8\eta[2]\epsilon[c1]^2\epsilon[r2]\epsilon[r3] - 2\eta[2]\epsilon[c1]^2\epsilon[r2]^2\epsilon[c3]^2 - 8\eta[2]\epsilon[c1]^2\epsilon[c2]^2\epsilon[r3]^2 - 8\eta[2]\epsilon[c1]^2\epsilon[c2]^2\epsilon[c3]^2 - 2\eta[2]\epsilon[c1]^2\epsilon[c2]\epsilon[r3]^2 - 8\eta[2]\epsilon[c1]^2\epsilon[c2]\epsilon[c3]^2 - 8\eta[3]\epsilon[r1]^2\epsilon[r2]\epsilon[r3] - 8\eta[3]\epsilon[r1]\epsilon[c2]^2\epsilon[r3]^2 - 8\eta[3]\epsilon[r1]^2\epsilon[r2]\epsilon[c3]^2 - 2\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[c3]^2 - 8\eta[3]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2 - 32\eta[3]\epsilon[r1]\epsilon[r2]\epsilon[c3]^2 \\
& C_{18}\epsilon[c1]\eta[1]\epsilon[c3]\eta[3]\epsilon[c2]^2 C_{72}\epsilon[c2]\eta[2]\epsilon[c3]\eta[3]\epsilon[r1] C_{18}\epsilon[c2]\eta[2]\epsilon[c3]\eta[3]\epsilon[r1]^2 C_{18}\epsilon[c2]\eta[2]\epsilon[c3]\eta[3]\epsilon[c1]^2 C_2\eta[2]\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2\eta[3] C_{72}\epsilon[c1]\eta[1]\epsilon[c2]\eta[2]\epsilon[r3] C_{18}\epsilon[c1]\eta[1]\epsilon[c2]\eta[2]\epsilon[r3]^2 C_{18}\epsilon[c1]\eta[1]\epsilon[c2]\eta[2]\epsilon[c3]^2 C_{72}\epsilon[c1]\eta[1]\epsilon[c3]\eta[3]\epsilon[r2] C_2\eta[2]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2\eta[1] C_8\eta[2]\epsilon[r1]\epsilon[r2]\epsilon[c3]^2\eta[3] - 2\eta[3]\epsilon[r1]^2\epsilon[c2]^2\epsilon[r3]^2 - 32\eta[3]\epsilon[r1]\epsilon[r2]\epsilon[r3] - 2\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2 - 8\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3] - 8\eta[3]\epsilon[r1]\epsilon[c2]^2\epsilon[r3] - 8\eta[3]\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2 - 8\eta[3]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2 - 8\eta[3]\epsilon[r1]\epsilon[r2]^2\epsilon[c3]^2 - 2\eta[3]\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]^2 - 8\eta[3]\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3] - 8\eta[3]\epsilon[c1]^2\epsilon[r2]\epsilon[r3]^2 - 8\eta[3]\epsilon[c1]^2\epsilon[r2]\epsilon[c3]^2 - 2\eta[3]\epsilon[c1]^2\epsilon[r2]\epsilon[c2]^2\epsilon[r3]^2 - 2\eta[3]\epsilon[c1]^2\epsilon[c2]^2\epsilon[c3]^2 C_2\eta[2]^2\epsilon[r1]^2\epsilon[r2]^2\epsilon[r3]^2 C_4\eta[2]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2 C_2\eta[2]^2\epsilon[c1]^2\epsilon[r2]^2\epsilon[r3]^2 C_{16}\eta[2]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2 - 2\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2 - 8\eta[2]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2 C_4\epsilon[c3]^2
\end{aligned}$$

[illegible]

$$\begin{aligned}
& [1]^2 \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[c3]^2 C^{16} \eta[1]^2 \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] C^4 \eta[1]^2 \epsilon[r1]^2 \epsilon[r2] \epsilon[r3]^2 C \eta[1]^2 \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[c3]^2 C^4 \eta[1]^2 \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] C \eta[1]^2 \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3]^2 C^8 \eta[2] \epsilon[r1] \epsilon[r2] \epsilon[r3] \eta[3] C^2 \eta[2] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[c3]^2 \eta[1] C^8 \eta[2] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3] \eta[1] C^2 \eta[2] \eta[3] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[c3]^2 C^2 \eta[2] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[c3]^2 \eta[3] C^2 \eta[3] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3] \eta[1] C^8 \eta[3] \epsilon[c1]^2 \epsilon[r2] \epsilon[r3] \eta[1] C^2 \eta[3] \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[r3] \eta[1] C^2 \eta[2] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3] \eta[3] C^2 \eta[3] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3] \eta[1] C^2 \eta[3] \epsilon[r1] \epsilon[r2]^2 \epsilon[r3] \eta[1] C^8 \eta[3] \epsilon[r1] \epsilon[r2] \epsilon[r3] \eta[1] C^2 \eta[3] \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3] \eta[2] C^2 \eta[3] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3] \eta[2] C^2 \eta[3] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] \eta[2] C^8 \eta[1] \epsilon[r1] \epsilon[r2] \epsilon[r3]^2 \eta[3] C^2 \eta[1] \epsilon[r1] \epsilon[r2]^2 \epsilon[c3]^2 \eta[3] C^8 \eta[1] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3] \eta[2] C^8 \eta[1] \epsilon[r1] \epsilon[r2]^2 \epsilon[r3] \eta[2] C^8 \eta[1] \epsilon[r1] \epsilon[r2]^2 \epsilon[r3] \eta[2] C^8 \eta[1] \epsilon[r1] \epsilon[r2]^2 \epsilon[r3]^2 \eta[3] C^2 \eta[2] \epsilon[r3]^2 \eta[1] \epsilon[r1]^2 \epsilon[c2]^2 - 8 \eta[1]^2 \epsilon[r1] \epsilon[r2]^2 - 32 \eta[1]^2 \epsilon[r1] \epsilon[r2] C^{16} \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[1]^2 \epsilon[r1] \epsilon[c2]^2 - 8 \eta[1]^2 \epsilon[r1] \epsilon[r3]^2 - 32 \eta[1]^2 \epsilon[r1] \epsilon[r3] - 8 \eta[1]^2 \epsilon[r1] \epsilon[c3]^2 C^8 \eta[2] \epsilon[r3]^2 \eta[1] C^{16} \epsilon[c2]^2 \eta[2]^2 \epsilon[r1] C^8 \eta[3] \epsilon[r2]^2 \eta[1] - 16 \eta[2] \epsilon[r1]^2 \eta[1] C^4 \epsilon[c2]^2 \eta[2]^2 \epsilon[c1]^2 - 16 \eta[3] \epsilon[r1]^2 \eta[1] C^4 \epsilon[c1]^2 \eta[1]^2 \epsilon[r3]^2 C^4 \epsilon[r1] \epsilon[c2]^2 \epsilon[r3]^2 C^4 \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[c3]^2 C^4 \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] C \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[c3]^2 C^4 \epsilon[r1] \epsilon[r2]^2 \epsilon[r3]^2 C \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[c3]^2 C^{16} \epsilon[r1] \epsilon[r2]^2 \epsilon[c3]^2 C \epsilon[r1]^2 \epsilon[r2]^2 \epsilon[r3]^2 C^{16} \epsilon[r1] \epsilon[r2]^2 \epsilon[r3] C \epsilon[c1]^2 \epsilon[r2]^2
\end{aligned}$$

$$\begin{aligned} &\epsilon[r_3]^2 C_{16} \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] C_4 \epsilon[r_1]^2 * \\ &\epsilon[r_2] \epsilon[r_3]^2 C_4 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 C_4 * \\ &\epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C_{16} \epsilon[r_1] \epsilon[r_2] * \\ &\epsilon[r_3]^2 C_4 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C_4 \epsilon[r_1] * \\ &\epsilon[c_2]^2 \epsilon[c_3]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3] \\ &^2 C_4 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 C_{16} \epsilon[c_1]^2 \epsilon[\\ &r_2] \epsilon[r_3] C_4 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \epsilon[c_1] \\ &^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C_4 \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[\\ &r_3] \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 -8\eta[1]\epsilon[r_1]^2 * \\ &\epsilon[r_2]^2 -32\eta[1]\epsilon[r_1]^2 \epsilon[r_2] -8\eta[1]\epsilon[r_1]^2 * \\ &\epsilon[c_2]^2 -8\eta[1]\epsilon[r_1] \epsilon[r_2]^2 -32\eta[1]\epsilon[r_1] * \\ &\epsilon[r_2] -8\eta[1]\epsilon[r_1] \epsilon[c_2]^2 -8\eta[1]\epsilon[c_1]^2 * \\ &\epsilon[r_2]^2 -32\eta[1]\epsilon[c_1]^2 \epsilon[r_2] -8\eta[1]\epsilon[c_1]^2 * \\ &\epsilon[c_2]^2 -8\eta[1]\epsilon[r_1] \epsilon[r_3]^2 C_2 \eta[2] \epsilon[r_3]^2 * \\ &\eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C_2 \eta[2] \epsilon[r_3]^2 \eta[1] \epsilon[\\ &c_1]^2 \epsilon[c_2]^2 C_2 \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[r_1]^2 \epsilon[\\ &c_2]^2 C_2 \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C_2 \eta[\\ &2] \epsilon[c_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C_2 \eta[1] \epsilon[r_1] \\ &* \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[2] C_2 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 * \\ &\epsilon[r_3]^2 \eta[3] C_8 \eta[2] \epsilon[r_3] \eta[1] \epsilon[r_1]^2 \epsilon[c_2] \\ &^2 C_8 \eta[2] \epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C_8 \eta[2] * \\ &\epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C_2 \eta[1] \epsilon[r_1] * \\ &\epsilon[c_2]^2 \epsilon[c_3]^2 \eta[2] C_2 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 * \\ &\epsilon[c_3]^2 \eta[3] C_2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[\\ &1] C_8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C_2 \eta[2] * \\ &\epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \eta[1] C_2 \eta[2] \epsilon[c_1]^2 * \\ &\epsilon[r_2] \epsilon[r_3]^2 \eta[1] C_8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] * \\ &\epsilon[r_3] \eta[1] C_2 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[1] \\ &C_2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C_2 \eta[2] * \\ &\epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[1] C_8 \eta[2] \epsilon[r_1] \epsilon[\\ &r_2] \epsilon[r_3] \eta[1] C_2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 * \\ &\eta[1] C_8 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C_8 \eta[3] \eta[\\ &2] \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C_2 \eta[3] \eta[2] \epsilon[r_1] \\ &^2 \epsilon[r_2]^2 \epsilon[c_3]^2 -4\eta[2] \epsilon[r_2] \epsilon[c_3]^2 \eta[1] \\ &-32\eta[1]^2 \epsilon[r_1] C_{16} \eta[1]^2 \epsilon[r_2] C_{16} \eta[1]^2 \epsilon[\\ &r_3] C_4 \eta[1]^2 \epsilon[r_2]^2 C_4 \eta[1]^2 \epsilon[r_3]^2 C_4 \eta[1]^2 * \\ &\epsilon[c_2]^2 C_4 \eta[1]^2 \epsilon[c_3]^2 C_4 \eta[2]^2 \epsilon[r_3]^2 * \\ &\epsilon[r_2]^2 -32\eta[1] \epsilon[r_1] \epsilon[r_3] -8\eta[1] \epsilon[r_1] * \end{aligned}$$

$$\begin{aligned}
& \epsilon[c_3]^{2C_{16}\eta[1]\epsilon[r_2]\epsilon[r_3]^{2C_{64}\eta[1]\epsilon[r_2]*} \\
& \epsilon[r_3]C_{16}\eta[1]\epsilon[r_2]\epsilon[c_3]^{2-8\eta[1]\epsilon[r_1]^2*} \\
& \epsilon[r_3]^{2-32\eta[1]\epsilon[r_1]^2\epsilon[r_3]-8\eta[1]\epsilon[r_1]^2*} \\
& \epsilon[c_3]^{2C_4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^{2C_{16}\eta[1]\epsilon[r_2]} \\
& ^2\epsilon[r_3]C_4\eta[1]\epsilon[r_2]^2\epsilon[c_3]^{2-8\eta[1]\epsilon[c_1]} \\
& ^2\epsilon[r_3]^{2-32\eta[1]\epsilon[c_1]^2\epsilon[r_3]-8\eta[1]\epsilon[c_1]} \\
& ^2\epsilon[c_3]^{2C_4\eta[1]\epsilon[c_2]^2\epsilon[r_3]^{2C_{16}\eta[1]*} \\
& \epsilon[c_2]^2\epsilon[r_3]C_4\eta[1]\epsilon[c_2]^2\epsilon[c_3]^{2-8\eta[2]*} \\
& \epsilon[r_1]^2\epsilon[r_2]^{2-8\eta[2]\epsilon[r_1]^2\epsilon[r_2]-8\eta[2]*} \\
& \epsilon[r_1]^2\epsilon[c_2]^{2-32\eta[2]\epsilon[r_1]\epsilon[r_2]^{2-32\eta[2]} \\
& ^2\epsilon[r_1]\epsilon[r_2]-32\eta[2]\epsilon[r_1]\epsilon[c_2]^{2-8\eta[2]*} \\
& \epsilon[c_1]^2\epsilon[r_2]^{2-8\eta[2]\epsilon[c_1]^2\epsilon[r_2]-8\eta[2]*} \\
& \epsilon[c_1]^2\epsilon[c_2]^{2C_{16}\eta[2]\epsilon[r_1]\epsilon[r_3]^{2C_{64}\eta[2]} \\
& ^2\epsilon[r_1]\epsilon[r_3]C_{16}\eta[2]\epsilon[r_1]\epsilon[c_3]^{2-8\eta[2]*} \\
& \epsilon[r_2]\epsilon[r_3]^{2-32\eta[2]\epsilon[r_2]\epsilon[r_3]-8\eta[2]*} \\
& \epsilon[r_2]\epsilon[c_3]^{2C_4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^{2C_{16}\eta[2]} \\
& *\epsilon[r_1]^2\epsilon[r_3]C_4\eta[2]\epsilon[r_1]^2\epsilon[c_3]^{2-8\eta[2]*} \\
& \epsilon[r_2]^2\epsilon[r_3]^{2-32\eta[2]\epsilon[r_2]^2\epsilon[r_3]-8\eta[2]*} \\
& \epsilon[r_2]^2\epsilon[c_3]^{2C_4\eta[2]\epsilon[c_1]^2\epsilon[r_3]^{2C_{16}\eta[2]} \\
& ^2\epsilon[c_1]^2\epsilon[r_3]C_4\eta[2]\epsilon[c_1]^2\epsilon[c_3]^{2-8\eta[2]} \\
& ^2\epsilon[c_2]^2\epsilon[r_3]^{2-32\eta[2]\epsilon[c_2]^2\epsilon[r_3]-8\eta[2]} \\
& ^2\epsilon[c_2]^2\epsilon[c_3]^{2C_4\eta[3]\epsilon[r_1]^2\epsilon[r_2]} \\
& ^{2C_{16}\eta[3]\epsilon[r_1]^2\epsilon[r_2]C_4\eta[3]\epsilon[r_1]^2\epsilon[r_2]} \\
& ^{2C_{16}\eta[3]\epsilon[r_1]\epsilon[r_2]^{2C_{64}\eta[3]\epsilon[r_1]\epsilon[r_2]} \\
& ^{2C_{16}\eta[3]\epsilon[r_1]\epsilon[c_2]^{2C_4\eta[3]\epsilon[c_1]^2\epsilon[r_2]} \\
& ^{2C_{16}\eta[3]\epsilon[c_1]^2\epsilon[r_2]C_4\eta[3]\epsilon[c_1]^2*} \\
& \epsilon[c_2]^{2-32\eta[3]\epsilon[r_1]\epsilon[r_3]^{2-32\eta[3]\epsilon[r_1]*} \\
& \epsilon[r_3]-32\eta[3]\epsilon[r_1]\epsilon[c_3]^{2-32\eta[3]\epsilon[r_2]*} \\
& \epsilon[r_3]^{2-32\eta[3]\epsilon[r_2]\epsilon[r_3]-32\eta[3]\epsilon[r_2]*} \\
& \epsilon[c_3]^{2-8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^{2-8\eta[3]\epsilon[r_1]} \\
& ^2\epsilon[r_3]-8\eta[3]\epsilon[r_1]^2\epsilon[c_3]^{2-8\eta[3]\epsilon[r_2]} \\
& ^2\epsilon[r_3]^{2-8\eta[3]\epsilon[r_2]^2\epsilon[r_3]-8\eta[3]\epsilon[r_2]} \\
& ^2\epsilon[c_3]^{2-8\eta[3]\epsilon[c_1]^2\epsilon[r_3]^{2-8\eta[3]\epsilon[c_1]} \\
& ^2\epsilon[r_3]-8\eta[3]\epsilon[c_1]^2\epsilon[c_3]^{2-8\eta[3]\epsilon[c_2]} \\
& ^2\epsilon[r_3]^{2-8\eta[3]\epsilon[c_2]^2\epsilon[r_3]-8\eta[3]\epsilon[c_2]} \\
& ^2\epsilon[c_2]^2\epsilon[c_3]^{2C_4\eta[3]^2\epsilon[c_2]^2\epsilon[r_3]^{2C_4\eta[3]} \\
& ^2\epsilon[r_1]^2\epsilon[r_3]^{2C_4\eta[3]^2\epsilon[r_2]^2\epsilon[r_3]} \\
& ^{2C_4\eta[3]^2\epsilon[c_1]^2\epsilon[r_3]^{2C_{16}\eta[3]^2\epsilon[r_1]*}
\end{aligned}$$

$$\begin{aligned} & \epsilon[r_3]^2 C^{16} \eta^3 \epsilon[r_2] \epsilon[r_3]^2 - 16 \eta^3 \epsilon[r_3] \\ & [r_3] \eta C^8 \eta^3 \eta^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 C^2 \eta^3 \\ & [r_3] \eta^2 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C^2 \eta^3 \eta^2 * \\ & \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C^2 \eta^3 \eta^2 \epsilon[c_1]^2 * \\ & \epsilon[r_2]^2 \epsilon[r_3]^2 C^8 \eta^3 \epsilon[r_2] \eta C^1 \epsilon[c_1]^2 * \\ & \epsilon[c_3]^2 C^2 \eta^3 \epsilon[r_2]^2 \eta C^1 \epsilon[c_1]^2 \epsilon[c_3] \\ & ^2 C^2 \eta^3 \epsilon[c_1]^2 \eta C^1 \epsilon[c_2]^2 \epsilon[r_3]^2 C^2 \eta^3 * \\ & \epsilon[c_1]^2 \eta C^1 \epsilon[c_2]^2 \epsilon[c_3]^2 C^8 \eta^2 \epsilon[r_1] * \\ & \epsilon[r_2] \epsilon[r_3]^2 \eta C^2 \eta^2 \epsilon[c_1]^2 \epsilon[r_2] * \\ & \epsilon[r_3]^2 \eta C^2 \eta^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] \eta C^3 \\ & C^2 \eta^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \eta C^3 C^8 \eta^3 * \\ & \epsilon[r_1]^2 \eta C^1 \epsilon[r_2] \epsilon[r_3]^2 C^8 \eta^3 \epsilon[c_1]^2 \eta C^1 \\ & \epsilon[r_2] \epsilon[r_3]^2 C^8 \eta^3 \eta^2 \epsilon[r_1] \epsilon[c_2]^2 * \\ & \epsilon[c_3]^2 C^2 \eta^3 \eta^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3] \\ & ^2 C^2 \eta^3 \eta^2 \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C^2 \eta^3 * \\ & \eta^2 \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 C^8 \eta^3 \epsilon[r_2]^2 * \\ & \eta^2 \epsilon[r_1] \epsilon[c_3]^2 C^2 \eta^3 \epsilon[r_1]^2 \epsilon[r_2]^2 \eta C^1 \\ & \epsilon[r_3]^2 C^2 \eta^3 \epsilon[r_1]^2 \epsilon[r_2]^2 \eta C^1 \epsilon[c_3] \\ & ^2 C^8 \eta^3 \epsilon[r_1]^2 \epsilon[r_2] \eta C^1 \epsilon[c_3]^2 C^2 \eta^3 * \\ & \epsilon[r_1]^2 \epsilon[c_2]^2 \eta C^1 \epsilon[r_3]^2 C^2 \eta^3 \epsilon[r_1]^2 * \\ & \epsilon[c_2]^2 \eta C^1 \epsilon[c_3]^2 C^2 \eta^3 \epsilon[c_1]^2 \epsilon[r_2]^2 * \\ & \eta C^1 \epsilon[r_3]^2 C^8 \eta^3 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3] \eta C^2 \\ & C^8 \eta^3 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3] \eta C^2 \eta^3 \epsilon[c_1] \\ & ^2 \epsilon[r_2]^2 \epsilon[r_3] \eta C^2 - 16 \eta^3 \epsilon[r_1] \eta C^1 - 16 * \\ & \eta^2 \epsilon[c_1]^2 \eta C^1 C^8 \eta^3 \epsilon[c_2]^2 \eta C^1 - 16 \eta^2 \epsilon[r_1] * \\ & \eta C^1 C^4 \epsilon[c_2]^2 \eta C^2 \epsilon[r_1]^2 C^4 \epsilon[c_1]^2 \eta C^2 * \\ & \epsilon[c_2]^2 C^{16} \epsilon[c_3]^2 \eta C^3 \epsilon[r_1] C^4 \epsilon[c_1]^2 * \\ & \eta C^1 \epsilon[c_3]^2 C^2 \eta C^1 \epsilon[c_2]^2 \epsilon[c_3]^2 C^2 \eta C^1 * \\ & \epsilon[c_2]^2 \epsilon[r_3]^2 C^4 \eta C^1 \epsilon[c_2]^2 \epsilon[r_3] C^1 \eta C^1 \\ & ^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \eta C^1 \epsilon[r_2]^2 \epsilon[r_3] C^1 \eta C^1 \\ & \epsilon[r_2]^2 \epsilon[r_3]^2 C^{16} \eta C^1 \epsilon[r_2] \epsilon[r_3] \\ & C^4 \eta C^1 \epsilon[r_2] \epsilon[c_3]^2 C^4 \eta C^1 \epsilon[r_2] \epsilon[r_3] \\ & ^2 C^{16} \epsilon[c_2]^2 \eta C^2 \epsilon[r_3] C^{16} \epsilon[c_1]^2 \eta C^2 * \\ & \epsilon[r_3] - 16 \eta^2 \epsilon[r_2] \eta C^3 - 16 \eta^2 \epsilon[r_3] \eta C^3 C^8 * \\ & \eta^2 \epsilon[r_1]^2 \eta C^3 - 16 \eta^2 \epsilon[r_3]^2 \eta C^3 C^{16} \eta C^1 * \\ & \epsilon[r_1]^2 \epsilon[r_2] C^4 \eta C^1 \epsilon[r_1]^2 \epsilon[c_2]^2 C^4 * \\ & \epsilon[c_3]^2 \eta C^3 \epsilon[c_2]^2 C^4 \eta C^1 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \end{aligned}$$

$$\begin{aligned}
& [r2]^2 - 16 \cdot \eta[1] \cdot \epsilon[r2]^2 \cdot \eta[2] - 16 \cdot \eta[1] \cdot \epsilon[r3]^2 \cdot \eta[3] - 16 \cdot \\
& \eta[1] \cdot \epsilon[c2]^2 \cdot \eta[2] - 16 \cdot \eta[1] \cdot \epsilon[c3]^2 \cdot \eta[3] C 16 \cdot \epsilon[c3] \\
& ^2 \cdot \eta[3]^2 \cdot \epsilon[r2] C 32 \cdot \eta[2] \cdot \epsilon[r3] \cdot \eta[1] C 4 \cdot \eta[2]^2 \cdot \epsilon[r1] \\
& ^2 \cdot \epsilon[r2]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[r1]^2 \cdot \epsilon[r2] C \eta[3]^2 \cdot \\
& \epsilon[r1]^2 \cdot \epsilon[c2]^2 C 4 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 - 4 \cdot \eta[2] \cdot \\
& \epsilon[r1] \cdot \epsilon[c3]^2 \cdot \eta[1] - 4 \cdot \eta[2] \cdot \epsilon[r2] \cdot \epsilon[r3]^2 \cdot \\
& \eta[1] - 16 \cdot \eta[2] \cdot \epsilon[r2] \cdot \epsilon[r3] \cdot \eta[1] - 32 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \\
& \epsilon[r2] \cdot \epsilon[r3] - 8 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3] \\
& - 8 \cdot \eta[3]^2 \cdot \epsilon[r1] \cdot \epsilon[c2]^2 \cdot \epsilon[r3] - 4 \cdot \eta[2] \cdot \epsilon[c1]^2 \cdot \\
& \epsilon[r3] \cdot \eta[3] - 4 \cdot \eta[2] \cdot \epsilon[c1]^2 \cdot \epsilon[c3]^2 \cdot \eta[3] C 4 \cdot \epsilon[r1] \\
& ^2 \cdot \eta[1]^2 \cdot \epsilon[r2]^2 \cdot \epsilon[r3] - 4 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r3] \\
& ^2 \cdot \eta[3] - 4 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r3] \cdot \eta[3] - 4 \cdot \eta[2] \cdot \epsilon[r1] \\
& ^2 \cdot \epsilon[c3]^2 \cdot \eta[3] - 4 \cdot \eta[2] \cdot \epsilon[c1]^2 \cdot \epsilon[r3]^2 \cdot \eta[3] \\
& C 8 \cdot \eta[2] \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \eta[1] C 8 \cdot \eta[2] \cdot \epsilon[r1] \cdot \epsilon[r2] \\
& \cdot \eta[1] C 8 \cdot \eta[2] \cdot \epsilon[r2] \cdot \epsilon[r3]^2 \cdot \eta[3] C 8 \cdot \eta[2] \cdot \epsilon[r2] \\
& \cdot \epsilon[r3] \cdot \eta[3] C 8 \cdot \eta[2] \cdot \epsilon[r2] \cdot \epsilon[c3]^2 \cdot \eta[3] - 8 \cdot \eta[1]^2 \cdot \\
& \epsilon[r1] \cdot \epsilon[r2] \cdot \epsilon[r3]^2 - 2 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[c2] \\
& ^2 \cdot \epsilon[c3]^2 C 16 \cdot \epsilon[c1]^2 \cdot \eta[1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3] - 2 \cdot \\
& \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[c3]^2 - 32 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \\
& \epsilon[r2] \cdot \epsilon[r3] - 8 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3] \\
& - 8 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[c2]^2 \cdot \epsilon[r3] - 2 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \\
& \epsilon[c2]^2 \cdot \epsilon[r3]^2 - 2 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[r2]^2 \cdot \epsilon[r3] \\
& ^2 - 8 \cdot \eta[1]^2 \cdot \epsilon[r1] \cdot \epsilon[r2] \cdot \epsilon[c3]^2 - 4 \cdot \eta[3] \cdot \epsilon[r2]^2 \cdot \\
& \epsilon[r3] \cdot \eta[1] C 8 \cdot \eta[3] \cdot \epsilon[c1]^2 \cdot \epsilon[r3] \cdot \eta[1] - 4 \cdot \eta[3] \cdot \\
& \epsilon[c2]^2 \cdot \epsilon[r3] \cdot \eta[1] - 16 \cdot \eta[3] \cdot \epsilon[c1]^2 \cdot \epsilon[r2] \cdot \\
& \eta[1] - 4 \cdot \eta[3] \cdot \epsilon[c1]^2 \cdot \epsilon[c2]^2 \cdot \eta[1] C 8 \cdot \eta[3] \cdot \epsilon[r1] \cdot \\
& \epsilon[r3] \cdot \eta[1] - 16 \cdot \eta[3] \cdot \epsilon[r2] \cdot \epsilon[r3] \cdot \eta[1] C 8 \cdot \eta[3] \cdot \\
& \epsilon[r1]^2 \cdot \epsilon[r3] \cdot \eta[1] - 4 \cdot \eta[3] \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 \cdot \eta[1] \\
& C \epsilon[c2]^2 \cdot \eta[2]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r3]^2 - 2 \cdot \eta[2]^2 \cdot \\
& \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \epsilon[c3]^2 - 8 \cdot \eta[2]^2 \cdot \epsilon[r1]^2 \cdot \epsilon[r2] \cdot \\
& \epsilon[r3] C 4 \cdot \epsilon[c1]^2 \cdot \eta[1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3]^2 - 16 \cdot \\
& \eta[1] \cdot \epsilon[r2] \cdot \epsilon[r3]^2 \cdot \eta[3] - 16 \cdot \eta[1] \cdot \epsilon[r2] \cdot \epsilon[c3] \\
& ^2 \cdot \eta[3] C 8 \cdot \eta[3] \cdot \epsilon[c2]^2 \cdot \epsilon[r3] \cdot \eta[2] C 8 \cdot \eta[1] \cdot \epsilon[r1] \cdot \\
& \epsilon[r3]^2 \cdot \eta[3] C 8 \cdot \eta[1] \cdot \epsilon[r1] \cdot \epsilon[c3]^2 \cdot \eta[3] C 4 \cdot \epsilon[r1] \\
& ^2 \cdot \eta[3]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r2] C 8 \cdot \eta[2] \cdot \eta[1] \cdot \epsilon[r1]^2 \cdot \\
& \epsilon[c2]^2 C 8 \cdot \eta[2] \cdot \eta[1] \cdot \epsilon[r1]^2 \cdot \epsilon[r2]^2 C \epsilon[c3] \\
& ^2 \cdot \eta[3]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r2]^2 C 4 \cdot \epsilon[c2]^2 \cdot \eta[2]^2 \cdot \\
& \epsilon[r1] \cdot \epsilon[c3]^2 - 2 \cdot \eta[2]^2 \cdot \epsilon[c1]^2 \cdot \epsilon[r2] \cdot \epsilon[r3]
\end{aligned}$$

$$\begin{aligned}
& [r_3]^2 - 8\eta^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3] - 2\eta^2 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 C^4 \epsilon[c_1]^2 \eta^2 \epsilon[c_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \eta^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 4\eta^2 \epsilon[r_1]^2 \epsilon[r_2] \eta^3 - 16\eta^2 \epsilon[r_1] \epsilon[r_2] \eta^3 - 2\eta^3 \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 32\eta \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] C^{16} \eta^3 \epsilon[r_1] \epsilon[r_2] C^4 \eta^3 \epsilon[r_1] \epsilon[c_2]^2 C \eta^3 \epsilon[c_1]^2 \epsilon[r_2]^2 C^4 \eta^3 \epsilon[c_1]^2 \epsilon[r_2] C \eta^3 \epsilon[c_1]^2 \epsilon[c_2]^2 - 32\eta^3 \epsilon[r_1] \epsilon[r_3] - 32\eta^3 \epsilon[r_2] \epsilon[r_3] C \eta^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C^4 \eta^2 \epsilon[r_1]^2 \epsilon[r_3] C \eta^2 \epsilon[c_1]^2 \epsilon[r_3]^2 C^4 \eta^2 \epsilon[c_1]^2 \epsilon[r_3] C \eta^2 \epsilon[c_1]^2 \epsilon[c_3]^2 C \eta^3 \epsilon[r_1]^2 \epsilon[r_2]^2 - 8\eta^2 \epsilon[r_1]^2 \epsilon[r_2] - 32\eta^2 \epsilon[r_1] \epsilon[r_2] - 8\eta^2 \epsilon[c_1]^2 \epsilon[r_2] C^4 \eta^2 \epsilon[r_1] \epsilon[r_3]^2 C^{16} \eta^2 \epsilon[r_1] \epsilon[r_3] C^4 \eta^2 \epsilon[r_1] \epsilon[c_3]^2 - 8\eta^2 \epsilon[r_2] \epsilon[r_3]^2 - 32\eta^2 \epsilon[r_2] \epsilon[r_3] - 8\eta^2 \epsilon[r_2] \epsilon[c_3]^2 C^{16} \epsilon[c_1]^2 \eta^2 \epsilon[r_2] - 16\eta^3 \epsilon[r_2]^2 \eta^2 - 16\eta^3 \epsilon[c_2]^2 \eta^2 C^4 \eta^2 \epsilon[r_1]^2 \epsilon[r_3]^2 C^4 \epsilon[c_3]^2 \eta^2 \epsilon[r_1]^2 C^4 \eta^2 \epsilon[r_1]^2 \epsilon[c_3]^2 C^{16} \eta^2 \epsilon[r_1]^2 \epsilon[r_3] C^{16} \eta^2 \epsilon[r_1] \epsilon[r_2]^2 C^4 \epsilon[c_1]^2 \eta^2 \epsilon[r_2]^2 C^{16} \eta^2 \epsilon[r_3] \epsilon[r_2]^2 C^{32} \eta^2 \epsilon[r_1] \eta^3 - 8\eta^3 \epsilon[c_2]^2 \epsilon[r_3] C^8 \eta^2 \epsilon[c_1]^2 \eta^3 - 16\eta^2 \epsilon[c_3]^2 \eta^3 C^{32} \eta^3 \epsilon[r_2] \eta^2 - 2\eta^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 8\eta \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 - 8\eta \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] - 2\eta^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 2\eta^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[r_3]^2 - 2\eta^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 - 32\eta \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] - 2\eta^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 - 8\eta \epsilon[r_1] \epsilon[r_2] \epsilon[c_2]^2 \epsilon[r_3] - 8\eta \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 - 2\eta^2 \epsilon[r_1] \epsilon[r_2]^2 \epsilon[c_3]^2 - 8\eta \epsilon[r_1] \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 8\eta \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 - 8\eta \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] - 2\eta^2 \epsilon[r_1] \epsilon[r_1]^2
\end{aligned}$$

$$\begin{aligned} & \epsilon[c2]^2 \epsilon[c3]^2 - 2 \eta[1] \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[c3]^2 - 8 \eta[1] \epsilon[c1]^2 \epsilon[r2] \epsilon[r3]^2 - 32 \eta[1] \epsilon[c1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[1] \epsilon[c1]^2 \epsilon[r2]^2 \epsilon[c3]^2 - 2 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 8 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r3] - 2 \eta[1] \epsilon[c1]^2 \epsilon[c2]^2 \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[r3] - 8 \eta[2] \epsilon[r1] \epsilon[c2]^2 \epsilon[r3]^2 - 2 \eta[2] \epsilon[r1]^2 \epsilon[r2] \epsilon[c3]^2 - 8 \eta[2] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3] - 8 \eta[2] \epsilon[r1]^2 \epsilon[c2]^2 \epsilon[r3]^2 - 8 \eta[3] \epsilon[r1]^2 \epsilon[r3] \end{aligned}$$

eps_c

$$\begin{aligned} & \left(9 \left(16 \epsilon_{cl} \eta_1 + 16 \epsilon_{c2} \eta_2 + 16 \epsilon_{c3} \eta_3 + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} + 16 \epsilon_{cl} \eta_1 \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \right. \right. \\ & + 4 \epsilon_{cl} \eta_1 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} + 16 \epsilon_{c2} \eta_2 \epsilon_{r3} + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \\ & + 4 \epsilon_{c2} \eta_2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{c3}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} + 16 \epsilon_{c3} \eta_3 \epsilon_{r2} \\ & + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{cl}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{c2}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \\ & + 16 \epsilon_{cl} \eta_1 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \\ & + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{cl} \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{r3} \\ & + 4 \epsilon_{c2} \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \\ & + 4 \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{r3} + \epsilon_{c2} \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{r2} + \epsilon_{c3} \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \\ & + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{r2} + 4 \epsilon_{c3} \eta_3 \epsilon_{r1} \epsilon_{c2}^2 + \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 + 4 \epsilon_{c3} \eta_3 \epsilon_{cl}^2 \epsilon_{r2} \\ & + \epsilon_{c3} \eta_3 \epsilon_{c2}^2 \epsilon_{r2}^2 \left. \right) \Big/ \left(64 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{c3}^2 + 64 \eta_1 + 64 \eta_2 + 64 \eta_3 + 64 \epsilon_{r1} \right. \\ & + 64 \epsilon_{r2} + 64 \epsilon_{r3} + 16 \epsilon_{cl}^2 + 16 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3} \\ & + 18 \epsilon_{cl} \eta_1 \epsilon_{c3} \eta_3 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2}^2 + 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 - 16 \eta_2 \epsilon_{r2} \eta_1 \\ & + 8 \eta_3 \epsilon_{r1}^2 \eta_1 \epsilon_{r3}^2 - 16 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 - 4 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \eta_2 - 8 \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \\ & - 2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\ & + 8 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1} \epsilon_{r2} \\ & + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{r2}^2 + 8 \eta_2 \eta_1 \epsilon_{cl}^2 \epsilon_{c2}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 \\ & - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_3 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{cl}^2 \epsilon_{r2} \eta_1 - 4 \eta_2 \epsilon_{r1} \epsilon_{r3}^2 \eta_1 \\ & - 16 \eta_2 \epsilon_{r1} \epsilon_{r3} \eta_1 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 + 8 \eta_3 \epsilon_{c2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{cl}^2 \epsilon_{r3} \eta_1 \\ & - 4 \eta_2 \epsilon_{cl}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r3}^2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 - 4 \eta_1 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 \\ & - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 - 4 \eta_1 \epsilon_{c2}^2 \epsilon_{c3}^2 \eta_2 \end{aligned} \quad (2.11)$$

$$\begin{aligned}
& -4\eta_1\epsilon_{c2}^2\epsilon_{c3}^2\eta_3-4\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\eta_2-4\eta_2\epsilon_{r1}^2\epsilon_{r3}^2\eta_1-16\eta_2\epsilon_{r1}^2\epsilon_{r3}\eta_1-4\eta_2\epsilon_{r1}^2 \\
& \epsilon_{c3}^2\eta_1+\epsilon_{c1}^2\eta_1^2\epsilon_{r2}^2\epsilon_{r3}^2+\epsilon_{c2}^2\eta_2^2\epsilon_{r1}^2\epsilon_{r3}^2+\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2+8\eta_3\eta_1\epsilon_{c1}^2\epsilon_{r3}^2+8\eta_3\eta_1\epsilon_{c1}^2 \\
& \epsilon_{c3}^2+16\eta_3^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2+4\eta_3^2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2+\eta_3^2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2-4\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_2-4\eta_3\epsilon_{r1}^2 \\
& \epsilon_{c2}^2\eta_2-2\eta_3^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-2\eta_3^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-8\eta_3^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}+8\eta_3\epsilon_{r1}^2\eta_1\epsilon_{c3}^2+4\eta_2^2 \\
& \epsilon_{c3}^2\epsilon_{r2}^2+4\eta_1^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-16\eta_3\epsilon_{r1}^2\epsilon_{r2}\eta_1-4\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\eta_1-4\eta_3\epsilon_{r1}\epsilon_{r2}^2\eta_1 \\
& -16\eta_3\epsilon_{r1}\epsilon_{r2}\eta_1+72\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3+72\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3-4\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\eta_1 \\
& +72\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2-2\eta_3^2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}+4\eta_3^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+8\eta_3\epsilon_{c2}^2\eta_2\epsilon_{c3}^2+16\epsilon_{c1}^2\eta_1^2 \\
& +16\eta_3^2\epsilon_{r1}+16\eta_3^2\epsilon_{r2}-32\eta_3^2\epsilon_{r3}+4\eta_3^2\epsilon_{r1}^2+4\eta_3^2\epsilon_{r2}^2+4\eta_3^2\epsilon_{c1}^2+4\eta_3^2\epsilon_{c2}^2 \\
& +16\eta_2^2\epsilon_{r1}-32\eta_2^2\epsilon_{r2}+16\eta_2^2\epsilon_{r3}+4\eta_2^2\epsilon_{r1}^2+4\eta_2^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{c1}^2+4\eta_2^2\epsilon_{c3}^2 \\
& +16\epsilon_{c3}^2\eta_3^2+4\epsilon_{r1}^2\epsilon_{r2}^2+16\epsilon_{r1}^2\epsilon_{r2}+4\epsilon_{r1}^2\epsilon_{c2}^2+16\epsilon_{r1}\epsilon_{r2}^2+64\epsilon_{r1}\epsilon_{r2}+16\epsilon_{r1}\epsilon_{c2}^2 \\
& +4\epsilon_{c1}^2\epsilon_{r2}^2+16\epsilon_{c1}^2\epsilon_{r2}+16\epsilon_{r1}^2+16\epsilon_{r2}^2+16\epsilon_{r3}^2+16\eta_1^2+16\eta_2^2+16\eta_3^2-2\eta_2 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-2\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{c3}^2-8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2-2\eta_2\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 \\
& -32\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2-32\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-2\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2-32\eta_2\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{r3}-2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-8\eta_2 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-8\eta_2\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2-2\eta_2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2-2\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3} \\
& -2\eta_2\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-2\eta_2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_2\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{c3}^2-8\eta_3 \\
& \epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}-8\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{r3}^2-8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{c3}^2-2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 \\
& -8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2-32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2+18\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{c2}^2+72\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1} \\
& +18\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{r1}^2+18\epsilon_{c2}\eta_2\epsilon_{c3}\eta_3\epsilon_{c1}^2+2\eta_2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2\eta_3 \\
& +72\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3}+18\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{r3}^2+18\epsilon_{c1}\eta_1\epsilon_{c2}\eta_2\epsilon_{c3}^2 \\
& +72\epsilon_{c1}\eta_1\epsilon_{c3}\eta_3\epsilon_{r2}+2\eta_2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2\eta_1+8\eta_2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2\eta_3-2\eta_3\epsilon_{r1}^2\epsilon_{c2}^2\epsilon_{r3}^2 \\
& -32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}-2\eta_3\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2-8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2-8\eta_3\epsilon_{r1} \\
& \epsilon_{c2}^2\epsilon_{r3}-8\eta_3\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_3\epsilon_{r1}\epsilon_{r2}^2\epsilon_{c3}^2-2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}-32\eta_3\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2-2\eta_3 \\
& \epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}-8\eta_3\epsilon_{r1}\epsilon_{c2}^2\epsilon_{c3}^2-2\eta_3\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{c3}^2-8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3}^2-8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{r3} \\
& -8\eta_3\epsilon_{c1}^2\epsilon_{r2}\epsilon_{c3}^2-2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}^2-2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{r3}-2\eta_3\epsilon_{c1}^2\epsilon_{c2}^2\epsilon_{c3}^2+\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{c3}^2 \\
& +\eta_2^2\epsilon_{r1}^2\epsilon_{r2}^2\epsilon_{r3}^2+4\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}^2+\eta_2^2\epsilon_{c1}^2\epsilon_{r2}^2\epsilon_{r3}^2+16\eta_2^2\epsilon_{r1}\epsilon_{r2}^2\epsilon_{r3}-2\eta_2^2\epsilon_{r1}^2\epsilon_{r2}\epsilon_{r3}^2 \\
& -8\eta_2^2\epsilon_{r1}\epsilon_{r2}\epsilon_{r3}^2+4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}^2\epsilon_{r2}+\epsilon_{c2}^2\eta_2^2\epsilon_{c1}^2\epsilon_{c3}^2+4\epsilon_{c3}^2\eta_3^2\epsilon_{r1}\epsilon_{c2}^2-8\eta_2^2\epsilon_{r1}\epsilon_{r2}\epsilon_{c3}^2
\end{aligned}$$

$$\begin{aligned}
& + 4 \epsilon_{cl}^2 \epsilon_{c2}^2 + 16 \epsilon_{r1} \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{c3}^2 + 16 \epsilon_{r2} \epsilon_{r3}^2 + 64 \epsilon_{r2} \epsilon_{r3} + 16 \epsilon_{r2} \\
& \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1}^2 \epsilon_{r3} + 4 \epsilon_{r1}^2 \epsilon_{c3}^2 + 4 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{cl}^2 \epsilon_{r3}^2 \\
& + 16 \epsilon_{cl}^2 \epsilon_{r3} + 4 \epsilon_{cl}^2 \epsilon_{c3}^2 + 4 \epsilon_{c2}^2 \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \epsilon_{r3} + 4 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1} + 64 \eta_1 \epsilon_{r2} \\
& + 64 \eta_1 \epsilon_{r3} - 32 \eta_1 \epsilon_{r1}^2 + 16 \eta_1 \epsilon_{r2}^2 + 16 \eta_1 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{cl}^2 + 16 \eta_1 \epsilon_{c2}^2 + 16 \eta_1 \epsilon_{c3}^2 \\
& + 64 \eta_2 \epsilon_{r1} - 32 \eta_2 \epsilon_{r2} + 64 \eta_2 \epsilon_{r3} + 16 \eta_2 \epsilon_{r1}^2 - 32 \eta_2 \epsilon_{r2}^2 + 16 \eta_2 \epsilon_{r3}^2 + 16 \eta_2 \epsilon_{cl}^2 \\
& - 32 \eta_2 \epsilon_{c2}^2 + 16 \eta_2 \epsilon_{c3}^2 + 64 \eta_3 \epsilon_{r1} + 64 \eta_3 \epsilon_{r2} - 32 \eta_3 \epsilon_{r3} + 16 \eta_3 \epsilon_{r1}^2 + 16 \eta_3 \epsilon_{r2}^2 \\
& - 32 \eta_3 \epsilon_{r3}^2 + 16 \eta_3 \epsilon_{cl}^2 + 16 \eta_3 \epsilon_{c2}^2 - 32 \eta_3 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 + 16 \eta_2^2 \epsilon_{r2}^2 + 16 \eta_{c2}^2 \eta_2^2 \\
& + 32 \eta_2 \eta_1 + 32 \eta_3 \eta_1 + 32 \eta_2 \eta_3 + 16 \eta_3^2 \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + 4 \eta_2^2 \epsilon_{r2}^2 \epsilon_{r1} \epsilon_{c3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{cl}^2 \epsilon_{c2}^2 + \epsilon_{cl}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \\
& \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 4 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{cl}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 \epsilon_{r3} + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \eta_2 + 4 \eta_3^2 \epsilon_{r3}^2 \epsilon_{r1}^2 \epsilon_{r2} + \\
& \eta_3^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + \eta_3^2 \epsilon_{r2}^2 \epsilon_{r3}^2 \epsilon_{r1}^2 - 4 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \eta_1 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_1 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{cl}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 2 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1} \\
& \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} \eta_1 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \eta_2 + 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \\
& \epsilon_{r3}^2 \eta_3 + 2 \eta_2 \epsilon_{r3}^2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r2} + 16 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \\
& \epsilon_{c2}^2 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r3}^2 - 32 \eta_1^2 \epsilon_{r1} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c3}^2 + 8 \eta_2 \epsilon_{r3}^2 \eta_1 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1} + 8 \eta_3 \\
& \epsilon_{r2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{cl}^2 - 16 \eta_3 \epsilon_{r1}^2 \eta_1 + 4 \epsilon_{cl}^2 \eta_1^2 \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \\
& \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 + \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 + \\
& \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 64 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} + \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} + \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \\
& + 4 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{r3} + 16 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 + 4 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} + 4 \epsilon_{r1} \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 + \epsilon_{cl}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{cl}^2 \epsilon_{r2} \epsilon_{c3}^2 + \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \epsilon_{cl}^2 \\
& \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{cl}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2
\end{aligned}$$

[illegible]

$$\begin{aligned}
& + 8 \eta_3 \epsilon_{r2}^2 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{c3}^2 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 \epsilon_{c3}^2 \\
& + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{r3}^2 + 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \eta_1 \epsilon_{c3}^2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 \epsilon_{r3}^2 + 8 \eta_3 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} \eta_2 \\
& + 8 \eta_3 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 2 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \eta_2 - 16 \eta_3 \epsilon_{r1} \eta_1 - 16 \eta_2 \epsilon_{c1}^2 \eta_1 + 8 \eta_2 \epsilon_{c3}^2 \eta_1 \\
& - 16 \eta_3 \epsilon_{c1}^2 \eta_1 + 8 \eta_3 \epsilon_{c2}^2 \eta_1 - 16 \eta_2 \epsilon_{r1} \eta_1 + 4 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r1}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 + 16 \epsilon_{c3}^2 \\
& \eta_3^2 \epsilon_{r1} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 + \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3}^2 + 4 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \eta_1^2 \epsilon_{r2}^2 \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& + \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3}^2 + 16 \eta_1^2 \epsilon_{r2} \epsilon_{r3} + 4 \eta_1^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 + 16 \epsilon_{c2}^2 \eta_2^2 \epsilon_{r3} + 16 \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{r3} - 16 \eta_2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \eta_3 - 16 \eta_2 \epsilon_{r3}^2 \eta_3 + 16 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2} + 4 \\
& \eta_1^2 \epsilon_{r1}^2 \epsilon_{c2}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c2}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 16 \eta_1 \epsilon_{r2}^2 \eta_2 - 16 \eta_1 \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{c2}^2 \eta_2 \\
& - 16 \eta_1 \epsilon_{c3}^2 \eta_3 + 16 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r2} + 32 \eta_2 \epsilon_{r3} \eta_1 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{r1}^2 \\
& \epsilon_{c2}^2 + 4 \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 4 \eta_2 \epsilon_{r1} \epsilon_{c3}^2 \eta_1 - 4 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_1 - 16 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_1 - 32 \\
& \eta_3^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \epsilon_{c3}^2 \eta_3 \\
& + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 \epsilon_{r3} - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3}^2 \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r3} \eta_3 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{c3}^2 \eta_3 - 4 \eta_2 \epsilon_{c1}^2 \\
& \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_1 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 + 8 \eta_2 \epsilon_{r2} \epsilon_{r3} \eta_3 + 8 \eta_2 \epsilon_{r2} \\
& \epsilon_{c3}^2 \eta_3 - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 32 \\
& \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 8 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1^2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_1^2 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3}^2 \\
& - 8 \eta_1^2 \epsilon_{r1} \epsilon_{r2} \epsilon_{c3}^2 - 4 \eta_3 \epsilon_{r2}^2 \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{c1}^2 \epsilon_{r3} \eta_1 - 4 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_1 - 16 \eta_3 \\
& \epsilon_{c1}^2 \epsilon_{r2} \eta_1 - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{c2}^2 \eta_1 + 8 \eta_3 \epsilon_{r1} \epsilon_{r3} \eta_1 - 16 \eta_3 \epsilon_{r2} \epsilon_{r3} \eta_1 + 8 \eta_3 \epsilon_{r1}^2 \epsilon_{r3} \eta_1 \\
& - 4 \eta_3 \epsilon_{c1}^2 \epsilon_{r2}^2 \eta_1 + \epsilon_{c2}^2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 - 2 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \epsilon_{r3}^2 \\
& - 16 \eta_1 \epsilon_{r2} \epsilon_{r3}^2 \eta_3 - 16 \eta_1 \epsilon_{r2} \epsilon_{c3}^2 \eta_3 + 8 \eta_3 \epsilon_{c2}^2 \epsilon_{r3} \eta_2 + 8 \eta_1 \epsilon_{r1} \epsilon_{r3}^2 \eta_3 + 8 \eta_1 \epsilon_{r1} \\
& \epsilon_{c3}^2 \eta_3 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 + 8 \eta_2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 + \epsilon_{c3}^2 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \epsilon_{c2}^2 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 2 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{c2}^2 \epsilon_{r3} + \epsilon_{c1}^2 \\
& \eta_1^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 4 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \eta_3 - 16 \eta_2 \epsilon_{r1} \epsilon_{r2} \eta_3 - 2 \eta_3 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 32 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} \\
& + 16 \eta_3^2 \epsilon_{r1} \epsilon_{r2} + 4 \eta_3^2 \epsilon_{r1} \epsilon_{c2}^2 + \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2}^2 + 4 \eta_3^2 \epsilon_{c1}^2 \epsilon_{r2} + \eta_3^2 \epsilon_{c1}^2 \epsilon_{c2}^2 - 32 \eta_3^2 \epsilon_{r1} \epsilon_{r3} \\
& - 32 \eta_3^2 \epsilon_{r2} \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r3} + \eta_2^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3}^2 + 4 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r3} + \\
& \eta_2^2 \epsilon_{c1}^2 \epsilon_{c3}^2 + \eta_3^2 \epsilon_{r1}^2 \epsilon_{r2}^2 - 8 \eta_2^2 \epsilon_{r1}^2 \epsilon_{r2} - 32 \eta_2^2 \epsilon_{r1} \epsilon_{r2} - 8 \eta_2^2 \epsilon_{c1}^2 \epsilon_{r2} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{r3}^2 + 16 \\
& \eta_2^2 \epsilon_{r1} \epsilon_{r3} + 4 \eta_2^2 \epsilon_{r1} \epsilon_{c3}^2 - 8 \eta_2^2 \epsilon_{r2} \epsilon_{r3}^2 - 32 \eta_2^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2^2 \epsilon_{r2} \epsilon_{c3}^2 + 16 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2} \\
& - 16 \eta_3 \epsilon_{r2}^2 \eta_2 - 16 \eta_3 \epsilon_{c2}^2 \eta_2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{r3}^2 + 4 \epsilon_{c3}^2 \eta_3^2 \epsilon_{r1}^2 + 4 \eta_1^2 \epsilon_{r1}^2 \epsilon_{c3}^2 + 16 \eta_1^2
\end{aligned}$$

$$\begin{aligned}
& \epsilon_{r1}^2 \epsilon_{r3} + 16 \eta_2^2 \epsilon_{r1} \epsilon_{r2}^2 + 4 \epsilon_{c1}^2 \eta_1^2 \epsilon_{r2}^2 + 16 \eta_2^2 \epsilon_{r3} \epsilon_{r2}^2 + 32 \eta_2 \epsilon_{r1} \eta_3 - 8 \eta_3^2 \epsilon_{c2}^2 \epsilon_{r3} \\
& + 8 \eta_2 \epsilon_{c1}^2 \eta_3 - 16 \eta_2 \epsilon_{c3}^2 \eta_3 + 32 \eta_3 \epsilon_{r2} \eta_1 - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 \\
& - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \\
& \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3} - 2 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{r2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3}^2 - 2 \eta_1 \epsilon_{r1} \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{r3} \\
& - 8 \eta_1 \epsilon_{r1} \epsilon_{r2} \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{r1}^2 \epsilon_{r2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{r2}^2 \epsilon_{c3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \\
& \epsilon_{r3}^2 - 32 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 2 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3}^2 - 8 \eta_1 \epsilon_{c1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 2 \eta_1 \epsilon_{c1}^2 \\
& \epsilon_{c2}^2 \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{r3} - 8 \eta_2 \epsilon_{r1} \epsilon_{c2}^2 \epsilon_{r3}^2 - 2 \eta_2 \epsilon_{r1}^2 \epsilon_{r2} \epsilon_{c3}^2 - 8 \eta_2 \epsilon_{r1}^2 \epsilon_{c2}^2 \epsilon_{r3} - 8 \eta_3^2 \\
& \epsilon_{r1}^2 \epsilon_{r3})
\end{aligned}$$

to string

$$\begin{aligned}
& "9*(16*\epsilon[c1]*\eta[1]C\ 16*\epsilon[c2]*\eta[2]C\ 16*\epsilon[c3]*\eta[3]C\ 16* \quad (2.12) \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]C\ 16*\epsilon[c1]*\eta[1]*\epsilon[r3]C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]^2C\ 4*\epsilon[c1]*\eta[1]*\epsilon[r3]^2C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[c2]^2C\ 4*\epsilon[c1]*\eta[1]*\epsilon[c3]^2C\ 16* \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]C\ 16*\epsilon[c2]*\eta[2]*\epsilon[r3]C\ 4* \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r3]^2C\ 4* \\
& \epsilon[c2]*\eta[2]*\epsilon[c1]^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[c3]^2C\ 16* \\
& \epsilon[c3]*\eta[3]*\epsilon[r1]C\ 16*\epsilon[c3]*\eta[3]*\epsilon[r2]C\ 4* \\
& \epsilon[c3]*\eta[3]*\epsilon[r1]^2C\ 4*\epsilon[c3]*\eta[3]*\epsilon[r2]^2C\ 4* \\
& \epsilon[c3]*\eta[3]*\epsilon[c1]^2C\ 4*\epsilon[c3]*\eta[3]*\epsilon[c2]^2C\ 4* \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]*\epsilon[r3]^2C\ 16*\epsilon[c1]*\eta[1]* \\
& \epsilon[r2]*\epsilon[r3]C\ 4*\epsilon[c1]*\eta[1]*\epsilon[r2]*\epsilon[r3]^2C \\
& \epsilon[c1]*\eta[1]*\epsilon[r2]^2*\epsilon[r3]^2C\ 4*\epsilon[c1]*\eta[1]* \\
& \epsilon[r2]^2*\epsilon[r3]C\ \epsilon[c1]*\eta[1]*\epsilon[r2]^2*\epsilon[c3] \\
& ^2C\ \epsilon[c1]*\eta[1]*\epsilon[c2]^2*\epsilon[r3]^2C\ 4*\epsilon[c1]*\eta[1]* \\
& \epsilon[c2]^2*\epsilon[r3]C\ \epsilon[c1]*\eta[1]*\epsilon[c2]^2*\epsilon[c3] \\
& ^2C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r1]*\epsilon[r3]^2C\ 16*\epsilon[c2]*\eta[2]* \\
& \epsilon[r1]*\epsilon[r3]C\ 4*\epsilon[c2]*\eta[2]*\epsilon[r1]*\epsilon[r3]^2C \\
& \epsilon[c2]*\eta[2]*\epsilon[r1]^2*\epsilon[r3]^2C\ 4*\epsilon[c2]*\eta[2]* \\
& \epsilon[r1]^2*\epsilon[r3]C\ \epsilon[c2]*\eta[2]*\epsilon[r1]^2*\epsilon[c3] \\
& ^2C\ \epsilon[c2]*\eta[2]*\epsilon[c1]^2*\epsilon[r3]^2C\ 4*\epsilon[c2]*\eta[2]* \\
& \epsilon[c1]^2*\epsilon[r3]C\ \epsilon[c2]*\eta[2]*\epsilon[c1]^2*\epsilon[c3] \\
& ^2C\ \epsilon[c3]*\eta[3]*\epsilon[r1]^2*\epsilon[r2]^2C\ 4*\epsilon[c3]*\eta[3]* \\
& \epsilon[r1]^2*\epsilon[r2]C\ \epsilon[c3]*\eta[3]*\epsilon[r1]^2*\epsilon[c2]
\end{aligned}$$

$$\begin{aligned}
& ^2C_4 \epsilon[c_3] \eta[3] \epsilon[r_1] \epsilon[r_2] ^2C_{16} \epsilon[c_3] \eta[3] \epsilon[r_1] \epsilon[r_2] C_4 \epsilon[c_3] \eta[3] \epsilon[r_1] \epsilon[c_2] ^2C \\
& \epsilon[c_3] \eta[3] \epsilon[c_1] ^2 \epsilon[r_2] ^2C_4 \epsilon[c_3] \eta[3] \epsilon[c_1] ^2 \epsilon[r_2] C \epsilon[c_3] \eta[3] \epsilon[c_1] ^2 \epsilon[c_2] ^2) \\
& / (64C_4 \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[c_3] ^2C_{64} \eta[1] C_{64} \eta[2] C_{64} \eta[3] C_{64} \epsilon[r_1] C_{64} \epsilon[r_2] C_{64} \epsilon[r_3] C_{16} \epsilon[c_1] \\
& ^2C_{16} \epsilon[c_2] ^2C_{16} \epsilon[c_3] ^2-8 \eta[3] ^2 \epsilon[c_1] ^2 \epsilon[r_3]-8 \eta[3] ^2 \epsilon[r_2] ^2 \epsilon[r_3] C_{18} \epsilon[c_1] \eta[1] \epsilon[c_3] \eta[3] \epsilon[r_2] ^2C_4 \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[r_3] ^2C_4 \epsilon[c_3] ^2 \eta[3] ^2 \epsilon[r_2] ^2C_4 \eta[2] ^2 \epsilon[c_1] ^2 \epsilon[r_2] ^2-16 \eta[2] \epsilon[r_2] \eta[1] C_8 \eta[3] \epsilon[r_1] ^2 \eta[1] \epsilon[r_3] ^2-16 \eta[3] \epsilon[r_1] \epsilon[r_2] ^2 \eta[2]-16 \eta[3] \epsilon[r_1] \epsilon[c_2] ^2 \eta[2]-4 \eta[3] \epsilon[c_1] ^2 \epsilon[r_2] ^2 \eta[2]-8 \eta[3] ^2 \epsilon[c_1] ^2 \epsilon[r_2] \epsilon[r_3]-2 \eta[3] ^2 \epsilon[c_1] ^2 \epsilon[c_2] ^2 \epsilon[r_3]-4 \eta[1] \epsilon[r_2] ^2 \epsilon[r_3] ^2 \eta[2]-4 \eta[1] \epsilon[r_2] ^2 \epsilon[r_3] ^2 \eta[3]-16 \eta[1] \epsilon[r_2] ^2 \epsilon[r_3] \eta[2] C_8 \eta[3] \epsilon[r_2] ^2 \epsilon[r_3] \eta[2] C_4 \epsilon[c_3] ^2 \eta[3] ^2 \epsilon[r_1] \epsilon[r_2] ^2C_{16} \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[r_1] \epsilon[r_3] C_4 \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[r_1] ^2 \epsilon[r_3] C_{16} \epsilon[c_3] ^2 \eta[3] ^2 \epsilon[r_1] \epsilon[r_2] C_8 \eta[2] \eta[1] \epsilon[c_1] ^2 \epsilon[r_2] ^2C_8 \eta[2] \eta[1] \epsilon[c_1] ^2 \epsilon[c_2] ^2-4 \eta[2] \epsilon[c_1] ^2 \epsilon[r_2] \eta[3]-16 \eta[2] \epsilon[r_1] \epsilon[r_3] ^2 \eta[3]-16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[3]-16 \eta[2] \epsilon[r_1] \epsilon[c_3] ^2 \eta[3] C \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[r_1] ^2 \epsilon[c_3] ^2C_8 \eta[2] \epsilon[c_1] ^2 \epsilon[r_2] \eta[1]-4 \eta[2] \epsilon[r_1] \epsilon[r_3] ^2 \eta[1]-16 \eta[2] \epsilon[r_1] \epsilon[r_3] \eta[1] C_4 \epsilon[c_3] ^2 \eta[3] ^2 \epsilon[c_1] ^2C_8 \eta[3] \epsilon[c_2] ^2 \eta[2] \epsilon[r_3] ^2-4 \eta[2] \epsilon[c_1] ^2 \epsilon[r_3] ^2 \eta[1]-16 \eta[2] \epsilon[c_1] ^2 \epsilon[r_3] \eta[1]-4 \eta[2] \epsilon[c_1] ^2 \epsilon[c_3] ^2 \eta[1] C_8 \eta[3] \epsilon[r_2] ^2 \eta[2] \epsilon[c_3] ^2C_8 \eta[3] \epsilon[r_2] ^2 \eta[2] \epsilon[r_3] ^2-4 \eta[1] \epsilon[r_2] ^2 \epsilon[c_3] ^2 \eta[2]-4 \eta[1] \epsilon[r_2] ^2 \epsilon[c_3] ^2 \eta[3]-4 \eta[1] \epsilon[c_2] ^2 \epsilon[r_3] ^2 \eta[2]-4 \eta[1] \epsilon[c_2] ^2 \epsilon[r_3] \eta[2]-4 \eta[1] \epsilon[c_2] ^2 \epsilon[c_3] ^2 \eta[3]-4 \eta[3] \epsilon[c_1] ^2 \epsilon[c_2] ^2 \eta[2]-4 \eta[2] \epsilon[r_1] ^2 \epsilon[r_3] ^2 \eta[1]-16 \eta[2] \epsilon[r_1] ^2 \epsilon[r_3] \eta[1]-4 \eta[2] \epsilon[r_1] ^2 \epsilon[c_3] ^2 \eta[1] C \epsilon[c_1] ^2 \eta[1] ^2 \epsilon[r_2] ^2 \epsilon[r_3] ^2C \epsilon[c_2] ^2 \eta[2] ^2 \epsilon[r_1] ^2
\end{aligned}$$

epsilon[r3]^2C epsilon[c3]^2*eta[3]^2*epsilon[r1]^2*epsilon[r2]^2C 8*eta[3]*eta[1]*epsilon[c1]^2*epsilon[c3]^2C 16*eta[3]^2*epsilon[r1]*epsilon[r2]*epsilon[r3]^2C 4*eta[3]^2*epsilon[c1]^2*epsilon[r2]*epsilon[r3]^2C eta[3]^2*epsilon[c1]^2*epsilon[c2]^2*epsilon[r3]^2-4*eta[3]*epsilon[r1]^2*epsilon[r2]^2*eta[2]-4*eta[3]*epsilon[r1]^2*epsilon[c2]^2*eta[2]-2*eta[3]^2*epsilon[c1]^2*epsilon[r2]^2*epsilon[r3]-2*eta[3]^2*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]-8*eta[3]^2*epsilon[r1]^2*epsilon[r2]*epsilon[r3]C 8*eta[3]*epsilon[r1]^2*eta[1]*epsilon[c3]^2C 4*eta[2]^2*epsilon[c3]^2*epsilon[r2]^2C 4*eta[1]^2*epsilon[r1]^2*epsilon[r2]*epsilon[c3]^2-16*eta[3]*epsilon[r1]^2*epsilon[r2]*eta[1]-4*eta[3]*epsilon[r1]^2*epsilon[c2]^2*eta[1]-4*eta[3]*epsilon[r1]*epsilon[r2]^2*eta[1]-16*eta[3]*epsilon[r1]*epsilon[r2]*eta[1]C 72*epsilon[c1]*eta[1]*epsilon[c3]*eta[3]C 72*epsilon[c2]*eta[2]*epsilon[c3]*eta[3]-4*eta[3]*epsilon[r1]^2*epsilon[r2]^2*eta[1]C 72*epsilon[c1]*eta[1]*epsilon[c2]*eta[2]-2*eta[3]^2*epsilon[r1]^2*epsilon[c2]^2*epsilon[r3]C 4*eta[3]^2*epsilon[r1]*epsilon[r2]^2*epsilon[r3]^2C 8*eta[3]*epsilon[c2]^2*eta[2]*epsilon[c3]^2C 16*epsilon[c1]^2*eta[1]^2C 16*eta[3]^2*epsilon[r1]C 16*eta[3]^2*epsilon[r2]-32*eta[3]^2*epsilon[r3]C 4*eta[3]^2*epsilon[r1]^2C 4*eta[3]^2*epsilon[r2]^2C 4*eta[3]^2*epsilon[c1]^2C 4*eta[3]^2*epsilon[c2]^2C 16*eta[2]^2*epsilon[r1]-32*eta[2]^2*epsilon[r2]C 16*eta[2]^2*epsilon[r3]C 4*eta[2]^2*epsilon[r1]^2C 4*eta[2]^2*epsilon[r3]^2C 4*eta[2]^2*epsilon[c1]^2C 4*eta[2]^2*epsilon[c3]^2C 16*epsilon[c3]^2*eta[3]^2C 4*epsilon[r1]^2*epsilon[r2]^2C 16*epsilon[r1]^2*epsilon[r2]C 4*epsilon[r1]^2*epsilon[c2]^2C 16*epsilon[r1]*epsilon[r2]^2C 64*epsilon[r1]*epsilon[r2]C 16*epsilon[r1]*epsilon[c2]^2C 4*epsilon[c1]^2*epsilon[r2]^2C 16*epsilon[c1]^2*epsilon[r2]C 16*epsilon[r1]^2C 16*epsilon[r2]^2C 16*epsilon[r3]^2C 16*eta[1]^2C 16*eta[2]^2C 16*eta[3]^2-2*eta[2]*epsilon[r1]^2*epsilon[r2]^2*epsilon[c3]^2-8*eta[2]*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]^2-2*eta[2]*epsilon[r1]^2*epsilon[c2]^2*epsilon[c3]^2-8*eta[2]*epsilon[r1]*epsilon[r2]*epsilon[c3]^2-2*eta[2]*epsilon[r1]^2*epsilon[c2]^2*epsilon[r3]^2-32*eta[2]*epsilon[r1]*epsilon[r2]*epsilon[r3]-2*eta[2]*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]^2-32*eta[2]*epsilon[r1]*epsilon[r2]^2*epsilon[r3]^2-32*eta[2]*epsilon[c1]^2*epsilon[r2]^2*epsilon[r3]^2-32*eta[2]*epsilon[r1]*epsilon[c2]^2*epsilon[r3]-2*eta[2]*epsilon[r1]^2*epsilon[r2]*epsilon[r3]^2-8*eta[2]*epsilon[r1]*epsilon[r2]^2*epsilon[c3]^2-8*eta[2]*epsilon[c1]^2*epsilon[r2]^2*epsilon[r3]-8*eta[2]*epsilon[r1]*epsilon[r2]*epsilon[r3]^2-8*eta[2]*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]-8*eta[2]*epsilon[r1]*epsilon[c2]^2*epsilon[c3]^2-2*eta

[illegible]

$$\begin{aligned}
& [2]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 C^4 \epsilon[c_1]^2 \epsilon[c_2] \\
& ^2 C^{16} \epsilon[r_1] \epsilon[r_3]^2 C^{64} \epsilon[r_1] \epsilon[r_3] C^{16} \epsilon[r_1] \epsilon[c_3]^2 C^{16} \epsilon[r_2] \epsilon[r_3]^2 C^{64} \epsilon[r_2] \epsilon[r_3] C^{16} \epsilon[r_2] \epsilon[c_3]^2 C^4 \epsilon[r_1]^2 \epsilon[r_3]^2 C^{16} \\
& \epsilon[r_1]^2 \epsilon[r_3] C^4 \epsilon[r_1]^2 \epsilon[c_3]^2 C^4 \epsilon[r_2] \\
& ^2 \epsilon[r_3]^2 C^{16} \epsilon[r_2]^2 \epsilon[r_3] C^4 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \epsilon[c_1]^2 \epsilon[r_3]^2 C^{16} \epsilon[c_1]^2 \epsilon[r_3] C^4 \\
& \epsilon[c_1]^2 \epsilon[c_3]^2 C^4 \epsilon[c_2]^2 \epsilon[r_3]^2 C^{16} \epsilon[c_2]^2 \epsilon[r_3] C^4 \epsilon[c_2]^2 \epsilon[c_3]^2 - 32 \eta[1] \epsilon[r_1] \\
& C^{64} \eta[1] \epsilon[r_2] C^{64} \eta[1] \epsilon[r_3] - 32 \eta[1] \epsilon[r_1] \\
& ^2 C^{16} \eta[1] \epsilon[r_2]^2 C^{16} \eta[1] \epsilon[r_3]^2 - 32 \eta[1] \epsilon[r_1] \\
& [c_1]^2 C^{16} \eta[1] \epsilon[c_2]^2 C^{16} \eta[1] \epsilon[c_3]^2 C^{64} \eta[2] * \\
& \epsilon[r_1] - 32 \eta[2] \epsilon[r_2] C^{64} \eta[2] \epsilon[r_3] C^{16} \eta[2] * \\
& \epsilon[r_1]^2 - 32 \eta[2] \epsilon[r_2]^2 C^{16} \eta[2] \epsilon[r_3]^2 C^{16} \eta[2] \\
& * \epsilon[c_1]^2 - 32 \eta[2] \epsilon[c_2]^2 C^{16} \eta[2] \epsilon[c_3]^2 C^{64} \eta[3] * \\
& \epsilon[r_1] C^{64} \eta[3] \epsilon[r_2] - 32 \eta[3] \epsilon[r_3] C^{16} \eta[3] * \\
& \epsilon[r_1]^2 C^{16} \eta[3] \epsilon[r_2]^2 - 32 \eta[3] \epsilon[r_3]^2 C^{16} \eta[3] \\
& * \epsilon[c_1]^2 C^{16} \eta[3] \epsilon[c_2]^2 - 32 \eta[3] \epsilon[c_3]^2 C^{16} \eta[1]^2 \epsilon[r_1]^2 C^{16} \eta[2]^2 \epsilon[r_2]^2 C^{16} \epsilon[c_2]^2 \eta[2] \\
& ^2 C^{32} \eta[2] \eta[1] C^{32} \eta[3] \eta[1] C^{32} \eta[2] \eta[3] C^{16} \eta[3]^2 * \\
& \epsilon[r_3]^2 - 32 \eta[2]^2 \epsilon[r_1] \epsilon[r_2] \epsilon[r_3] C^4 \eta[2] \\
& ^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C^4 \eta[2]^2 \epsilon[r_2]^2 * \\
& \epsilon[r_1] \epsilon[c_3]^2 C \epsilon[c_3]^2 \eta[3]^2 \epsilon[c_1]^2 \epsilon[c_2]^2 C \epsilon[c_1]^2 \eta[1]^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C \eta[3]^2 * \\
& \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C \epsilon[c_3]^2 \eta[3]^2 * \\
& \epsilon[r_1]^2 \epsilon[c_2]^2 C^4 \epsilon[c_2]^2 \eta[2]^2 \epsilon[r_1] \epsilon[r_3]^2 C^4 \epsilon[c_1]^2 \eta[1]^2 \epsilon[r_2] \epsilon[c_3]^2 C \epsilon[c_1] \\
& ^2 \eta[1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \eta[2]^2 \epsilon[c_1]^2 * \\
& \epsilon[r_2]^2 \epsilon[r_3] C \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^4 \eta[3]^2 \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C^4 \epsilon[c_2] \\
& ^2 \eta[2]^2 \epsilon[c_1]^2 \epsilon[r_3] C^8 \eta[1] \epsilon[r_1] \epsilon[r_2] \\
& ^2 \eta[2] C^8 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 \eta[2] C^4 \eta[3]^2 \epsilon[r_3]^2 \epsilon[r_1]^2 \epsilon[r_2] C \eta[3]^2 \epsilon[r_1]^2 \epsilon[c_2]^2 * \\
& \epsilon[r_3]^2 C \eta[3]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 \epsilon[r_1]^2 - 4 \eta[3] * \epsilon[r_1] \epsilon[c_2]^2 \eta[1] C^4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2] \\
& ^2 \epsilon[r_3] C \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3]^2 C \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[c_3]^2 C^{16} \eta[1]^2 \epsilon[r_1] \\
& ^2 \epsilon[r_2] \epsilon[r_3] C^4 \eta[1]^2 \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]
\end{aligned}$$

[illegible]

$$\begin{aligned}
& \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 16 \epsilon[r_1] \epsilon[r_2] * \\
& \epsilon[r_3]^2 C 4 \epsilon[r_1]^2 \epsilon[r_2]^2 \epsilon[r_3] C 4 \epsilon[r_1] * \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 C \epsilon[c_1]^2 \epsilon[r_2]^2 \epsilon[c_3] \\
& ^2 C 4 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[r_3]^2 C 16 \epsilon[c_1]^2 \epsilon[r_2] \\
& \epsilon[r_3] C 4 \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 C \epsilon[c_1] \\
& ^2 \epsilon[c_2]^2 \epsilon[r_3]^2 C 4 \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[r_3] C \epsilon[c_1]^2 \epsilon[c_2]^2 \epsilon[c_3]^2 - 8 \eta[1] \epsilon[r_1]^2 * \\
& \epsilon[r_2]^2 - 32 \eta[1] \epsilon[r_1]^2 \epsilon[r_2] - 8 \eta[1] \epsilon[r_1]^2 * \\
& \epsilon[c_2]^2 - 8 \eta[1] \epsilon[r_1] \epsilon[r_2]^2 - 32 \eta[1] \epsilon[r_1] * \\
& \epsilon[r_2] - 8 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 - 8 \eta[1] \epsilon[c_1]^2 * \\
& \epsilon[r_2]^2 - 32 \eta[1] \epsilon[c_1]^2 \epsilon[r_2] - 8 \eta[1] \epsilon[c_1]^2 * \\
& \epsilon[c_2]^2 - 8 \eta[1] \epsilon[r_1] \epsilon[r_3]^2 C 2 \eta[2] \epsilon[r_3]^2 * \\
& \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 2 \eta[2] \epsilon[r_3]^2 \eta[1] \epsilon[r_3]^2 * \\
& \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C 2 \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[r_1]^2 \epsilon[r_3]^2 * \\
& \eta[1] \epsilon[c_2]^2 C 2 \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 2 \eta[2] \\
& \eta[2] \epsilon[c_3]^2 \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C 2 \eta[1] \epsilon[r_1] \\
& * \epsilon[c_2]^2 \epsilon[r_3]^2 \eta[2] C 2 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 * \\
& \epsilon[r_3]^2 \eta[3] C 8 \eta[2] \epsilon[r_3] \eta[1] \epsilon[r_1]^2 \epsilon[c_2] \\
& ^2 C 8 \eta[2] \epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[r_2]^2 C 8 \eta[2] * \\
& \epsilon[r_3] \eta[1] \epsilon[c_1]^2 \epsilon[c_2]^2 C 2 \eta[1] \epsilon[r_1] * \\
& \epsilon[c_2]^2 \epsilon[c_3]^2 \eta[2] C 2 \eta[1] \epsilon[r_1] \epsilon[c_2]^2 * \\
& \epsilon[c_3]^2 \eta[3] C 2 \eta[3] \epsilon[r_1]^2 \epsilon[c_2]^2 \epsilon[r_3] \eta[1] C 8 \eta[3] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C 2 \eta[2] * \\
& \epsilon[c_1]^2 \epsilon[r_2] \epsilon[c_3]^2 \eta[1] C 2 \eta[2] \epsilon[c_1]^2 * \\
& \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C 8 \eta[2] \epsilon[c_1]^2 \epsilon[r_2] * \\
& \epsilon[r_3] \eta[1] C 2 \eta[2] \epsilon[r_1] \epsilon[r_2] \epsilon[r_3]^2 \eta[1] \\
& C 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3]^2 \eta[1] C 2 \eta[2] * \\
& \epsilon[r_1] \epsilon[r_2] \epsilon[c_3]^2 \eta[1] C 8 \eta[2] \epsilon[r_1] \epsilon[r_2] \\
& \epsilon[r_3] \eta[1] C 2 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[c_3]^2 * \\
& \eta[1] C 8 \eta[2] \epsilon[r_1]^2 \epsilon[r_2] \epsilon[r_3] \eta[1] C 8 \eta[3] \eta[2] \\
& \epsilon[r_1] \epsilon[c_2]^2 \epsilon[r_3]^2 C 2 \eta[3] \eta[2] \epsilon[r_1] \\
& ^2 \epsilon[r_2]^2 \epsilon[c_3]^2 - 4 \eta[2] \epsilon[r_2] \epsilon[c_3]^2 \eta[1] \\
& - 32 \eta[1]^2 \epsilon[r_1] C 16 \eta[1]^2 \epsilon[r_2] C 16 \eta[1]^2 \epsilon[r_3] C 4 \eta[1]^2 \epsilon[r_2]^2 C 4 \eta[1]^2 \epsilon[r_3]^2 C 4 \eta[2]^2 \epsilon[r_3]^2 * \\
& \epsilon[r_2]^2 - 32 \eta[1] \epsilon[r_1] \epsilon[r_3] - 8 \eta[1] \epsilon[r_1] * \\
& \epsilon[c_3]^2 C 16 \eta[1] \epsilon[r_2] \epsilon[r_3]^2 C 64 \eta[1] \epsilon[r_2] * \\
& \epsilon[r_3] C 16 \eta[1] \epsilon[r_2] \epsilon[c_3]^2 - 8 \eta[1] \epsilon[r_1]^2 *
\end{aligned}$$

$$\begin{aligned}
& \epsilon[r_3]^2 - 32\eta[1]\epsilon[r_1]^2\epsilon[r_3] - 8\eta[1]\epsilon[r_1]^2\epsilon[c_3]^2 C_4\eta[1]\epsilon[r_2]^2\epsilon[r_3]^2 C_{16}\eta[1]\epsilon[r_2]^2\epsilon[r_3] C_4\eta[1]\epsilon[r_2]^2\epsilon[c_3]^2 - 8\eta[1]\epsilon[c_1]^2\epsilon[r_3]^2 - 32\eta[1]\epsilon[c_1]^2\epsilon[r_3] - 8\eta[1]\epsilon[c_1]^2\epsilon[c_3]^2 C_4\eta[1]\epsilon[c_2]^2\epsilon[r_3]^2 C_{16}\eta[1]\epsilon[c_2]^2\epsilon[r_3] C_4\eta[1]\epsilon[c_2]^2\epsilon[c_3]^2 - 8\eta[2]\epsilon[r_1]^2\epsilon[r_2]^2 - 8\eta[2]\epsilon[r_1]^2\epsilon[r_2] - 8\eta[2]\epsilon[r_1]^2\epsilon[c_2]^2 - 32\eta[2]\epsilon[r_1]\epsilon[r_2]^2 - 32\eta[2]\epsilon[r_1]\epsilon[r_2] - 32\eta[2]\epsilon[r_1]\epsilon[c_2]^2 - 8\eta[2]\epsilon[c_1]^2\epsilon[r_2]^2 - 8\eta[2]\epsilon[c_1]^2\epsilon[c_3]^2 C_{16}\eta[2]\epsilon[r_1]\epsilon[r_3]^2 C_{64}\eta[2]\epsilon[r_1]\epsilon[r_3] C_{16}\eta[2]\epsilon[r_1]\epsilon[c_3]^2 - 8\eta[2]\epsilon[r_2]\epsilon[r_3]^2 - 32\eta[2]\epsilon[r_2]\epsilon[r_3] - 8\eta[2]\epsilon[r_2]\epsilon[c_3]^2 C_4\eta[2]\epsilon[r_1]^2\epsilon[r_3]^2 C_{16}\eta[2]\epsilon[r_1]^2\epsilon[r_3] C_4\eta[2]\epsilon[r_1]^2\epsilon[c_3]^2 - 8\eta[2]\epsilon[r_2]^2\epsilon[r_3]^2 - 32\eta[2]\epsilon[r_2]^2\epsilon[r_3] - 8\eta[2]\epsilon[r_2]^2\epsilon[c_3]^2 C_4\eta[2]\epsilon[c_1]^2\epsilon[r_3]^2 C_{16}\eta[2]\epsilon[c_1]^2\epsilon[r_3] C_4\eta[2]\epsilon[c_1]^2\epsilon[c_3]^2 - 8\eta[2]\epsilon[c_2]^2\epsilon[r_3]^2 - 32\eta[2]\epsilon[c_2]^2\epsilon[r_3] - 8\eta[2]\epsilon[c_2]^2\epsilon[c_3]^2 C_4\eta[3]\epsilon[r_1]^2\epsilon[r_2]^2 C_{16}\eta[3]\epsilon[r_1]\epsilon[r_2]^2 C_{64}\eta[3]\epsilon[r_1]\epsilon[r_2] C_{16}\eta[3]\epsilon[r_1]\epsilon[c_2]^2 C_4\eta[3]\epsilon[c_1]^2\epsilon[r_2]^2 C_{16}\eta[3]\epsilon[c_1]^2\epsilon[r_2] C_4\eta[3]\epsilon[c_1]^2\epsilon[c_2]^2 - 32\eta[3]\epsilon[r_1]\epsilon[r_3]^2 - 32\eta[3]\epsilon[r_1]\epsilon[r_3] - 32\eta[3]\epsilon[r_1]\epsilon[c_3]^2 - 32\eta[3]\epsilon[r_2]\epsilon[r_3]^2 - 32\eta[3]\epsilon[r_2]\epsilon[r_3] - 32\eta[3]\epsilon[r_2]\epsilon[c_3]^2 - 8\eta[3]\epsilon[r_1]^2\epsilon[r_3]^2 - 8\eta[3]\epsilon[r_1]^2\epsilon[c_3]^2 - 8\eta[3]\epsilon[r_2]^2\epsilon[r_3]^2 - 8\eta[3]\epsilon[r_2]^2\epsilon[c_3]^2 - 8\eta[3]\epsilon[c_1]^2\epsilon[r_3]^2 - 8\eta[3]\epsilon[c_1]^2\epsilon[c_3]^2 - 8\eta[3]\epsilon[c_2]^2\epsilon[r_3]^2 - 8\eta[3]\epsilon[c_2]^2\epsilon[c_3]^2 C_4\eta[3]^2\epsilon[c_2]^2\epsilon[r_3]^2 C_4\eta[3]^2\epsilon[r_1]^2\epsilon[r_3]^2 C_4\eta[3]^2\epsilon[r_2]^2\epsilon[r_3]^2 C_4\eta[3]^2\epsilon[c_1]^2\epsilon[r_3]^2 C_{16}\eta[3]^2\epsilon[r_1]\epsilon[r_3]^2 C_{16}\eta[3]^2\epsilon[r_2]\epsilon[r_3]^2 - 16\eta[3]\epsilon[r_3]\eta[1] C_8\eta[3]\eta[2]\epsilon[r_1]\epsilon[r_2]^2\epsilon[r_3]^2 C_2\eta[3]
\end{aligned}$$

$$\begin{aligned}
& [3]^* \eta[2]^* \epsilon[r1]^2 * \epsilon[c2]^2 * \epsilon[r3]^2 C^2 \eta[3]^* \eta[2]^* \\
& \epsilon[r1]^2 * \epsilon[r2]^2 * \epsilon[r3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 * \\
& \epsilon[r2]^2 * \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[r2]^* \eta[1]^* \epsilon[c1]^2 * \\
& \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[r2]^2 * \eta[1]^* \epsilon[c1]^2 * \epsilon[c3]^2 \\
& ^2 C^2 \eta[3]^* \epsilon[c1]^2 * \eta[1]^* \epsilon[c2]^2 * \epsilon[r3]^2 C^2 \eta[3]^* \\
& \epsilon[c1]^2 * \eta[1]^* \epsilon[c2]^2 * \epsilon[c3]^2 C^8 \eta[2]^* \epsilon[r1]^* \\
& \epsilon[r2]^* \epsilon[r3]^2 * \eta[3] C^2 \eta[2]^* \epsilon[c1]^2 * \epsilon[r2]^* \epsilon[r3]^* \eta[3] \\
& C^2 \eta[2]^* \epsilon[c1]^2 * \epsilon[r2]^* \epsilon[c3]^2 * \eta[3] C^8 \eta[3]^* \\
& \epsilon[r1]^2 * \eta[1]^* \epsilon[r2]^* \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[c1]^2 * \eta[1]^* \\
& \epsilon[r2]^* \epsilon[r3]^2 C^8 \eta[3]^* \eta[2]^* \epsilon[r1]^* \epsilon[c2]^2 * \\
& \epsilon[c3]^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 * \epsilon[r2]^2 * \epsilon[c3]^2 \\
& ^2 C^2 \eta[3]^* \eta[2]^* \epsilon[c1]^2 * \epsilon[c2]^2 * \epsilon[r3]^2 C^2 \eta[3]^* \\
& \eta[2]^* \epsilon[c1]^2 * \epsilon[c2]^2 * \epsilon[c3]^2 C^8 \eta[3]^* \epsilon[r2]^2 * \\
& \eta[2]^* \epsilon[r1]^* \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[r1]^2 * \epsilon[r2]^2 * \eta[1]^* \\
& \epsilon[r3]^2 C^2 \eta[3]^* \epsilon[r1]^2 * \epsilon[r2]^2 * \eta[1]^* \epsilon[c3]^2 \\
& ^2 C^8 \eta[3]^* \epsilon[r1]^2 * \epsilon[r2]^* \eta[1]^* \epsilon[c3]^2 C^2 \eta[3]^* \\
& \epsilon[r1]^2 * \epsilon[c2]^2 * \eta[1]^* \epsilon[r3]^2 C^2 \eta[3]^* \epsilon[r1]^2 * \\
& \epsilon[c2]^2 * \eta[1]^* \epsilon[c3]^2 C^2 \eta[3]^* \epsilon[c1]^2 * \epsilon[r2]^2 * \\
& \eta[1]^* \epsilon[r3]^2 C^8 \eta[3]^* \epsilon[r1]^* \epsilon[r2]^2 * \epsilon[r3]^* \eta[2] \\
& C^8 \eta[3]^* \epsilon[r1]^* \epsilon[c2]^2 * \epsilon[r3]^* \eta[2] C^2 \eta[3]^* \epsilon[c1]^2 * \\
& \epsilon[r2]^2 * \epsilon[r3]^* \eta[2] - 16^* \eta[3]^* \epsilon[r1]^* \eta[1] - 16^* \\
& \eta[2]^* \epsilon[c1]^2 * \eta[1] C^8 \eta[2]^* \epsilon[c3]^2 * \eta[1] - 16^* \eta[3]^* \\
& \epsilon[c1]^2 * \eta[1] C^8 \eta[3]^* \epsilon[c2]^2 * \eta[1] - 16^* \eta[2]^* \epsilon[r1]^* \\
& \eta[1] C^4 \epsilon[c2]^2 * \eta[2]^2 * \epsilon[r1]^2 C^4 \epsilon[c1]^2 * \eta[1]^2 * \\
& \epsilon[c2]^2 C^16 \epsilon[c3]^2 * \eta[3]^2 * \epsilon[r1] C^4 \epsilon[c1]^2 * \\
& \eta[1]^2 * \epsilon[c3]^2 C^2 \eta[1]^2 * \epsilon[c2]^2 * \epsilon[c3]^2 C^2 \eta[1]^2 * \\
& \epsilon[c2]^2 * \epsilon[r3]^2 C^4 \eta[1]^2 * \epsilon[c2]^2 * \epsilon[r3] C^2 \eta[1]^2 * \\
& \epsilon[r2]^2 * \epsilon[c3]^2 C^4 \eta[1]^2 * \epsilon[r2]^2 * \epsilon[r3] C^2 \eta[1]^2 * \\
& \epsilon[r2]^2 * \epsilon[r3]^2 C^16 \eta[1]^2 * \epsilon[r2]^* \epsilon[r3] \\
& C^4 \eta[1]^2 * \epsilon[r2]^* \epsilon[c3]^2 C^4 \eta[1]^2 * \epsilon[r2]^* \epsilon[r3] \\
& ^2 C^16 \epsilon[c2]^2 * \eta[2]^2 * \epsilon[r3] C^16 \epsilon[c1]^2 * \eta[1]^2 * \\
& \epsilon[r3] - 16^* \eta[2]^* \epsilon[r2]^* \eta[3] - 16^* \eta[2]^* \epsilon[r3]^* \eta[3] C^8 * \\
& \eta[2]^* \epsilon[r1]^2 * \eta[3] - 16^* \eta[2]^* \epsilon[r3]^2 * \eta[3] C^16 \eta[1]^2 * \\
& \epsilon[r1]^2 * \epsilon[r2] C^4 \eta[1]^2 * \epsilon[r1]^2 * \epsilon[c2]^2 C^4 * \\
& \epsilon[c3]^2 * \eta[3]^2 * \epsilon[c2]^2 C^4 \eta[1]^2 * \epsilon[r1]^2 * \epsilon[r2]^2 * \\
& - 16^* \eta[1]^* \epsilon[r2]^2 * \eta[2] - 16^* \eta[1]^* \epsilon[r3]^2 * \eta[3] - 16^* \\
& \eta[1]^* \epsilon[c2]^2 * \eta[2] - 16^* \eta[1]^* \epsilon[c3]^2 * \eta[3] C^16 \epsilon[c3]
\end{aligned}$$

$$\begin{aligned}
&^2\eta[3]^2\epsilon[r2]C\ 32\eta[2]\epsilon[r3]\eta[1]C\ 4\eta[2]^2\epsilon[r1]^2\epsilon[r2]^2C\ 4\eta[3]^2\epsilon[r1]^2\epsilon[r2]C\ \eta[3]^2\epsilon[r1]^2\epsilon[r2]^2C\ 4\eta[3]^2\epsilon[r1]\epsilon[r2]^2-4\eta[2]\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2\eta[1]-4\eta[2]\epsilon[r2]\epsilon[r3]^2\eta[1]-16\eta[2]\epsilon[r2]\epsilon[r3]\eta[1]-32\eta[3]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]-8\eta[3]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]-8\eta[3]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]^2\eta[3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]^2\eta[3]C\ 4\epsilon[r1]^2\eta[1]^2\epsilon[r2]^2\epsilon[r3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]^2\eta[3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]\eta[3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]\eta[3]-4\eta[2]\epsilon[r1]^2\epsilon[r3]^2\eta[3]C\ 8\eta[2]\epsilon[r1]^2\epsilon[r2]\eta[1]C\ 8\eta[2]\epsilon[r1]\epsilon[r2]\eta[1]C\ 8\eta[2]\epsilon[r2]\epsilon[r3]^2\eta[3]C\ 8\eta[2]\epsilon[r2]\epsilon[r3]\eta[3]C\ 8\eta[2]\epsilon[r2]\epsilon[r3]^2\eta[3]-8\eta[1]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2-2\eta[1]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2-2\eta[1]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2-32\eta[1]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]-8\eta[1]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]-8\eta[1]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]-2\eta[1]^2\epsilon[r1]\epsilon[r2]^2\epsilon[r3]^2-8\eta[1]^2\epsilon[r1]\epsilon[r2]\epsilon[r3]^2-4\eta[3]\epsilon[r2]^2\epsilon[r3]\eta[1]C\ 8\eta[3]\epsilon[r1]^2\epsilon[r3]\eta[1]-4\eta[3]\epsilon[r2]^2\epsilon[r3]\eta[1]-16\eta[3]\epsilon[r1]^2\epsilon[r2]\eta[1]-4\eta[3]\epsilon[r1]^2\epsilon[r2]^2\eta[1]C\ 8\eta[3]\epsilon[r1]\epsilon[r3]\eta[1]-16\eta[3]\epsilon[r2]\epsilon[r3]\eta[1]C\ 8\eta[3]\epsilon[r1]^2\epsilon[r3]\eta[1]-4\eta[3]\epsilon[r1]^2\epsilon[r2]^2\eta[1]C\ \epsilon[r2]^2\eta[2]^2\epsilon[r1]^2\epsilon[r3]^2-2\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2-8\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]C\ 4\epsilon[r1]^2\eta[1]^2\epsilon[r2]\epsilon[r3]^2-16\eta[1]\epsilon[r2]\epsilon[r3]^2\eta[3]-16\eta[1]\epsilon[r2]\epsilon[r3]^2\eta[3]^2\eta[3]C\ 8\eta[3]\epsilon[r2]^2\epsilon[r3]\eta[2]C\ 8\eta[1]\epsilon[r1]\epsilon[r3]^2\eta[3]C\ 4\epsilon[r3]^2\eta[3]^2\epsilon[r1]^2\epsilon[r2]C\ 8\eta[2]\eta[1]\epsilon[r1]^2\epsilon[r2]^2C\ \epsilon[r3]^2\eta[3]^2\epsilon[r1]^2\epsilon[r2]^2C\ 4\epsilon[r2]^2\eta[2]^2\epsilon[r1]\epsilon[r3]^2-2\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2-8\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]-2\eta[2]^2\epsilon[r1]^2\epsilon[r2]\epsilon[r3]^2C\ 4\epsilon[r1]^2\eta[1]^2
\end{aligned}$$

epsilon[c2]^2*epsilon[r3]C epsilon[c1]^2*eta[1]^2*epsilon[c2]^2*epsilon
[c3]^2-4*eta[2]*epsilon[r1]^2*epsilon[r2]*eta[3]-16*eta[2]*epsilon[r1]*
epsilon[r2]*eta[3]-2*eta[3]*epsilon[r1]^2*epsilon[c2]^2*epsilon[c3]^2-32*
eta[1]*epsilon[r1]^2*epsilon[r2]*epsilon[r3]C 16*eta[3]^2*epsilon[r1]*
epsilon[r2]C 4*eta[3]^2*epsilon[r1]*epsilon[c2]^2C eta[3]^2*epsilon[c1]
^2*epsilon[r2]^2C 4*eta[3]^2*epsilon[c1]^2*epsilon[r2]C eta[3]^2*epsilon
[c1]^2*epsilon[c2]^2-32*eta[3]^2*epsilon[r1]*epsilon[r3]-32*eta[3]^2*
epsilon[r2]*epsilon[r3]C eta[2]^2*epsilon[r1]^2*epsilon[r3]^2C 4*eta[2]^2*
epsilon[r1]^2*epsilon[r3]C eta[2]^2*epsilon[r1]^2*epsilon[c3]^2C eta[2]
^2*epsilon[c1]^2*epsilon[r3]^2C 4*eta[2]^2*epsilon[c1]^2*epsilon[r3]C eta
[2]^2*epsilon[c1]^2*epsilon[c3]^2C eta[3]^2*epsilon[r1]^2*epsilon[r2]^2
-8*eta[2]^2*epsilon[r1]^2*epsilon[r2]-32*eta[2]^2*epsilon[r1]*epsilon[r2]
-8*eta[2]^2*epsilon[c1]^2*epsilon[r2]C 4*eta[2]^2*epsilon[r1]*epsilon[r3]
^2C 16*eta[2]^2*epsilon[r1]*epsilon[r3]C 4*eta[2]^2*epsilon[r1]*epsilon
[c3]^2-8*eta[2]^2*epsilon[r2]*epsilon[r3]^2-32*eta[2]^2*epsilon[r2]*
epsilon[r3]-8*eta[2]^2*epsilon[r2]*epsilon[c3]^2C 16*epsilon[c1]^2*eta[1]
^2*epsilon[r2]-16*eta[3]*epsilon[r2]^2*eta[2]-16*eta[3]*epsilon[c2]^2*eta
[2]C 4*eta[1]^2*epsilon[r1]^2*epsilon[r3]^2C 4*epsilon[c3]^2*eta[3]^2*
epsilon[r1]^2C 4*eta[1]^2*epsilon[r1]^2*epsilon[c3]^2C 16*eta[1]^2*
epsilon[r1]^2*epsilon[r3]C 16*eta[2]^2*epsilon[r1]*epsilon[r2]^2C 4*
epsilon[c1]^2*eta[1]^2*epsilon[r2]^2C 16*eta[2]^2*epsilon[r3]*epsilon[r2]
^2C 32*eta[2]*epsilon[r1]*eta[3]-8*eta[3]^2*epsilon[c2]^2*epsilon[r3]C 8*
eta[2]*epsilon[c1]^2*eta[3]-16*eta[2]*epsilon[c3]^2*eta[3]C 32*eta[3]*
epsilon[r2]*eta[1]-2*eta[1]*epsilon[r1]*epsilon[c2]^2*epsilon[r3]^2-8*eta
[1]*epsilon[r1]^2*epsilon[r2]*epsilon[c3]^2-8*eta[1]*epsilon[r1]^2*
epsilon[c2]^2*epsilon[r3]-2*eta[1]*epsilon[r1]^2*epsilon[r2]^2*epsilon
[c3]^2-2*eta[1]*epsilon[r1]*epsilon[r2]^2*epsilon[r3]^2-2*eta[1]*epsilon
[r1]^2*epsilon[c2]^2*epsilon[c3]^2-8*eta[1]*epsilon[r1]*epsilon[r2]*
epsilon[c3]^2-2*eta[1]*epsilon[r1]^2*epsilon[c2]^2*epsilon[r3]^2-32*eta
[1]*epsilon[r1]*epsilon[r2]*epsilon[r3]-2*eta[1]*epsilon[r1]^2*epsilon[r2]
^2*epsilon[r3]^2-8*eta[1]*epsilon[r1]*epsilon[r2]^2*epsilon[r3]-2*eta[1]*
epsilon[c1]^2*epsilon[r2]^2*epsilon[r3]^2-8*eta[1]*epsilon[r1]*epsilon
[c2]^2*epsilon[r3]-8*eta[1]*epsilon[r1]^2*epsilon[r2]*epsilon[r3]^2-2*eta
[1]*epsilon[r1]*epsilon[r2]^2*epsilon[c3]^2-8*eta[1]*epsilon[c1]^2*
epsilon[r2]^2*epsilon[r3]-8*eta[1]*epsilon[r1]*epsilon[r2]*epsilon[r3]^2
-8*eta[1]*epsilon[r1]^2*epsilon[r2]^2*epsilon[r3]-2*eta[1]*epsilon[r1]*
epsilon[c2]^2*epsilon[c3]^2-2*eta[1]*epsilon[c1]^2*epsilon[r2]^2*epsilon
[c3]^2-8*eta[1]*epsilon[c1]^2*epsilon[r2]*epsilon[r3]^2-32*eta[1]*epsilon

```
[c1]^2*epsilon[r2]*epsilon[r3]-8*eta[1]*epsilon[c1]^2*epsilon[r2]*epsilon
[c3]^2-2*eta[1]*epsilon[c1]^2*epsilon[c2]^2*epsilon[r3]^2-8*eta[1]*
epsilon[c1]^2*epsilon[c2]^2*epsilon[r3]-2*eta[1]*epsilon[c1]^2*epsilon
[c2]^2*epsilon[c3]^2-8*eta[2]*epsilon[r1]^2*epsilon[r2]*epsilon[r3]-8*eta
[2]*epsilon[r1]*epsilon[c2]^2*epsilon[r3]^2-2*eta[2]*epsilon[r1]^2*
epsilon[r2]*epsilon[c3]^2-8*eta[2]*epsilon[r1]^2*epsilon[c2]^2*epsilon
[r3]-8*eta[3]^2*epsilon[r1]^2*epsilon[r3])"
```

Let's code the expression in Python now.

Numerical application

$$\varepsilon_1 := -9.04639807977 + I \cdot 28.8437268491; \eta_1 := (1e0 - \eta_2 - \eta_3);$$

$$\begin{aligned} & -9.04639807977 + 28.8437268491 I \\ & 0.7308 \end{aligned} \quad (2.13)$$

$$\varepsilon_2 := -0.62993 + I \cdot 24.915; \eta_2 := 16.87e-2; \#Cr$$

$$\begin{aligned} & -0.62993 + 24.915 I \\ & 0.1687 \end{aligned} \quad (2.14)$$

$$\varepsilon_3 := -17.8077402901e0 + I \cdot 28.97412511e0; \eta_3 := 10.05e-2; \#Ni$$

$$\begin{aligned} & -17.8077402901 + 28.97412511 I \\ & 0.1005 \end{aligned} \quad (2.15)$$

$$\varepsilon_{r1} := \text{Re}(\varepsilon_1) : \varepsilon_{c1} := \text{Im}(\varepsilon_1) : \varepsilon_{r2} := \text{Re}(\varepsilon_2) : \varepsilon_{c2} := \text{Im}(\varepsilon_2) : \varepsilon_{r3} := \text{Re}(\varepsilon_3) : \varepsilon_{c3}$$

$$:= \text{Im}(\varepsilon_3) :$$

$$\begin{aligned} & eps_r; \\ & -7.977157246 \end{aligned} \quad (2.16)$$

$$\begin{aligned} & eps_c \\ & 28.75781330 \end{aligned} \quad (2.17)$$

restart

Formula used by Inam Mirza.

$$\varepsilon_{eff} := \varepsilon_m \cdot \left(1 + \frac{3 \cdot f1 \cdot \text{beta}}{1 - f1 \cdot \text{beta}} \right); \text{beta} := \frac{(\varepsilon_{np} - \varepsilon_m)}{\varepsilon_{np} - 2 \cdot \varepsilon_m}$$

$$\begin{aligned} & 1.625 + \frac{(3.475945863 + 1.205338949 I) f1}{1 + (-0.7130145361 - 0.2472490151 I) f1} \\ & 0.7130145361 + 0.2472490151 I \end{aligned} \quad (1)$$

$$\varepsilon_{np} := 0 - I \cdot 2.8; fI := -2.8 I \tag{2}$$

$$\varepsilon_m := 1.625 - 0 \cdot I \tag{3}$$

$$\begin{aligned} & evalc(\varepsilon_{eff}) \\ & 1.625 + \frac{3.475945863 \, fI \, (1 - 0.7130145361 \, fI)}{(1 - 0.7130145361 \, fI)^2 + 0.06113207547 \, fI^2} \\ & - \frac{0.2980188680 \, fI^2}{(1 - 0.7130145361 \, fI)^2 + 0.06113207547 \, fI^2} \\ & + I \left(\frac{1.205338949 \, fI \, (1 - 0.7130145361 \, fI)}{(1 - 0.7130145361 \, fI)^2 + 0.06113207547 \, fI^2} \right. \\ & \left. + \frac{0.8594241912 \, fI^2}{(1 - 0.7130145361 \, fI)^2 + 0.06113207547 \, fI^2} \right) \tag{4} \end{aligned}$$